Gauge-Invariant Low-Rank Optimization: Balanced Quotient Geometry, Convergence, and Generalization

Anonymous Authors

Abstract

Low-rank factors carry a general-linear symmetry that makes optimizer state depend on the chosen representative. We construct the balanced quotient geometry, derive its product-space metric and gradient, and combine coordinate transport with a Sylvester projection of persistent first-moment state. A second-order pullback identity yields the sharp Hessian bound $2bL_f + \|\nabla f\|_2$; exact residual-gap and deep-chain spectra lead to Schur momentum rates and a Lyapunov-certified local basin. Hessian whitening gives finite-sample risk bounds with quadratic-model coefficient $1/2$, and an identifiable homoscedastic sensing model has leading risk $\sigma_\varepsilon^2 r(m + d - r)/(2n)$, matching its Fisher information. Experiments include 208 matched-budget task-search attempts and 36 frozen-configuration initialization checks: across three model initializations, mean validation-selected test loss decreases by 22.6% on Cora and 82.2% on synthetic SDF fitting relative to AdamW. Twelve-gauge sensing tests attain product-trajectory disagreement at most 5.2×10^{-5}, and complete loss trajectories connect the geometric analysis with measured optimization behavior.

1. Introduction

Low-rank adaptation represents a trainable update by $X = BA$, with $B \in \mathbb{R}^{m\times r}$ and $A \in \mathbb{R}^{r\times d}$ (Hu et al., 2022). The identity $(BQ)(Q^{-1}A) = BA$ introduces an exact $\mathrm{GL}(r)$ symmetry. A loss depending only on X is insensitive to this change of factors, whereas a coordinate-dependent optimizer can distinguish the representatives. This raises two separate questions: how to define a genuine quotient optimization algorithm, and whether its geometric corrections improve completed task outcomes.

We construct the geometry on the balanced factor manifold modulo its residual orthogonal symmetry. This is necessary because the ordinary Euclidean metric on arbitrary factors is not invariant under $\mathrm{GL}(r)$. The resulting quotient is the rank-r product manifold equipped with an explicit metric. Its gradient is a state-dependent product preconditioner, not the ambient matrix gradient. Balancing achieves orbit-optimal factor norms and condition numbers; it does not automatically improve the product condition number or the task loss.

Our analysis distinguishes the raw gradient from the actual update. For every product-only minibatch loss, the raw factor gradient is horizontal exactly. Vertical update energy arises from optimizer transformations and state, rather than from sampling that product loss. Removing this energy preserves the product differential but changes the finite step through mixed horizontal–vertical terms. We derive these terms explicitly. A time-independent trajectory discrepancy bound additionally needs contraction of the actual update map, including state. Strong convexity of the ambient product objective alone does not supply that contraction; a rank-one counterexample demonstrates the distinction.

The main second-order step is a pullback cancellation: Boumal's second-order retraction theorem (Boumal, 2023, Proposition 5.45) reduces the quotient Hessian to a factor Hessian restricted to horizontal lifts. This gives $\|\operatorname{Hess}_g f\| \le 2bL_f + \|\nabla f\|_2$, with residual coefficient one attained by a rank-one example. For Frobenius reconstruction, an Eckart–Young decomposition gives the complete residual-gap spectrum, including the $\sigma_i \pm \tau_k$ modes. The exact state Jacobian then yields the Schur stability interval, optimal local momentum factor, and a Lyapunov-certified nonlinear basin. The joint deep-chain normal-solve spectrum remains distinct from this task-Hessian spectrum.

Statistical bounds are organized around covariance and curvature in the identifiable quotient coordinates. Laurent–Massart and Hsu–Kakade–Zhang concentration applied after Hessian whitening gives a local risk bound with a sharp quadratic-model coefficient $1/2$. Under the declared regular homoscedastic sensing model, influence functions and averaged iterates have leading prediction risk $\sigma_\varepsilon^2 r(m + d - r)/(2n)$; the same constant appears in the Cramér–Rao inequality for locally unbiased tangent estima-

.**AUTHORERR: Missing \icmlcorrespondingauthor.**

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tors. The design covariance cancels from this prediction-risk constant, whereas it remains in product Frobenius error. Global capacity bounds, nonconvex stationarity bounds, and the premises needed to enter the local statistical regime are stated separately.

Our completed controlled studies evaluate graph prediction, synthetic implicit representations, matrix completion, and gauge-perturbed sensing. They include 208 matched-budget search attempts and 36 frozen-configuration initialization checks. With fixed data and three model initializations, validation-selected test loss improves over AdamW by 22.6% on Cora and 82.2% on SDF fitting. Balance-only isolates the contribution of section choice, while the sensing controls directly measure horizontal memory and product-trajectory consistency. Complete baseline outcomes, selection rules, loss trajectories, and numerical-stop records accompany the main comparisons. The mathematical analysis and the empirical measurements use the same declared first-moment update.

2. Related Work

LoRA and its variants improve fine-tuning efficiency by constraining trainable updates to a low-rank form (Hu et al., 2022). Subsequent methods adapt the rank budget, initialization, or learning-rate scaling, including AdaLoRA, DoRA, LoRA+, GaLore, PiSSA, and LoRA-GA (Zhang et al., 2023; Liu et al., 2024; Hayou et al., 2024; Zhao et al., 2024; Meng et al., 2024; Wang et al., 2024). These methods address capacity allocation, optimization scaling, or memory, but they generally optimize a particular factor representative. Our focus is orthogonal: we study the exact $\mathrm{GL}(r)$ symmetry of the factorization and ask which optimizer components are invariant to this symmetry.

Riemannian optimization on fixed-rank matrices provides the mathematical background for quotient geometries, horizontal spaces, and retractions (Absil et al., 2008; Mishra et al., 2014; Boumal, 2023; Vandereycken, 2013; Bonnabel, 2013). Riemannion move the trainable object toward the product-space fixed-rank manifold, where parametrization ambiguity is removed by construction (Bogachev et al., 2025). Intrinsic Muon is closer to a norm-ball or product-space optimization view, emphasizing matrix-norm geometry and a linear-minimization-oracle style direction (Li et al., 2026). Our construction also belongs to fixed-rank Riemannian optimization. We specify the metric induced by the balanced factor manifold, retain the factor interface, and analyze the connection together with persistent first-moment state. Table 1 compares these mechanisms under explicit feature definitions.

This distinction matters for both method and interpretation. Embedded product formulations remove factor ambiguity at the variable level; quotient formulations instead retain a choice of representative and a horizontal lift. Neither mechanism by itself establishes that removing the vertical component of an adaptive factor update improves a task metric. Our connection-based view retains the factors, decomposes optimizer updates into horizontal and vertical components via a Sylvester equation, and then tests balance retraction, horizontal projection, and horizontal momentum separately. The resulting message is more diagnostic than a generic Riemannian-optimizer comparison: balance is theorem-enforced to fix the representative conditioning, while the empirical value of horizontal projection depends on the regime.

There is also a broad literature on symmetries, implicit bias, and deep linear dynamics (Badrinarayanan et al., 2015; Saxe et al., 2014; Chou et al., 2020; Arora et al., 2019; Soltanolkotabi et al., 2025; Hardt et al., 2016). These works show that parameterization can shape optimization even when the represented function class is unchanged. We add a concrete diagnostic for LoRA: the raw gradient is horizontal, but adaptive optimizer transformations can introduce vertical update energy. The vertical component is invisible to the product differential but contributes mixed terms at finite step size. Its trajectory effect depends on the actual state dynamics.

3. Problem Setup

Write $\mathcal{K}_{a,b} = \{X : \mathrm{rank}(X) = r,\ a \le \sigma_r(X) \le \sigma_1(X) \le b\}$, with $0 < a \le b$. We study product losses $F(B, A) = f(BA)$ on full-rank factors. For a tangent $u = (u_B, u_A)$, the product differential is $D\pi[u] = u_B A + B u_A$, whose kernel is $\mathcal{V} = \{(B\Omega, -\Omega A)\}$. At balanced factors $B^\top B = AA^\top = H$, the horizontal lift satisfies $B^\top u_B - u_A A^\top = 0$. Its orthogonal projection is computed by

$$\begin{aligned} H\Omega + \Omega H &= B^\top u_B - u_A A^\top, \\ \Pi_H u &= (u_B - B\Omega, u_A + \Omega A). \end{aligned} \tag{1}$$

Away from balance, the exact Euclidean projector uses $B^\top B$ and $AA^\top$ on the two different sides of the Sylvester equation. This unbalanced projection must not be mistaken for a globally invariant Euclidean $\mathrm{GL}(r)$ connection.

For $X = U\Sigma V^\top$, balanced representatives are $(U\Sigma^{1/2}R, R^\top\Sigma^{1/2}V^\top)$ with $R \in \mathrm{O}(r)$. They attain $\kappa(B) = \kappa(A) = \sqrt{\kappa(X)}$ at the point where balance is applied. The update energy statistic is $\rho_{\mathrm{upd}} = \|u - \Pi_H u\|^2 / \|u\|^2$, measured in the same factor coordinates in which u is defined. A denominator based on the target matrix's condition number is a different statistic from one based on the current product.

Table 1. **Mechanism-level comparison.** Keywords summarize the cited construction. ✓: explicitly included; ✗: not part of that construction. Geometry includes embedded fixed-rank and quotient formulations. Retr. means a regular rank-r retraction, not merely a rank upper bound. State means explicit transport of a persistent tangent state. Diag. means raw-gradient versus update-direction vertical-energy diagnostics. Gen. means a stated statistical generalization bound. Symbols do not rank task performance.

Method	Keywords	Geo-metry	Retr.	State	Diag.	Gen.
Low-rank adaptation and initialization						
LoRA (Hu et al., 2022)	Factor update	✗	✗	✗	✗	✗
LoRA+ (Hayou et al., 2024)	Factor LR ratio	✗	✗	✗	✗	✗
AdaLoRA (Zhang et al., 2023)	Adaptive rank budget	✗	✗	✗	✗	✗
PiSSA (Meng et al., 2024)	Principal-SVD initialization	✗	✗	✗	✗	✗
Fixed-rank geometry and intrinsic optimization						
Fixed-rank geometry (Mishra et al., 2014)	Quotient metric / lift	✓	✓	✓[a]	✗	✗
Riemannion (Bogachev et al., 2025)	Embedded tangent / spectral step	✓	✓	✓	✗	✗
Intrinsic Muon (Li et al., 2026)	Quotient metric / intrinsic LMO	✓	✓	✗[b]	✗	✗
QuotientOpt (ours)	**Balance / Sylvester / horizontal memory**	✓	✓	✓	✓	✓[c]

[a] Geometric tangent/direction transport, including conjugate-gradient directions; not a claim that every algorithm in the cited framework uses EMA momentum. [b] The reported iMuon momentum implementation uses factor buffers (its Appendix K); the intrinsic direction's quotient symmetry is a separate property. This mark does not deny its geometric or convergence results. [c] Conditional stability and capacity bounds; neither an unconditional test-accuracy guarantee nor a claim that balancing shrinks the predictor class. LoRA+'s feature-learning scaling theory is distinct from the Gen. criterion.

4. QuotientOpt and Mathematical Guarantees

Table 2 specifies the update. Balance precedes differentiation, and persistent state is expressed in the same representative as the new gradient. The complete iteration and expanded guarantees are in Appendix A.

Theorem 1 (Balanced quotient geometry)**.** *Let* $X = U\Sigma V^\top \in \mathcal{K}_{a,b}$ *and define*

$$K_X(G) = U\Sigma U^\top G + GV\Sigma V^\top, \tag{2}$$

$$g_X(Z,Y) = \left\langle Z, K_X^\dagger Y \right\rangle_F, \tag{3}$$

$$\operatorname{grad}_g f(X) = K_X(\nabla f(X)). \tag{4}$$

Then $(2b)^{-1}\|Z\|_F^2 \le \|Z\|_g^2 \le a^{-1}\|Z\|_F^2$*. At a nondegenerate quotient critical point, the full factor Hessian has* r^2 *gauge zero directions. Moreover,*

$$\|\operatorname{Hess}_g f(X)\|_{g\to g} \le 2b\|\nabla^2 f(X)\|_{F\to F} + \|\nabla f(X)\|_2. \tag{5}$$

The residual coefficient one is attained.

Boumal's second-order retraction theorem (Boumal, 2023, Proposition 5.45) gives the pullback cancellation. Horizontal lifts, the connection, metric differential, and Levi–Civita calculations are developed in Appendices B–C. For $u = h + (\bar{B}\Omega, -\Omega\bar{A})$, the finite-step identity is

$$X^+_{\text{unproj}} - X^+_{\text{proj}} = \eta^2\big(-h_B\Omega\bar{A} + \bar{B}\Omega h_A - \bar{B}\Omega^2\bar{A}\big). \tag{6}$$

Theorem 2 (Local momentum and deep-chain spectra)**.** *At a regular quotient minimum with Hessian spectrum* $[\mu, \Lambda]$*, the reduced position–memory modes satisfy*

$$z^2 - \{1 + \beta - \eta(1-\beta)\lambda\}z + \beta = 0. \tag{7}$$

The exact stability interval and optimal interval-wide parameters are

$$0 < \eta(1-\beta)\Lambda < 2(1+\beta), \tag{8}$$

$$q_\star = \frac{\sqrt{\Lambda} - \sqrt{\mu}}{\sqrt{\Lambda} + \sqrt{\mu}}, \tag{9}$$

$$\beta_\star = q_\star^2, \qquad \eta_\star = (\mu\Lambda)^{-1/2}. \tag{10}$$

For the balanced depth-L chain, the joint vertical normal operator has

$$\lambda_{abk} = \sigma_a^{2/L} + \sigma_b^{2/L} - 2(\sigma_a\sigma_b)^{1/L}\cos(k\pi/L). \tag{11}$$

Schur modal analysis supplies the rate, and a discrete Lyapunov equation certifies the nonlinear basin. An Eckart–Young decomposition gives the reconstruction spectrum $\{\sigma_i + \sigma_j, \sigma_i \pm \tau_k, \sigma_i\}$. Product and normal-solve spectra, stochastic bounds, averaged iterates, and numerical-error formulas are derived in Appendices D–F and Appendix E.

Theorem 3 (Whitened local risk)**.** *Under the relative-Hessian, score-tail, and localization conditions in Ap-*

Table 2. **QuotientOpt update, in execution order.** $X_t = B_t A_t$, $G_t = \nabla f_{I_t}(X_t)$, $\beta \in [0,1)$, $\eta_t > 0$. Exact horizontal projection uses $\varepsilon = 0$; the implemented regularized solve has residual $2\varepsilon\Omega_t$. Balance-only retains stages 1–4 and 6 with $M_t = U_t$.

Step	Operation	Update / invariant
1	Balance	$(\bar{B}_t, \bar{A}_t) = (B_t Q_t, Q_t^{-1} A_t)$; $\bar{B}_t^\top \bar{B}_t = \bar{A}_t \bar{A}_t^\top = H_t$.
2	Transport memory	$(\widetilde{M}_B, \widetilde{M}_A) = (M_{B,t-1} Q_t, Q_t^{-1} M_{A,t-1})$.
3	Differentiate	$(g_B, g_A) = (G_t \bar{A}_t^\top, \bar{B}_t^\top G_t)$; $\bar{B}_t^\top g_B - g_A \bar{A}_t^\top = 0$.
4	Accumulate	$(U_B, U_A) = \beta(\widetilde{M}_B, \widetilde{M}_A) + (1-\beta)(g_B, g_A)$.
5	**Horizontal memory**	$(H_t + \varepsilon I)\Omega_t + \Omega_t(H_t + \varepsilon I) = \bar{B}_t^\top U_B - U_A \bar{A}_t^\top$; $(M_B, M_A) = (U_B - \bar{B}_t \Omega_t, U_A + \Omega_t \bar{A}_t)$.
6	Factor step	$(B_{t+1}, A_{t+1}) = (\bar{B}_t - \eta_t M_B, \bar{A}_t - \eta_t M_A)$.

Table 3. **Held-out loss and validation-target iteration counts.** Cora and SDF: mean ± sample SD over model initializations 42/43/44, fixed data/shuffle seed 42 and frozen parent-selected learning rates. CiteSeer and matrix: seed 42, equal 13-setting budgets. Test loss uses a validation-selected checkpoint. Matrix entries give the first validation hit of 10^{-4} relative initial MSE; the three-initialization records are in Table 23. Bold marks the smallest reported mean or seed-42 count in each column.

Method	Cora CE ↓	CiteSeer CE ↓	SDF MSE (10^{-5}) ↓	Matrix epoch ↓
AdamW	1.1376 ± 0.0793	1.3762	6.21 ± 0.17	138
SGD-M	$\mathbf{0.8741 \pm 0.0562}$	1.2508	26.19 ± 6.24	> 2000
Balance-only	0.8807 ± 0.0551	1.2168	1.18 ± 0.07	117
QuotientOpt	0.8807 ± 0.0552	**1.2163**	$\mathbf{1.10 \pm 0.26}$	**107**

pendix I.1, define

$$\mathcal{T}_u(\mathsf{V}) = \operatorname{tr}\mathsf{V} + 2\sqrt{\operatorname{tr}(\mathsf{V}^2)u} + 2\|\mathsf{V}\|_2 u. \tag{12}$$

With probability at least $1 - q_H - e^{-u}$,

$$R(\widehat{X}_n) - R(X_\star) \le \frac{1+\epsilon_H}{2n(1-\delta_H)^2}\mathcal{T}_u(\mathsf{V}). \tag{13}$$

For an identifiable homoscedastic sensing model under the stated expectation conditions,

$$n\,\mathbb{E}[R(\widehat{X}_n) - R(X_\star)] \longrightarrow \frac{\sigma_\varepsilon^2}{2} r(m+d-r). \tag{14}$$

Hessian whitening precedes the Laurent–Massart and Hsu–Kakade–Zhang quadratic-form bounds (Laurent & Massart, 2000; Hsu et al., 2012). The quadratic-model coefficient $1/2$ is exact; the asymptotic constant agrees with the identifiable Cramér–Rao bound. Stability, capacity, intrinsic-chart, network, margin, local-risk, and influence-function derivations are given in Appendices H–I.

5. Experiments

Study design. We compare AdamW, SGD with momentum (SGD-M), balance-only, and QuotientOpt. Balance-only uses the same section change, transported first moment, and factor step as QuotientOpt, omitting the Sylvester projection. Geometric runs use normalized EMA momentum, $\beta = 0.9$, regularization $\varepsilon = 10^{-6}$, and balance interval one; SGD-M uses its library momentum convention. Equal tuning budgets apply to all four methods. Shared non-factor optimization and all data protocols are specified in Appendix K.

The additional four-task search contains 208 attempts: 13 optimizer settings per method and task, with 1000 epochs for graphs and 2000 for regression. Of these, 186 complete the full horizon and 22 terminate numerically. We then freeze parent-selected learning rates and complete 36 initialization checks, varying model initialization over 42, 43, and 44 while holding data and shuffle seed 42 fixed. All 244 additional attempts have therefore finished. Every stopped run and finite prefix is recorded in Appendix O; the original exhaustive proposal grid is tracked separately.

Selection protocol. The earliest minimum-validation-loss checkpoint selects the model; the lowest validation loss selects the configuration, with deterministic LR and epoch tie breaks. Test metrics enter neither decision. Cora and CiteSeer use public transductive splits, a rank-4 GCN input adapter, and cross-entropy (CE). The synthetic sphere SDF uses a rank-8 SIREN and MSE. Matrix completion uses a 128×128 planted rank-8 matrix of condition 100 with disjoint 50/10/40% train/validation/test masks. The original accuracy-selected Cora analysis and all seven learning rates are retained in Appendix M.

Graph prediction and implicit representations. Across three fixed-data model initializations, QuotientOpt reduces Cora test CE from AdamW's 1.1376 to 0.8807, a 22.6% mean reduction, with lower loss in all three pairs. SGD-M

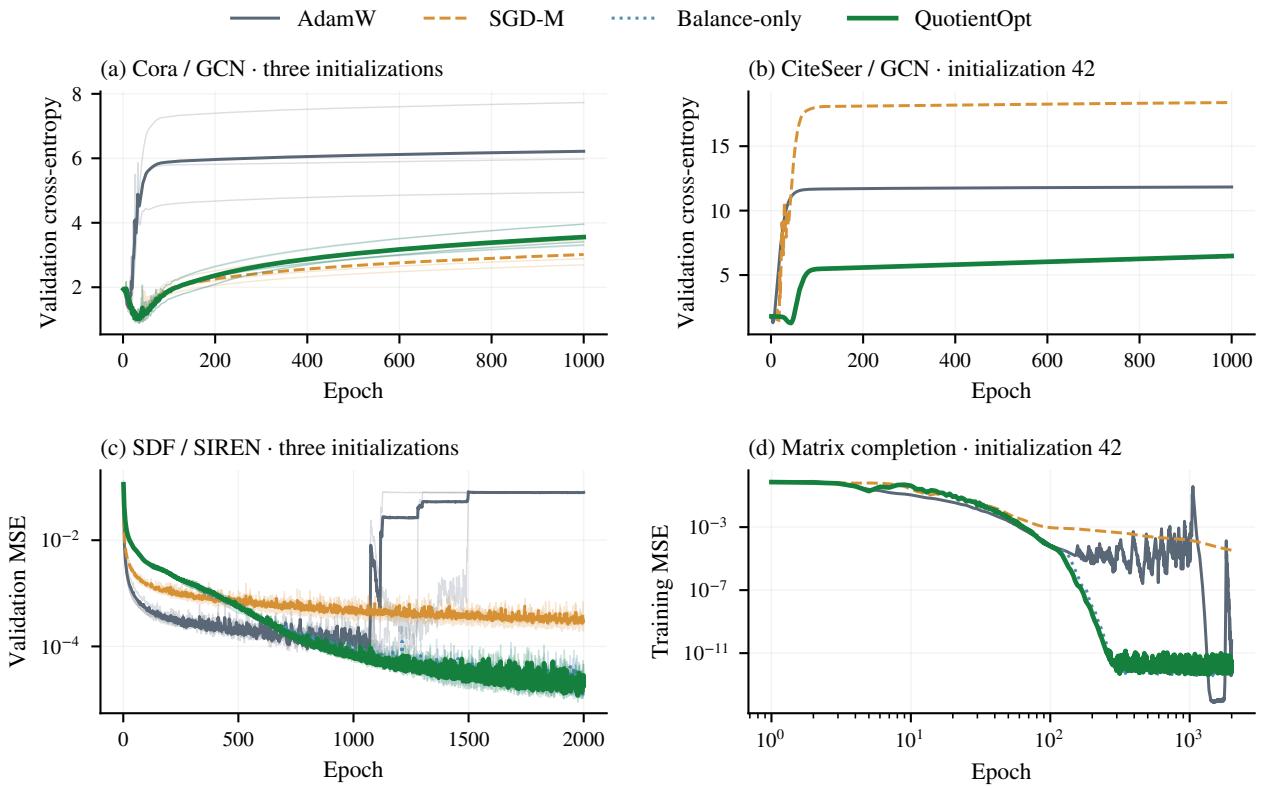


Figure 1. **Complete loss trajectories across three task families.** All methods and recorded epochs are retained, without smoothing. (a,c) Thin lines show three model initializations and thick lines their arithmetic mean, using fixed data and frozen parent-selected learning rates. (b) CiteSeer: actual minimum-validation-CE selected runs over equal 13-setting budgets, seed 42. (d) Matrix completion: actual runs selected by earliest validation hit of the 10^{-4} relative-MSE target; SGD-M uses its minimum-validation-loss completed run because it never hits this target. Both axes in (d) are logarithmic. Test outcomes and iteration counts are in Table 3; complete records appear in Appendices O–Q.

achieves 0.8741 and balance-only 0.8807. On CiteSeer at seed 42, QuotientOpt gives test CE 1.2163, versus 1.3762 for AdamW and 1.2508 for SGD-M; balance-only gives 1.2168. The corresponding accuracies are 65.1%, 45.7%, 54.2%, and 65.1%.

For SDF fitting, three-initialization mean test MSE is 1.10×10^{-5} for QuotientOpt, 6.21×10^{-5} for AdamW, 2.62×10^{-4} for SGD-M, and 1.18×10^{-5} for balance-only: mean reductions of 82.2%, 95.8%, and 6.2%. Paired lower-loss counts are $3/3$, $3/3$, and $2/3$. The joint 13-setting seed-42 selection gives test MSE 9.36×10^{-6} versus AdamW's 6.03×10^{-5}; this comparison also holds when finite checkpoints from subsequently stopped trials are eligible. Table 3 reports sample SDs conditional on the fixed data split and frozen optimizer configurations.

Matrix-completion iteration counts. With a common initial validation scale and equal 13-setting budgets, the seed-42 relative-MSE target 10^{-4} is first attained at epochs 107, 117, and 138 for QuotientOpt, balance-only, and AdamW; SGD-M does not reach it by epoch 2000. This gives 22.5% fewer validation-target iterations than AdamW in the seed-42 tuning comparison. Across the frozen-configuration initialization checks, target attainment is $2/3$ for QuotientOpt and $3/3$ for AdamW; Table 23 gives every count. Final MSEs, additional targets, Burgers/DeepONet outcomes, and all stopped runs appear in Table 17 and Appendix O.

Poisson/PINN fixed-setting result. In a completed four-method cohort of the original proposal grid, the rank-16 Poisson/PINN configuration at learning rate 10^{-3} and seed 42 reaches 2000 epochs for every method. Its final held-out relative L_2 errors are 0.05384 for QuotientOpt, 0.05577 for balance-only, 0.06223 for AdamW, and 0.07767 for SGD-M. Thus the QuotientOpt endpoint is 13.5% below AdamW's and 3.5% below balance-only's at this one matched setting. This is a fixed-configuration comparison, not a completed full-grid selection; all 26 complete four-method cohorts, including unfavorable ones, are reported in Table 25.

Table 4. **Frozen Poisson/PINN initialization check.** Validation-MSE-selected held-out relative L_2 error, rank 16, 2000 epochs, learning rate 10^{-3}. Data and shuffle seed are fixed at 42; columns vary only model initialization. Mean ± sample SD is conditional on that fixed data. All four methods use the same frozen settings.

Method	42	43	44	Mean ± SD
AdamW	0.06143	0.05875	0.06472	0.06163 ± 0.00299
SGD-M	0.07777	0.06438	0.07450	0.07221 ± 0.00698
Balance-only	0.05022	0.04655	0.05255	0.04977 ± 0.00302
QuotientOpt	0.04994	0.04663	0.05255	**0.04971 ± 0.00297**

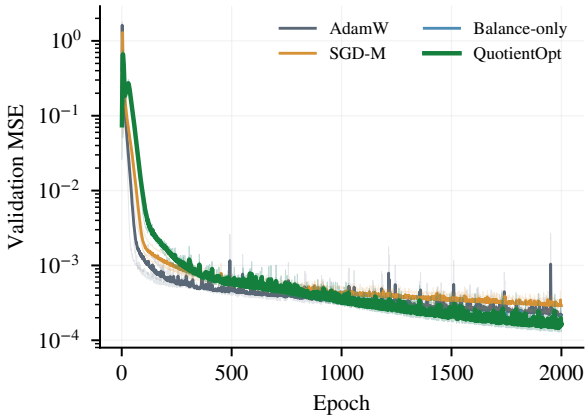


Figure 2. **Poisson/PINN validation loss.** Rank 16, fixed learning rate 10^{-3}, common data and shuffle seed 42. Thin lines show every model initialization (42, 43, 44); thick lines are their arithmetic mean at each epoch. All 2000 validation-MSE observations per run are shown without smoothing. Held-out outcomes are in Table 4.

Poisson/PINN across model initializations. Keeping the same data, shuffle, and rank-16 optimizer settings, we freeze the original learning rate 10^{-3} and run model initializations 43 and 44 for all four methods. Together with the pre-existing initialization 42, each method has three completed 2000-epoch runs. Selecting checkpoints only by validation MSE, the mean held-out relative L_2 error is 0.04971 ± 0.00297 for QuotientOpt, 0.06163 ± 0.00299 for AdamW, 0.07221 ± 0.00698 for SGD-M, and 0.04977 ± 0.00302 for balance-only, where the dispersion is sample SD. QuotientOpt reduces the mean by 19.4% versus AdamW and 31.2% versus SGD-M, with lower test error in all three paired initializations against each. Against balance-only, the mean difference is 0.13% and the paired count is $1/3$; these outcomes do not isolate an accuracy gain from horizontal projection. Table 4 and Figure 2 give the full paired comparison, and Table 27 retains each selected and final-horizon endpoint.

Adjusted-run recovery. All 22 numerical stops in the additional task search were retried from their original initial models with lower factor learning rates. Each adjusted run reaches 2000 epochs. Within its separate retry budget, validation MSE selects the configuration and checkpoint. On Burgers/DeepONet, QuotientOpt's selected test MSE decreases from 4.21×10^{-3} in its original 13-setting search to 2.47×10^{-3} among seven adjusted runs, a 41.5% reduction. On SDF/SIREN, SGD-M's test MSE decreases from 2.42×10^{-4} to 1.09×10^{-4} among six adjusted runs, a 55.0% reduction. The original matched-budget comparisons in Table 17 remain the primary task results. The other three retry groups complete the horizon with higher validation-selected test MSE than their original group best: SDF balance-only rises from 1.10×10^{-5} to 1.35×10^{-4}, SDF QuotientOpt from 9.36×10^{-6} to 1.40×10^{-4}, and matrix-completion SGD-M from 1.17×10^{-8} to 2.48×10^{-8}. AdamW's original Burgers test MSE, 8.66×10^{-5}, is still below the adjusted QuotientOpt result. Table 5 reports every retry group, and Appendix P lists all 22 outcomes.

Gauge robustness and horizontal state. Twelve scalar or dense gauges act on the same initial product, with 1000 updates per attempt. Table 6 separately reports product disagreement, update-direction vertical energy, and identity-gauge reconstruction error. QuotientOpt has maximum completed final disagreement 5.20×10^{-5} and mean vertical energy 1.20×10^{-14}; balance-only gives 5.40×10^{-5} and 1.91×10^{-6}. AdamW has smaller identity-gauge reconstruction error and maximum gauge disagreement 1.07. These measurements connect horizontal memory to representation consistency while distinguishing it from objective minimization. Complete sensing-loss histories appear in Figure 7; all gauges and replay controls are retained in Appendices M and N.

RU01 proposal-grid audit and focused continuation. At the November 4, 2026, 13:05 EST scope decision, 158 formal runs had completed their prescribed horizons, including 115 main configurations; 26 task/rank/setting cohorts had all four methods. Five fixed-step gauge runs retained failure status, and four active runs were paused with checkpoints preserved. The nineteen completions since the earlier 139-run snapshot span all seven task families. At one newly paired Cora/GCN hidden-adapter setting ($r = 4$, learning rate 10^{-3}), the final test accuracies are 77.7% for QuotientOpt and 77.1% for AdamW. SGD-M and balance-only were still pending for that setting. In the newly complete CIFAR-10/MLP rank-32 cohort, QuotientOpt reaches 47.24% final test accuracy versus 49.42% for AdamW, 47.82% for SGD-M, and 47.23% for balance-only. These are fixed-setting final endpoints, not winners from the full validation-selected search. In rank-16 matrix sensing, the complete cohort has final reconstruction errors 0.92767 for QuotientOpt and 0.02616 for AdamW. All nineteen new records, including unfavorable results, appear in Table 26; all complete four-method cohorts remain in Table 25. The remaining

Table 5. **Adjusted fixed-step follow-up after numerical stops.** Every one of the 22 retried configurations reaches 2000 epochs from the bitwise-identical parent initial model. Only the factor learning rate changes. Within each task/method row, the retry setting and checkpoint minimize validation MSE. Original test MSE is the best of the preceding equal 13-setting search, also selected by validation MSE. Relative reduction is $100(1-\mathrm{retry}/\mathrm{original})\%$; a negative value means higher test MSE after retry. The retry counts are a separate tuning budget. Bold indicates a lower test MSE within the same method. All 22 outcomes are in Table 20.

Task / method	Completed	Retry LR	Original test MSE ↓	Retry test MSE ↓	Relative reduction
Burgers / DeepONet / QuotientOpt	7/7	0.00255	0.0042	**0.0025**	+41.5%
SDF / SIREN / SGD-M	6/6	0.0085	2.42×10^{-4}	$\mathbf{1.09\times10^{-4}}$	+55.0%
SDF / SIREN / Balance-only	3/3	0.85	1.10×10^{-5}	1.35×10^{-4}	-1125.4%
SDF / SIREN / QuotientOpt	3/3	0.55	9.36×10^{-6}	1.40×10^{-4}	-1391.8%
Matrix completion / SGD-M	3/3	0.255	1.17×10^{-8}	2.48×10^{-8}	-111.6%

Table 6. **Twelve-gauge sensing control.** Rank 4, 1000 updates, seed 42. $\Delta_T = \|X_T^{(Q)} - X_T^{(I)}\|_F / \|X_*\|_F$; the maximum uses completed gauges. $\bar{\rho}_{\mathrm{upd}}$ is mean update-direction vertical energy. Identity error reports final reconstruction without gauge perturbation. Failed gauges remain counted. All individual gauges and recovery controls are in Appendices M and N.

Method	Finished	Max. Δ_T ↓	$\bar{\rho}_{\mathrm{upd}}$ ↓	Identity error ↓
AdamW	12/12	1.07×10^{0}	1.35×10^{-1}	**0.0460**
SGD-M	7/12	8.86×10^{-1}	5.65×10^{-5}	0.3797
Balance-only	12/12	5.40×10^{-5}	1.91×10^{-6}	0.9994
QuotientOpt	12/12	$\mathbf{5.20\times10^{-5}}$	$\mathbf{1.20\times10^{-14}}$	0.9994

exhaustive queue is replaced by a frozen-configuration Poisson/PINN follow-up: two new model initializations crossed with all four methods at the previous rank-16, learning-rate-10^{-3} setting. All eight new runs reach 2000 epochs and are tracked separately from this snapshot.

6. Conclusion

Balanced quotient geometry gives an explicit product metric and a coordinate-consistent first-moment update. Second-order pullback cancellation, residual-gap spectra, Schur analysis, and Hessian whitening yield sharp local curvature, momentum, and statistical constants under the stated assumptions. Completed controls connect horizontal memory to gauge consistency. Graph prediction and SDF fitting improve validation-selected held-out loss relative to AdamW under the reported protocols. Full baseline comparisons and initialization records identify the contribution of balancing and projection separately. Complete derivations, optimizer settings, loss histories, and run records accompany these results.

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A. Expanded Algorithm and Guarantee Statements

Definition 1 (Iteration and ablation)**.** For $p_t = (B_t, A_t)$, $\beta \in [0, 1)$, $\eta_t > 0$, let

$$(\bar{B}_t, \bar{A}_t) = \mathrm{Bal}(B_t, A_t) = (B_t Q_t, Q_t^{-1} A_t), \tag{15}$$
$$H_t = \bar{B}_t^\top \bar{B}_t = \bar{A}_t \bar{A}_t^\top, \tag{16}$$
$$\widetilde{M}_{B,t-1} = M_{B,t-1} Q_t, \tag{17}$$
$$\widetilde{M}_{A,t-1} = Q_t^{-1} M_{A,t-1}, \tag{18}$$
$$g_{B,t} = \nabla f_{I_t}(X_t) \bar{A}_t^\top, \tag{19}$$
$$g_{A,t} = \bar{B}_t^\top \nabla f_{I_t}(X_t), \tag{20}$$
$$U_{B,t} = \beta \widetilde{M}_{B,t-1} + (1-\beta) g_{B,t}, \tag{21}$$
$$U_{A,t} = \beta \widetilde{M}_{A,t-1} + (1-\beta) g_{A,t}, \tag{22}$$
$$C_t = \bar{B}_t^\top U_{B,t} - U_{A,t} \bar{A}_t^\top, \tag{23}$$
$$H_{\varepsilon,t} = H_t + \varepsilon I, \tag{24}$$
$$H_{\varepsilon,t} \Omega_t + \Omega_t H_{\varepsilon,t} = C_t, \tag{25}$$
$$M_{B,t} = U_{B,t} - \bar{B}_t \Omega_t, \tag{26}$$
$$M_{A,t} = U_{A,t} + \Omega_t \bar{A}_t, \tag{27}$$
$$B_{t+1} = \bar{B}_t - \eta_t M_{B,t}, \tag{28}$$
$$A_{t+1} = \bar{A}_t - \eta_t M_{A,t}. \tag{29}$$

The balance-only ablation sets $(M_{B,t}, M_{A,t}) = (U_{B,t}, U_{A,t})$. The analyzed first-moment iteration is distinct from an elementwise second-moment adaptive heuristic.

The proof uses the second-order pullback-Hessian theorem (Boumal, 2023, Proposition 5.45) to combine the connection terms exactly. Schur modal analysis and a discrete Lyapunov certificate then control the actual momentum state. For statistics, whitening precedes the Laurent–Massart quadratic-form bound, and the resulting leading constant is compared with Fisher information.

Theorem 4 (Quotient metric, stationarity, and finite-step decomposition)**.** *For $X = U\Sigma V^\top \in \mathcal{K}_{a,b}$,*

$$K_X(G) = U\Sigma U^\top G + GV\Sigma V^\top, \qquad g_X(Z, Y) = \left\langle Z, K_X^\dagger Y \right\rangle_F, \tag{30}$$
$$\mathrm{grad}_g\, f(X) = K_X(\nabla f(X)), \qquad (2b)^{-1} \|Z\|_F^2 \le \|Z\|_g^2 \le a^{-1} \|Z\|_F^2, \tag{31}$$
$$\bar{B}^\top g_B - g_A \bar{A}^\top = 0. \tag{32}$$

For $u = h + (\bar{B}\Omega, -\Omega\bar{A})$,

$$X^+_{\mathrm{unproj}} - X^+_{\mathrm{proj}} = \eta^2(-h_B \Omega \bar{A} + \bar{B}\Omega h_A - \bar{B}\Omega^2 \bar{A}). \tag{33}$$

For a unit horizontal lift ξ and $G = \nabla f(X)$,

$$\mathrm{Hess}_g f[L_p\xi, L_p\xi] = \left\langle L_p\xi, \nabla^2 f[L_p\xi] \right\rangle_F + 2 \left\langle G, \xi_B \xi_A \right\rangle_F, \tag{34}$$
$$\|\mathrm{Hess}_g\, f(X)\|_{g\to g} \le 2b \|\nabla^2 f(X)\|_{F\to F} + \|G\|_2. \tag{35}$$

The residual coefficient one is attained (Appendix C.3). At a nondegenerate quotient critical point the full factor Hessian has exactly r^2 gauge zero directions. Complete derivations are in Appendices B–C.

Theorem 5 (Stochastic and momentum convergence)**.** *Under the regularity, pullback, oracle, and stepsize assumptions in Appendix E, a biased quotient-gradient update satisfies*

$$\frac{\sum_{t<T} \eta_t \mathbb{E}\, \|g_t\|_g^2}{S_T} \le \frac{2\Delta_0 + 4\sum_{t<T} \eta_t \mathbb{E}\, \|b_t\|_g^2 + L_R \sum_{t<T} \eta_t^2 \nu_t^2}{S_T}, \quad S_T = \sum_{t<T} \eta_t. \tag{36}$$

For transported normalized momentum satisfying Eq. (444)*,*

$$\frac{1}{T}\sum_{t<T}\mathbb{E}\,\|g_t\|_g^2 \le \frac{2\Delta_0}{\eta T} + \frac{2q_0}{(1-\beta)T} + 2(1-\beta)\nu^2. \tag{37}$$

For noiseless RIP sensing in an invariant geodesically convex basin with $k_-(X) \ge a$ and $\|X - X_*\|_2 + \delta\|X - X_*\|_F \le \omega < a(1-\delta)$,

$$\mu_{\mathrm{RIP}} = a(1-\delta) - \omega, \qquad f(X_T) \le (1 - \eta\mu_{\mathrm{RIP}})^T f(X_0). \tag{38}$$

The noisy, high-probability, numerical-error, and admissible-basin statements are proved separately in Appendix E.

Theorem 6 (Joint deep-chain spectrum)**.** *For the canonical balanced depth-L chain, its joint vertical normal operator has eigenvalues*

$$\lambda_{abk} = \sigma_a^{2/L} + \sigma_b^{2/L} - 2(\sigma_a\sigma_b)^{1/L}\cos(k\pi/L), \tag{39}$$

$$\kappa(\mathcal{V}_W^*\mathcal{V}_W) = \kappa(X)^{2/L}\cot^2(\pi/(2L)). \tag{40}$$

Its product preconditioner has core eigenvalues

$$\chi_{ab}^{(L)} = \sum_{i=1}^{L} \sigma_a^{2(L-i)/L}\sigma_b^{2(i-1)/L}. \tag{41}$$

The normal solve and product-optimization spectra are different objects. Appendix D gives diagonalization, damping error, complexity, and aligned-mode hitting-time proofs.

Theorem 7 (Sharp local momentum factor)**.** *At a regular quotient minimum with Hessian spectrum in $[\mu, \Lambda]$, the reduced position–memory Jacobian has characteristic polynomials*

$$z^2 - \{1 + \beta - \eta(1-\beta)\lambda\}z + \beta, \quad \lambda \in \mathrm{spec}(\mathrm{Hess}_g f). \tag{42}$$

Its exact stability condition and optimal interval-wide factor are

$$0 < \eta(1-\beta)\Lambda < 2(1+\beta), \tag{43}$$

$$q_\star = \frac{\sqrt{\Lambda} - \sqrt{\mu}}{\sqrt{\Lambda} + \sqrt{\mu}}, \qquad \beta_\star = q_\star^2, \qquad \eta_\star = (\mu\Lambda)^{-1/2}. \tag{44}$$

A discrete Lyapunov equation gives the invariant nonlinear contraction ball in Eq. (581)*. For Frobenius reconstruction at the best rank-r approximation to Y, with a strict retained/omitted singular-value gap and a nonempty off-core tangent block,*

$$\mu = \sigma_r(Y) - \sigma_{r+1}(Y), \qquad \Lambda = 2\sigma_1(Y). \tag{45}$$

These extrema follow from the complete spectrum $\{\sigma_i + \sigma_j, \sigma_i \pm \tau_k, \sigma_i\}$ in Appendix F.

Theorem 8 (Statistical guarantees under declared sampling assumptions)**.** *For adjacent IID datasets and the actual state-map regularity in Eq.* (667)*,*

$$\epsilon_{\mathrm{stab}} \le \frac{2C_\ell C_U}{n}\sum_{t<T}\eta_t\prod_{k>t}(1 + L_U\eta_k). \tag{46}$$

For a bounded product class and empirical suboptimality ϵ_{opt},

$$R(X_T) - \inf_{\mathcal{C}} R \le \epsilon_{\mathrm{opt}} + 2U_n, \tag{47}$$

$$\mathbb{E}R(X_T) - \inf_{\mathcal{C}} R \le \epsilon_{\mathrm{opt}} + \epsilon_{\mathrm{stab}}. \tag{48}$$

Here U_n may be the nuclear-norm, intrinsic-chart, network, or margin bound in Appendix H. A squared-gradient guarantee alone does not supply ϵ_{opt}.

Theorem 9 (Whitened local risk and its efficiency constant)**.** *Under the relative-Hessian, score-tail, and localization conditions of Appendix I.1, with probability at least* $1 - q_H - e^{-u}$*,*

$$R(\widehat{X}_n) - R(X_\star) \le \frac{1+\epsilon_H}{2n(1-\delta_H)^2}\{\operatorname{tr} \mathsf{V} + 2\sqrt{\operatorname{tr}(\mathsf{V}^2)u} + 2\|\mathsf{V}\|_2 u\}. \tag{49}$$

For a correctly specified identifiable sensing model with homoscedastic noise variance σ_ε^2*, the local influence and averaging results give, under their stated expectation conditions,*

$$n\,\mathbb{E}[R(\widehat{X}_n) - R(X_\star)] \longrightarrow \frac{\sigma_\varepsilon^2}{2} r(m+d-r). \tag{50}$$

The same constant is the Cramér–Rao lower bound for locally unbiased estimators under quadratic tangent loss. The finite-sample coefficient $1/2$ *is an equality for a quadratic chart model with exact reference Hessian (Appendix I).*

B. Regular Quotients, Balanced Lifts, and Exact Finite-Step Identities

B.1. Spaces, actions, and local coordinates

Definition 2.

$$1 \le r \le \min(m,d), \qquad 0 < a \le b < \infty, \tag{51}$$

$$\mathcal{P} = \mathbb{R}_*^{m\times r} \times \mathbb{R}_*^{r\times d}, \tag{52}$$

$$\pi(B,A) = BA, \tag{53}$$

$$\Phi_Q(B,A) = (BQ, Q^{-1}A), \qquad Q \in \mathrm{GL}(r), \tag{54}$$

$$\mathcal{M}_r = \{X : \operatorname{rank} X = r\}, \qquad \mathcal{Q} = \mathcal{P}/\mathrm{GL}(r), \tag{55}$$

$$\mathcal{S} = \{(B,A) \in \mathcal{P} : B^\top B = AA^\top\}, \tag{56}$$

$$\mathcal{K}_{a,b} = \{X \in \mathcal{M}_r : a \le \sigma_r(X),\ \sigma_1(X) \le b\}. \tag{57}$$

The two starred factor spaces have full column and row rank, respectively.

Theorem 10 (Principal quotient and product identification)**.**

$$\pi(p) = \pi(p') \iff \exists! Q \in \mathrm{GL}(r) : p' = \Phi_Q(p), \tag{58}$$

$$\mathcal{Q} \simeq \mathcal{M}_r, \qquad \dim \mathcal{Q} = r(m+d-r), \tag{59}$$

$$D\pi_p[\xi] = \xi_B A + B\xi_A, \qquad \ker D\pi_p = \{(B\Omega, -\Omega A) : \Omega \in \mathbb{R}^{r\times r}\}. \tag{60}$$

The action is free and proper.

Proof.

$$BA = B'A' \Longrightarrow \operatorname{range} B = \operatorname{range} B' \Longrightarrow B' = BQ, \quad Q = B^\dagger B' \in \mathrm{GL}(r), \tag{61}$$

$$BA = BQA' \Longrightarrow A = QA' \Longrightarrow A' = Q^{-1}A, \tag{62}$$

$$BQ = B \Longrightarrow Q = B^\dagger B = I_r. \tag{63}$$

For $p_k \to p \in \mathcal{P}$ and $\Phi_{Q_k} p_k \to p' \in \mathcal{P}$,

$$Q_k = B_k^\dagger B_k' \to B^\dagger B', \tag{64}$$

$$Q_k^{-1} = (B_k')^\dagger B_k \to (B')^\dagger B, \qquad (B^\dagger B')((B')^\dagger B) = I_r. \tag{65}$$

For $|I| = |J| = r$ and $\det X_{I,J} \ne 0$,

$$s_{I,J}(X) = (X_{:,J}, X_{I,J}^{-1} X_{I,:}), \qquad \pi(s_{I,J}(X)) = X. \tag{66}$$

These maps give smooth local inverse quotient maps. Furthermore,

$$D\pi_p[\xi] = 0 \Longrightarrow (I - BB^\dagger)\xi_B A = 0 \tag{67}$$
$$\Longrightarrow \xi_B = B\Omega, \quad \Omega = B^\dagger \xi_B \tag{68}$$
$$\Longrightarrow B(\Omega A + \xi_A) = 0 \Longrightarrow \xi_A = -\Omega A, \tag{69}$$
$$\dim \mathcal{Q} = mr + rd - r^2. \tag{70}$$

□

Proposition 1 (Explicit regular rank chart)**.** *Fix $X = U\Sigma V^\top$ and orthogonal completions $U_\perp, V_\perp$. For $C \in \mathrm{GL}(r)$, $D \in \mathbb{R}^{r\times(d-r)}$, $E \in \mathbb{R}^{(m-r)\times r}$,*

$$\Psi(C, D, E) = [U, U_\perp] \begin{pmatrix} C & D \\ E & EC^{-1}D \end{pmatrix} [V, V_\perp]^\top. \tag{71}$$

This is a rank-r chart near $(\Sigma, 0, 0)$ with $r^2 + r(d-r) + (m-r)r = r(m+d-r)$ coordinates.

Proof.

$$\begin{pmatrix} C & D \\ E & EC^{-1}D \end{pmatrix} = \begin{pmatrix} I \\ EC^{-1} \end{pmatrix} C \begin{pmatrix} I & C^{-1}D \end{pmatrix}, \tag{72}$$
$$\operatorname{rank} \begin{pmatrix} C & D \\ E & F \end{pmatrix} = r \iff F - EC^{-1}D = 0. \tag{73}$$

Writing $\Delta = (\Delta C, \Delta D, \Delta E)$ and $J = C^{-1}$,

$$D(EC^{-1}D)[\Delta] = \Delta E J D + E J \Delta D - E J \Delta C J D, \tag{74}$$
$$D(C^{-1})[K] = -JKJ, \qquad D^2(C^{-1})[K, L] = JKJLJ + JLJKJ. \tag{75}$$

At $(\Sigma, 0, 0)$,

$$D\Psi[\Delta] = [U, U_\perp] \begin{pmatrix} \Delta C & \Delta D \\ \Delta E & 0 \end{pmatrix} [V, V_\perp]^\top, \tag{76}$$
$$D^2\Psi[\Delta, \Gamma] = U_\perp(\Delta E \Sigma^{-1} \Gamma D + \Gamma E \Sigma^{-1} \Delta D) V_\perp^\top. \tag{77}$$

□

B.2. Orbit optimality and residual symmetry

Theorem 11 (Balanced representatives)**.** *For $X = U\Sigma V^\top \in \mathcal{M}_r$,*

$$\pi^{-1}(X) \cap \mathcal{S} = \{(U\Sigma^{1/2}R, R^\top \Sigma^{1/2} V^\top) : R \in \mathrm{O}(r)\}, \tag{78}$$
$$\min_{BA=X} \tfrac{1}{2}(\|B\|_F^2 + \|A\|_F^2) = \|X\|_*, \tag{79}$$
$$\min_{BA=X} \max(\|B\|_2^2, \|A\|_2^2) = \|X\|_2, \tag{80}$$
$$\min_{BA=X} \sqrt{\kappa(B)\kappa(A)} = \sqrt{\kappa(X)}, \tag{81}$$
$$B^\top B = AA^\top \Longrightarrow \kappa(B) = \kappa(A) = \sqrt{\kappa(X)}. \tag{82}$$

Proof. Set $B_0 = U\Sigma^{1/2}$, $A_0 = \Sigma^{1/2}V^\top$. For $B^\top B = AA^\top = H$ and a second balanced pair $(BQ, Q^{-1}A)$, put

$P = QQ^\top$. Then

$$Q^\top HQ = Q^{-1}HQ^{-\top} \Longrightarrow PHP = H, \tag{83}$$

$$(H^{1/2}PH^{1/2})^2 = H^2 \Longrightarrow H^{1/2}PH^{1/2} = H \Longrightarrow P = I, \tag{84}$$

$$\|X\|_* \le \|B\|_F \|A\|_F \le \tfrac{1}{2}(\|B\|_F^2 + \|A\|_F^2), \tag{85}$$

$$\|X\|_2 \le \|B\|_2 \|A\|_2 \le \max(\|B\|_2^2, \|A\|_2^2), \tag{86}$$

$$\sigma_1(X) \le \sigma_1(B)\sigma_1(A), \qquad \sigma_r(X) \ge \sigma_r(B)\sigma_r(A), \tag{87}$$

$$\kappa(X) \le \kappa(B)\kappa(A), \qquad \sigma_i(B_0) = \sigma_i(A_0) = \sqrt{\sigma_i(X)}, \tag{88}$$

$$\tfrac{1}{2}(\|B_0\|_F^2 + \|A_0\|_F^2) = \sum_i \sigma_i(X). \tag{89}$$

□

Lemma 1 (Balanced tangent equations)**.**

$$C_p(\xi) = B^\top \xi_B - \xi_A A^\top, \qquad V_p(\Omega) = (B\Omega, -\Omega A), \tag{90}$$

$$T_p\mathcal{S} = \{\xi : \operatorname{sym} C_p(\xi) = 0\}, \tag{91}$$

$$\mathcal{V}_p^O = \{V_p(\Omega) : \Omega^\top = -\Omega\}, \tag{92}$$

$$\mathcal{H}_p = T_p\mathcal{S} \cap (\mathcal{V}_p^O)^\perp = \ker C_p. \tag{93}$$

The induced Euclidean metric on $\mathcal{S}$ descends to $\mathcal{S}/\mathrm{O}(r) \simeq \mathcal{M}_r$.

Proof.

$$D(B^\top B - AA^\top)[\xi] = C_p(\xi) + C_p(\xi)^\top, \tag{94}$$

$$\langle \xi, V_p(\Omega)\rangle_F = \langle C_p(\xi), \Omega\rangle_F, \tag{95}$$

$$\xi \perp \mathcal{V}_p^O \iff \operatorname{skew} C_p(\xi) = 0, \tag{96}$$

$$\operatorname{sym} C_p(\xi) = \operatorname{skew} C_p(\xi) = 0 \iff C_p(\xi) = 0, \tag{97}$$

$$D(B^\top B - AA^\top)[V_p(\Omega)] = 2(H\Omega + \Omega H), \quad \Omega^\top = \Omega, \tag{98}$$

$$\lambda_{\min}(\Omega \mapsto H\Omega + \Omega H) = 2\lambda_{\min}(H) > 0, \tag{99}$$

$$\|\xi_B R\|_F^2 + \left\|R^\top \xi_A\right\|_F^2 = \|\xi_B\|_F^2 + \|\xi_A\|_F^2, \quad R^\top R = I. \tag{100}$$

□

B.3. Minimum-energy lift and product-space metric

Proposition 2 (Exact projector)**.** *For arbitrary full-rank factors, define*

$$\mathscr{L}_p(\Omega) = (B^\top B)\Omega + \Omega(AA^\top), \tag{101}$$

$$\Omega_u = \mathscr{L}_p^{-1} C_p(u), \qquad \Pi_p u = u - V_p\Omega_u. \tag{102}$$

Then

$$\Pi_p^2 = \Pi_p = \Pi_p^*, \qquad \operatorname{range}\Pi_p = \ker C_p, \qquad D\pi_p\Pi_p = D\pi_p, \tag{103}$$

$$\operatorname{lift}_p(Z) = \operatorname*{argmin}_{D\pi_p\xi = Z} \tfrac{1}{2}(\|\xi_B\|_F^2 + \|\xi_A\|_F^2). \tag{104}$$

The quotient interpretation is at $p \in \mathcal{S}$.

Proof.

$$V_p^* = C_p, \tag{105}$$
$$C_p V_p = \mathscr{L}_p = \mathscr{L}_p^* \succ 0, \tag{106}$$
$$\Pi_p = I - V_p (V_p^* V_p)^{-1} V_p^*, \tag{107}$$
$$\Pi_p^2 = I - 2V_p \mathscr{L}_p^{-1} C_p + V_p \mathscr{L}_p^{-1} (C_p V_p) \mathscr{L}_p^{-1} C_p = \Pi_p, \tag{108}$$
$$D\pi_p V_p(\Omega) = B\Omega A - B\Omega A = 0, \tag{109}$$
$$D\pi_p \xi = D\pi_p \xi^\star \Longrightarrow \xi - \xi^\star \in \mathcal{V}_p, \tag{110}$$
$$\xi^\star \in \mathcal{H}_p \Longrightarrow \|\xi\|_F^2 = \|\xi^\star\|_F^2 + \|\xi - \xi^\star\|_F^2 . \tag{111}$$

□

Theorem 12 (Metric and full tangent spectrum)**.** *For $X = U\Sigma V^\top$, put*

$$P_X(Z) = UU^\top Z + ZVV^\top - UU^\top ZVV^\top, \tag{112}$$
$$T_X \mathcal{M}_r = \{Z : (I - UU^\top) Z (I - VV^\top) = 0\}, \tag{113}$$
$$K_X(G) = U\Sigma U^\top G + GV\Sigma V^\top, \qquad g_X(Z,Y) = \left\langle Z, K_X^\dagger Y \right\rangle_F . \tag{114}$$

For $G = \nabla f(X)$,

$$\mathrm{lift}_p(\mathrm{grad}_g f) = (GA^\top, B^\top G), \qquad \mathrm{grad}_g f(X) = K_X(G), \tag{115}$$
$$\left\|\mathrm{grad}_g f\right\|_g^2 = \left\|GA^\top\right\|_F^2 + \left\|B^\top G\right\|_F^2 , \tag{116}$$
$$\mathrm{spec}(K_X|_{T_X}) = \{\sigma_i + \sigma_j\}_{i,j \le r} \cup \{\sigma_i\}_{i \le r}^{d-r} \cup \{\sigma_j\}_{j \le r}^{m-r}. \tag{117}$$

For $X \in \mathcal{K}_{a,b}$,

$$(2b)^{-1} \|Z\|_F^2 \le \|Z\|_g^2 \le a^{-1} \|Z\|_F^2 , \tag{118}$$
$$a \|P_X G\|_F^2 \le \left\|\mathrm{grad}_g f\right\|_g^2 \le 2b \|P_X G\|_F^2 . \tag{119}$$

Proof. Let $L_p = D\pi_p$. Then

$$\langle L_p \xi, G \rangle_F = \left\langle \xi_B, GA^\top \right\rangle_F + \left\langle \xi_A, B^\top G \right\rangle_F , \tag{120}$$
$$L_p^* G = (GA^\top, B^\top G), \tag{121}$$
$$L_p L_p^* G = BB^\top G + GA^\top A = K_X(G), \tag{122}$$
$$C_p(L_p^* G) = 0, \tag{123}$$
$$\mathrm{lift}_p Z = L_p^* K_X^\dagger Z, \tag{124}$$
$$\langle \mathrm{lift}_p Z, \mathrm{lift}_p Y \rangle_F = \left\langle Z, K_X^\dagger Y \right\rangle_F , \tag{125}$$
$$g_X(K_X G, Z) = \langle P_X G, Z \rangle_F = Df(X)[Z]. \tag{126}$$

In the orthogonal U, V block bases,

$$Z \equiv \begin{pmatrix} Z_{11} & Z_{12} \\ Z_{21} & 0 \end{pmatrix}, \tag{127}$$
$$K_X Z \equiv \begin{pmatrix} \Sigma Z_{11} + Z_{11}\Sigma & \Sigma Z_{12} \\ Z_{21}\Sigma & 0 \end{pmatrix}, \tag{128}$$
$$g_X(Z,Z) = \sum_{i,j \le r} \frac{|Z_{11,ij}|^2}{\sigma_i + \sigma_j} + \sum_{i,j} \frac{|Z_{12,ij}|^2}{\sigma_i} + \sum_{i,j} \frac{|Z_{21,ij}|^2}{\sigma_j}. \tag{129}$$

All denominators lie in $[a, 2b]$; empty blocks are omitted. □

Corollary 1 (Samplewise horizontality)**.**

$$f_z = f_z(BA) \Longrightarrow \nabla(f_z \circ \pi) = (G_z A^\top, B^\top G_z), \tag{130}$$

$$C_p(\nabla(f_z \circ \pi)) = 0, \qquad \rho_{\mathrm{grad}} = 0. \tag{131}$$

No expectation or minibatch averaging is required.

B.4. Retraction, mixed terms, and coordinate transport

Proposition 3 (Second-order balanced quotient retraction)**.** *For $p \in \mathcal{S}$, $\xi \in \mathcal{H}_p$, define*

$$R_X(\xi) = (B + \xi_B)(A + \xi_A), \tag{132}$$

$$\|\xi_B\|_2 < \sigma_r(B), \quad \|\xi_A\|_2 < \sigma_r(A) \Longrightarrow R_X(\xi) \in \mathcal{M}_r. \tag{133}$$

Then

$$R_X(0) = X, \qquad DR_X(0)[\xi] = L_p\xi, \tag{134}$$

$$R_X(-\eta\xi) = X - \eta L_p\xi + \eta^2 \xi_B\xi_A, \qquad \|\xi_B\xi_A\|_F \le \tfrac{1}{2}\|\xi\|_F^2, \tag{135}$$

$$\nabla^g_{\dot\gamma}\dot\gamma(0) = 0, \qquad \gamma(t) = R_X(t\xi). \tag{136}$$

Proof.

$$\sigma_r(B + \xi_B) \ge \sigma_r(B) - \|\xi_B\|_2 > 0, \tag{137}$$

$$\sigma_r(A + \xi_A) \ge \sigma_r(A) - \|\xi_A\|_2 > 0, \tag{138}$$

$$\begin{aligned}(B + t\xi_B)^\top(B + t\xi_B) - (A + t\xi_A)(A + t\xi_A)^\top \\ = t^2(\xi_B^\top\xi_B - \xi_A\xi_A^\top).\end{aligned} \tag{139}$$

The local balancing equation gives

$$Q(0) = I, \qquad Q'(0) = 0, \tag{140}$$

$$s(t) = \Phi_{Q(t)}(B + t\xi_B, A + t\xi_A) \in \mathcal{S}, \tag{141}$$

$$s'(0) = \xi, \qquad s''(0) = V_p(Q''(0)), \tag{142}$$

$$\left(\nabla^g_{\dot\gamma}\dot\gamma(0)\right)^\uparrow = \Pi_p s''(0) = 0. \tag{143}$$

□

Proposition 4 (Exact horizontal–vertical discrepancy)**.** *For $u = h + V_p\Omega$, $h = \Pi_p u$,*

$$\begin{aligned}(B - \eta u_B)(A - \eta u_A) - (B - \eta h_B)(A - \eta h_A) \\ = \eta^2(-h_B\Omega A + B\Omega h_A - B\Omega^2 A),\end{aligned} \tag{144}$$

$$\|\Delta X_{\mathrm{unproj}} - \Delta X_{\mathrm{proj}}\|_F \le \eta^2\{\sqrt{2b}\,\|h\|_F\,\|\Omega\|_F + b\,\|\Omega\|_F^2\}. \tag{145}$$

The inequality assumes $p \in \mathcal{S}$, $\sigma_1(X) \le b$.

Proof.

$$L_p V_p\Omega = 0, \tag{146}$$

$$(h_B + B\Omega)(h_A - \Omega A) - h_B h_A = -h_B\Omega A + B\Omega h_A - B\Omega^2 A, \tag{147}$$

$$\|h_B\Omega A\|_F + \|B\Omega h_A\|_F \le \sqrt{b}(\|h_B\|_F + \|h_A\|_F)\,\|\Omega\|_F \tag{148}$$

$$\le \sqrt{2b}\,\|h\|_F\,\|\Omega\|_F, \tag{149}$$

$$\left\|B\Omega^2 A\right\|_F \le b\left\|\Omega^2\right\|_F \le b\,\|\Omega\|_F^2. \tag{150}$$

□

Proposition 5 (Damped and inexact connection)**.** *For $aI \preceq H \preceq bI$, let*

$$H\Omega_0 + \Omega_0 H = C, \tag{151}$$

$$(H + \varepsilon I)\widehat{\Omega} + \widehat{\Omega}(H + \varepsilon I) = C + E, \quad \varepsilon \geq 0, \tag{152}$$

$$\widehat{h} = u - V_p\widehat{\Omega}, \qquad h_0 = \Pi_p u. \tag{153}$$

Then

$$\left\|\widehat{\Omega} - \Omega_0\right\|_F \leq \frac{\varepsilon \left\|C\right\|_F}{2a(a+\varepsilon)} + \frac{\left\|E\right\|_F}{2(a+\varepsilon)}, \tag{154}$$

$$\left\|\widehat{h} - h_0\right\|_F \leq \sqrt{2b}\left\{\frac{\varepsilon \left\|C\right\|_F}{2a(a+\varepsilon)} + \frac{\left\|E\right\|_F}{2(a+\varepsilon)}\right\}, \tag{155}$$

$$C_p(\widehat{h}) = 2\varepsilon\widehat{\Omega} - E. \tag{156}$$

Proof. With $H = S\,\mathrm{diag}(\lambda_i)S^\top$,

$$(S^\top \Omega_0 S)_{ij} = \frac{(S^\top C S)_{ij}}{\lambda_i + \lambda_j}, \tag{157}$$

$$(S^\top \widehat{\Omega} S)_{ij} = \frac{(S^\top (C+E) S)_{ij}}{\lambda_i + \lambda_j + 2\varepsilon}, \tag{158}$$

$$\left|\frac{1}{\lambda_i + \lambda_j + 2\varepsilon} - \frac{1}{\lambda_i + \lambda_j}\right| \leq \frac{\varepsilon}{2a(a+\varepsilon)}, \tag{159}$$

$$\left\|V_p D\right\|_F^2 = \mathrm{tr}(D^\top H D) + \mathrm{tr}(D H D^\top) \leq 2b\left\|D\right\|_F^2, \tag{160}$$

$$C_p(\widehat{h}) = C - H\widehat{\Omega} - \widehat{\Omega}H = 2\varepsilon\widehat{\Omega} - E. \tag{161}$$

□

Proposition 6 (Vector action and covector action)**.** *For $p' = \Phi_Q p$,*

$$D\Phi_Q(\xi) = (\xi_B Q, Q^{-1}\xi_A), \tag{162}$$

$$L_{p'} D\Phi_Q \xi = L_p \xi, \tag{163}$$

$$\left\|D\Phi_Q \xi\right\|_F \leq \max(\left\|Q\right\|_2, \left\|Q^{-1}\right\|_2)\left\|\xi\right\|_F, \tag{164}$$

$$\nabla_B(f \circ \pi)(p') = \nabla_B(f \circ \pi)(p) Q^{-\top}, \tag{165}$$

$$\nabla_A(f \circ \pi)(p') = Q^\top \nabla_A(f \circ \pi)(p). \tag{166}$$

For $Q \in \mathrm{O}(r)$,

$$\left\|D\Phi_Q \xi\right\|_F = \left\|\xi\right\|_F, \qquad \Pi_{p'} D\Phi_Q = D\Phi_Q \Pi_p. \tag{167}$$

Proof.

$$D\pi_{p'} D\Phi_Q \xi = \xi_B Q Q^{-1} A + B Q Q^{-1} \xi_A = L_p \xi, \tag{168}$$

$$G(A')^\top = G A^\top Q^{-\top}, \qquad (B')^\top G = Q^\top B^\top G, \tag{169}$$

$$C_{p'}(D\Phi_Q \xi) = Q^\top C_p(\xi) Q, \quad Q^\top Q = I, \tag{170}$$

$$\Omega_{D\Phi_Q \xi} = Q^\top \Omega_\xi Q. \tag{171}$$

□

B.5. Critical points and Hessian inertia

Theorem 13 (Critical equivalence and gauge nullity)**.**

$$D(f|_{\mathcal{M}_r})(X) = 0 \iff \nabla(f \circ \pi)(p) = 0 \iff \operatorname{grad}_g f(X) = 0, \tag{172}$$

$$\operatorname{Inertia}(\operatorname{Hess}_g f(X)) = \operatorname{Inertia}(D^2(f \circ \Psi)(\theta)), \tag{173}$$

$$\ker \nabla^2(f \circ \pi)(p) = \mathcal{V}_p \tag{174}$$

at a nondegenerate quotient critical point.

Proof.

$$D(f \circ \pi)_p = Df_X \circ L_p, \qquad L_p : \mathcal{H}_p \xrightarrow{\simeq} T_X\mathcal{M}_r, \tag{175}$$

$$F(\theta, \omega) = f(\Psi(\theta)), \qquad D^2F = \begin{pmatrix} D^2(f \circ \Psi) & 0 \\ 0 & 0_{r^2} \end{pmatrix}, \tag{176}$$

$$\operatorname{Inertia}(\nabla^2 F) = \operatorname{Inertia}(D^2(f \circ \Psi)) + (0, 0, r^2). \tag{177}$$

At criticality, coordinate changes act by congruence. For arbitrary p,

$$D^2(f \circ \pi)_p[\xi, \zeta] = \left\langle L_p\xi, \nabla^2 f(X)[L_p\zeta] \right\rangle_F + \left\langle \nabla f(X), \xi_B\zeta_A + \zeta_B\xi_A \right\rangle_F . \tag{178}$$

□

C. Metric Differentiation, Levi–Civita Connection, and Hessian Calculus

C.1. Projector and square-root differentials

Definition 3. For $X = U\Sigma V^\top \in \mathcal{K}_{a,b}$,

$$S_L(X) = (XX^\top)^{1/2} = U\Sigma U^\top, \qquad S_R(X) = (X^\top X)^{1/2} = V\Sigma V^\top, \tag{179}$$

$$K_X(Y) = S_L(X)Y + YS_R(X), \qquad A_X = K_X^\dagger, \quad A_X|_{T_X} = K_X|_{T_X}^{-1}. \tag{180}$$

The square roots are restricted to the constant-rank stratum.

Lemma 2 (Differential of the tangent projector)**.** *For $Z \in T_X\mathcal{M}_r$, let*

$$Z = [U, U_\perp] \begin{pmatrix} Z_{11} & Z_{12} \\ Z_{21} & 0 \end{pmatrix} [V, V_\perp]^\top. \tag{181}$$

Then

$$DP_U[Z] = U_\perp Z_{21}\Sigma^{-1}U^\top + U\Sigma^{-1}Z_{21}^\top U_\perp^\top, \tag{182}$$

$$DP_V[Z] = V_\perp Z_{12}^\top \Sigma^{-1}V^\top + V\Sigma^{-1}Z_{12}V_\perp^\top, \tag{183}$$

$$DP_X[Z](Y) = DP_U[Z]Y(I - P_V) + (I - P_U)YDP_V[Z], \tag{184}$$

$$\|DP_X[Z](Y)\|_F \le \frac{4}{a} \|Z\|_F \|Y\|_F . \tag{185}$$

Proof. For a differentiable rank-r path $X(t)$,

$$P_U X = X, \qquad (I - P_U)\dot{P}_U X = (I - P_U)\dot{X}, \tag{186}$$

$$P_U\dot{P}_U P_U = 0, \qquad (I - P_U)\dot{P}_U(I - P_U) = 0, \tag{187}$$

$$(I - P_U)\dot{P}_U U = U_\perp Z_{21}\Sigma^{-1}, \tag{188}$$

$$\dot{P}_U = (I - P_U)\dot{P}_U P_U + P_U\dot{P}_U(I - P_U). \tag{189}$$

Apply the same identities to $X^\top$. Differentiate $P_X Y = P_U Y + Y P_V - P_U Y P_V$ and use

$$\|DP_U[Z]\|_2 \le 2a^{-1}\|Z_{21}\|_F, \qquad \|DP_V[Z]\|_2 \le 2a^{-1}\|Z_{12}\|_F, \tag{190}$$

$$\|Z_{21}\|_F + \|Z_{12}\|_F \le \sqrt{2}\,\|Z\|_F. \tag{191}$$

□

Theorem 14 (Exact square-root and metric differentials)**.** *Write $D_L = DS_L(X)[Z]$, $D_R = DS_R(X)[Z]$. In the U, V block bases,*

$$D_L \equiv \begin{pmatrix} T_L & Z_{21}^\top \\ Z_{21} & 0 \end{pmatrix}, \qquad \Sigma T_L + T_L \Sigma = Z_{11}\Sigma + \Sigma Z_{11}^\top, \tag{192}$$

$$D_R \equiv \begin{pmatrix} T_R & Z_{12} \\ Z_{12}^\top & 0 \end{pmatrix}, \qquad \Sigma T_R + T_R \Sigma = Z_{11}^\top\Sigma + \Sigma Z_{11}, \tag{193}$$

$$(T_L)_{ij} = \frac{Z_{11,ij}\sigma_j + \sigma_i Z_{11,ji}}{\sigma_i + \sigma_j}, \tag{194}$$

$$(T_R)_{ij} = \frac{Z_{11,ji}\sigma_j + \sigma_i Z_{11,ij}}{\sigma_i + \sigma_j}, \tag{195}$$

$$DK_X[Z](Y) = D_L Y + Y D_R, \tag{196}$$

$$\|DK_X[Z](Y)\|_F \le \frac{4b}{a}\|Z\|_F \|Y\|_F. \tag{197}$$

Proof. Differentiate $S_L^2 = XX^\top$, $S_R^2 = X^\top X$:

$$S_L D_L + D_L S_L = ZX^\top + XZ^\top, \tag{198}$$

$$S_R D_R + D_R S_R = Z^\top X + X^\top Z, \tag{199}$$

$$ZX^\top + XZ^\top \equiv \begin{pmatrix} Z_{11}\Sigma + \Sigma Z_{11}^\top & \Sigma Z_{21}^\top \\ Z_{21}\Sigma & 0 \end{pmatrix}. \tag{200}$$

The constant-rank square-root path has zero normal–normal first derivative. Consequently

$$\|T_L\|_F \le \frac{b}{a}\|Z_{11}\|_F, \qquad \|D_L\|_F^2 = \|T_L\|_F^2 + 2\|Z_{21}\|_F^2 \le \frac{4b^2}{a^2}\|Z\|_F^2, \tag{201}$$

$$\|D_R\|_F \le \frac{2b}{a}\|Z\|_F, \qquad \|D_L Y + Y D_R\|_F \le (\|D_L\|_F + \|D_R\|_F)\|Y\|_F. \tag{202}$$

□

Proposition 7 (Pseudoinverse derivative on the moving tangent bundle)**.** *For $P = P_X$, $K = K_X$, $A = K_X^\dagger$, and $\dot K = DK_X[Z]$,*

$$DA_X[Z] = -A\dot K A + A^2\dot K(I - P) + (I - P)\dot K A^2, \tag{203}$$

$$P\,DA_X[Z]\,P = -A(P\dot K P)A. \tag{204}$$

For the embedded connection $\nabla_Z^E Y = P_X DY[Z]$,

$$(\nabla_Z^E K)(Y) = P_X DK_X[Z](Y), \tag{205}$$

$$(\nabla_Z^E A)(Y) = -A_X(\nabla_Z^E K)(A_X Y), \tag{206}$$

$$\big\|(\nabla_Z^E A)(Y)\big\|_F \le \frac{4b}{a^3}\|Z\|_F \|Y\|_F. \tag{207}$$

Proof. In the orthogonal decomposition $T_X \oplus T_X^{\perp}$,

$$K = \begin{pmatrix} K_T & 0 \\ 0 & 0 \end{pmatrix}, \qquad A = \begin{pmatrix} K_T^{-1} & 0 \\ 0 & 0 \end{pmatrix}, \tag{208}$$

$$\dot{A}_{TT} = -K_T^{-1}\dot{K}_{TT}K_T^{-1}, \qquad \dot{A}_{TN} = K_T^{-2}\dot{K}_{TN}, \qquad \dot{A}_{NT} = \dot{K}_{NT}K_T^{-2}. \tag{209}$$

These identities follow by differentiating $KA = AK = P$, $A = A^*$, $K = K^*$, and $A(I-P) = 0$. For a tangent field Y,

$$\nabla_Z^E(KY) - K\nabla_Z^E Y = P_X DK_X[Z](Y) + P_X K_X DY[Z] - K_X P_X DY[Z] \tag{210}$$

$$= P_X DK_X[Z](Y), \tag{211}$$

$$\nabla_Z^E(AK) = 0, \tag{212}$$

$$\nabla_Z^E A = -A(\nabla_Z^E K)A. \tag{213}$$

Finally $\|A|_{T_X}\|_2 \le a^{-1}$ and Eq. (197) imply the bound. □

C.2. Levi–Civita connection and an explicit Hessian bound

Definition 4. For tangent fields Y, Z, define

$$B_X(Z,Y) = (I - P_X)DY[Z] = DP_X[Z](Y), \tag{214}$$

$$\mathfrak{D}_X(Z,Y) = \nabla_Z^g Y - \nabla_Z^E Y, \tag{215}$$

$$\langle \mathfrak{Q}_X(Z,Y), W\rangle_F = \left\langle Z, (\nabla_W^E A)Y\right\rangle_F, \qquad W \in T_X\mathcal{M}_r. \tag{216}$$

Theorem 15 (Closed connection tensor)**.**

$$2g_X(\mathfrak{D}_X(Z,Y),W) = \left\langle Y, (\nabla_Z^E A)W\right\rangle_F + \left\langle Z, (\nabla_Y^E A)W\right\rangle_F - \left\langle Z, (\nabla_W^E A)Y\right\rangle_F, \tag{217}$$

$$\mathfrak{D}_X(Z,Y) = \tfrac{1}{2}K_X\big[(\nabla_Z^E A)Y + (\nabla_Y^E A)Z - \mathfrak{Q}_X(Z,Y)\big], \tag{218}$$

$$\|B_X(Z,Y)\|_F \le \frac{4}{a}\|Z\|_F\|Y\|_F, \tag{219}$$

$$\|\mathfrak{D}_X(Z,Y)\|_F \le \frac{12b^2}{a^3}\|Z\|_F\|Y\|_F. \tag{220}$$

Proof. For $\nabla^g = \nabla^E + \mathfrak{D}$, metric compatibility and zero torsion give

$$Zg(Y,W) = g(\nabla_Z^E Y, W) + g(Y, \nabla_Z^E W) + \left\langle Y, (\nabla_Z^E A)W\right\rangle_F, \tag{221}$$

$$\mathfrak{D}(Z,Y) = \mathfrak{D}(Y,Z), \tag{222}$$

$$2g(\mathfrak{D}(Z,Y),W) = (\nabla_Z^E g)(Y,W) + (\nabla_Y^E g)(Z,W) - (\nabla_W^E g)(Z,Y). \tag{223}$$

Set $C_A = 4b/a^3$. Then

$$|\langle \mathfrak{Q}(Z,Y), W\rangle_F| \le C_A\|Z\|_F\|Y\|_F\|W\|_F, \tag{224}$$

$$\|\mathfrak{D}(Z,Y)\|_F \le \tfrac{1}{2}(2b)(3C_A)\|Z\|_F\|Y\|_F = \frac{12b^2}{a^3}\|Z\|_F\|Y\|_F. \tag{225}$$

□

C.3. The pullback cancellation and a sharp Hessian bound

Theorem 16 (Second-order pullback identity)**.** *Let $p = (B, A)$ be balanced, $L_p = D\pi_p$, and let ξ, ζ be the minimum-energy horizontal lifts of Z, Y. For $G = \nabla f(X)$,*

$$\mathrm{Hess}_g f(X)[Z,Y] = \left\langle L_p\xi, \nabla^2 f(X)[L_p\zeta]\right\rangle_F + \langle G, \xi_B\zeta_A + \zeta_B\xi_A\rangle_F. \tag{226}$$

If k_-, k_+ are the smallest and largest tangent eigenvalues of K_X, and $\ell I \preceq \nabla^2 f(X) \preceq L_f I$ with $\ell \geq 0$, then

$$(\ell k_- - \|G\|_2)I \preceq \operatorname{Hess}_g f(X) \preceq (L_f k_+ + \|G\|_2)I. \tag{227}$$

More generally,

$$\|\operatorname{Hess}_g f(X)\|_{g\to g} \leq k_+ \|\nabla^2 f(X)\|_{F\to F} + \|G\|_2 \leq 2bL_f + \|G\|_2. \tag{228}$$

Proof. Proposition 3 provides a second-order retraction. The pullback-Hessian theorem (Boumal, 2023, Propositions 5.44–5.45) therefore gives

$$\operatorname{Hess}_g f[Z,Z] = \frac{d^2}{dt^2} f((B+t\xi_B)(A+t\xi_A))\Big|_{t=0} \tag{229}$$

$$= \left\langle L_p\xi, \nabla^2 f[L_p\xi]\right\rangle_F + 2\left\langle G, \xi_B\xi_A\right\rangle_F. \tag{230}$$

Polarization proves the bilinear identity. Spectral–nuclear duality and the arithmetic–geometric mean inequality give

$$|2\left\langle G, \xi_B\xi_A\right\rangle_F| \leq 2\|G\|_2\|\xi_B\xi_A\|_* \tag{231}$$

$$\leq 2\|G\|_2\|\xi_B\|_F\|\xi_A\|_F \leq \|G\|_2\|\xi\|^2, \tag{232}$$

$$k_-\|\xi\|^2 \leq \|L_p\xi\|_F^2 \leq k_+\|\xi\|^2, \qquad \|\xi\|^2 = \|Z\|_g^2. \tag{233}$$

Apply Rayleigh–Ritz and Weyl's eigenvalue perturbation inequality (Bhatia, 1997, Chapter III). The connection and second-fundamental-form terms in the product coordinate formula are combined before taking norms. Their cancellation removes the inverse-rank-margin factors from this Hessian bound. □

Proposition 8 (Sharpness of the residual coefficient)**.** *The coefficient one multiplying $\|G\|_2$ in Eq.* (228) *cannot be uniformly decreased.*

Proof. Take $r = 1$, $m = d = 2$, $B = A^\top = \sqrt{\sigma}\, e_1$, and $G = \tau e_2 e_2^\top$. The unit horizontal lift

$$\xi_B = e_2/\sqrt{2}, \qquad \xi_A = \pm e_2^\top/\sqrt{2} \tag{234}$$

satisfies $2\left\langle G, \xi_B\xi_A\right\rangle_F = \pm\tau = \pm\|G\|_2$. For a linear ambient objective, its full quotient Hessian therefore attains both endpoints of the residual bound. □

Theorem 17 (Hessian: coordinate formula and pullback bound)**.** *For $G = \nabla f(X)$,*

$$\operatorname{Hess}_g f(X)[Z] = P_X\{DK_X[Z](G) + K_X(\nabla^2 f(X)[Z])\} + \mathfrak{D}_X(Z, K_X G), \tag{235}$$

$$\operatorname{Hess}_g f(X)[Z,Y] = \left\langle Z, \nabla^2 f(X)[Y]\right\rangle_F + \left\langle G, B_X(Z,Y) - \mathfrak{D}_X(Z,Y)\right\rangle_F. \tag{236}$$

If $\|\nabla^2 f\|_{F\to F} \leq L_f$, then

$$\|\operatorname{Hess}_g f(X)\|_{g\to g} \leq 2bL_f + \|\nabla f(X)\|_2 =: L_g. \tag{237}$$

Proof.

$$\operatorname{Hess}_g f[Z] = \nabla_Z^g(K_X G) = \nabla_Z^E(K_X G) + \mathfrak{D}_X(Z, K_X G), \tag{238}$$

$$\operatorname{Hess}_g f[Z,Y] = Z(Df[Y]) - Df[\nabla_Z^g Y] \tag{239}$$

$$= \left\langle Z, \nabla^2 f[Y]\right\rangle_F + \left\langle G, DY[Z] - \nabla_Z^g Y\right\rangle_F \tag{240}$$

$$= \left\langle Z, \nabla^2 f[Y]\right\rangle_F + \left\langle G, B_X(Z,Y) - \mathfrak{D}_X(Z,Y)\right\rangle_F, \tag{241}$$

$$|\operatorname{Hess}_g f[Z,Y]| \leq (2bL_f + \|G\|_2)\,\|Z\|_g\,\|Y\|_g. \tag{242}$$

The final estimate uses Eq. (226) and its residual operator bound rather than separate norms of the connection tensors. □

C.4. Higher derivatives without internal singular-value gaps

Proposition 9 (Resolvent differentiation)**.** *Let Γ enclose $[a^2, b^2]$, exclude 0, and stay in the analytic domain of $z^{1/2}$. Put*

$$M = XX^\top, \qquad R(z) = (zI - M)^{-1}, \qquad M_Z = ZX^\top + XZ^\top. \tag{243}$$

A local smooth extension agreeing with S_L on $\mathcal{M}_r$ satisfies

$$S_L(X) = \frac{1}{2\pi \mathrm{i}} \oint_\Gamma z^{1/2} R(z)\, dz, \tag{244}$$

$$DS_L[Z] = \frac{1}{2\pi \mathrm{i}} \oint_\Gamma z^{1/2} R(z) M_Z R(z)\, dz, \tag{245}$$

$$D^2 S_L[Z,Y] = \frac{1}{2\pi \mathrm{i}} \oint_\Gamma z^{1/2} \big[R(ZY^\top + YZ^\top)R + RM_Z RM_Y R + RM_Y RM_Z R \big]\, dz. \tag{246}$$

These expressions require separation from the zero spectrum, not $\sigma_i \neq \sigma_j$ for positive singular values.

Proof.

$$D(zI - M)^{-1}[Z] = RM_Z R, \tag{247}$$

$$D^2(zI - M)^{-1}[Z,Y] = R(ZY^\top + YZ^\top)R + RM_Z RM_Y R + RM_Y RM_Z R. \tag{248}$$

For $\delta_\Gamma = \operatorname{dist}(\Gamma, \operatorname{spec}(M)) > 0$,

$$\sup_{z \in \Gamma} \|R(z)\|_2 \leq \delta_\Gamma^{-1}, \qquad \|M_Z\|_2 \leq 2b \|Z\|_F, \tag{249}$$

$$\big\|D^2 S_L[Z,Y]\big\|_F \leq \frac{|\Gamma| \sup_\Gamma |z^{1/2}|}{2\pi} \left(2\delta_\Gamma^{-2} + 8b^2 \delta_\Gamma^{-3}\right) \|Z\|_F \|Y\|_F. \tag{250}$$

Differentiation under the contour integral follows from uniform resolvent bounds. □

Proposition 10 (Coordinate metric and curvature)**.** *For chart coordinates θ^i and $J_i = \partial_i \Psi$,*

$$g_{ij} = \langle J_i, A_X J_j \rangle_F, \tag{251}$$

$$\partial_k g_{ij} = \langle \partial_k J_i, A_X J_j \rangle_F + \langle J_i, DA_X[J_k] J_j \rangle_F + \langle J_i, A_X \partial_k J_j \rangle_F, \tag{252}$$

$$\Gamma^k_{ij} = \tfrac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}), \tag{253}$$

$$R^k{}_{\ell ij} = \partial_i \Gamma^k_{j\ell} - \partial_j \Gamma^k_{i\ell} + \Gamma^k_{im} \Gamma^m_{j\ell} - \Gamma^k_{jm} \Gamma^m_{i\ell}, \tag{254}$$

$$(\operatorname{Hess}_g f)_{ij} = \partial_{ij}(f \circ \Psi) - \Gamma^k_{ij} \partial_k (f \circ \Psi). \tag{255}$$

Repeated indices are summed.

C.5. Balanced gradient flow and discrete imbalance

Proposition 11. *For $\dot{B} = -GA^\top$, $\dot{A} = -B^\top G$,*

$$\frac{d}{dt}(B^\top B - AA^\top) = -AG^\top B - B^\top GA^\top + B^\top GA^\top + AG^\top B = 0, \tag{256}$$

$$\dot{X} = -BB^\top G - GA^\top A = -K_X G \quad \text{on } \mathcal{S}, \tag{257}$$

$$\frac{d}{dt} f(X) = -\left\|GA^\top\right\|_F^2 - \left\|B^\top G\right\|_F^2 = -\left\|\operatorname{grad}_g f\right\|_g^2. \tag{258}$$

For a horizontal discrete step $p^+ = p - \eta v$,

$$(B^+)^\top B^+ - A^+ (A^+)^\top = \eta^2 (v_B^\top v_B - v_A v_A^\top), \tag{259}$$

$$\left\|(B^+)^\top B^+ - A^+ (A^+)^\top\right\|_F \leq \eta^2 \|v\|_F^2, \tag{260}$$

$$\|Q_{\mathrm{bal}} - I\|_F + \left\|Q_{\mathrm{bal}}^{-1} - I\right\|_F \leq C_{\mathrm{bal}} \eta^2 \|v\|_F^2 \tag{261}$$

for a smooth near-identity local balancing gauge. The last bound follows from the inverse Sylvester derivative and the implicit function theorem.

C.6. Differentiating balance and the actual persistent-state map

Definition 5 (A smooth SPD balancing gauge)**.** For full-rank factors, set

$$G_B = B^\top B, \qquad G_A = AA^\top, \tag{262}$$

$$T = (G_B^{1/2} G_A G_B^{1/2})^{1/2}, \qquad P = G_B^{-1/2} T G_B^{-1/2}, \qquad Q = P^{1/2}, \tag{263}$$

$$\mathcal{B}(B, A) = (BQ, Q^{-1}A). \tag{264}$$

All square roots in this definition are positive definite.

Proposition 12 (Exact balance differential)**.** *For $\delta p = (\delta B, \delta A)$,*

$$PG_BP = G_A, \qquad Q^\top G_B Q = Q^{-1} G_A Q^{-\top}, \tag{265}$$

$$\delta G_B = B^\top \delta B + \delta B^\top B, \tag{266}$$

$$\delta G_A = \delta A A^\top + A \delta A^\top, \tag{267}$$

$$(\delta P) G_B P + P G_B (\delta P) = \delta G_A - P(\delta G_B)P, \tag{268}$$

$$C = G_B^{1/2} (\delta P) G_B^{1/2}, \tag{269}$$

$$CT + TC = G_B^{1/2} \{\delta G_A - P(\delta G_B)P\} G_B^{1/2}, \tag{270}$$

$$Q\,\delta Q + \delta Q\, Q = \delta P, \tag{271}$$

$$(D\mathcal{B}_p[\delta p])_B = \delta B Q + B \delta Q, \tag{272}$$

$$(D\mathcal{B}_p[\delta p])_A = Q^{-1}\delta A - Q^{-1}\delta Q Q^{-1} A. \tag{273}$$

Thus two positive-definite Sylvester solves determine the gauge differential without differentiating individual eigenvectors.

Proof.

$$PG_BP = G_B^{-1/2} T^2 G_B^{-1/2} = G_A, \tag{274}$$

$$Q^2 G_B Q^2 = G_A \Longrightarrow Q G_B Q = Q^{-1} G_A Q^{-1}, \tag{275}$$

$$D(PG_BP) = (\delta P) G_B P + P(\delta G_B) P + P G_B (\delta P), \tag{276}$$

$$D(Q^2) = Q\delta Q + \delta Q Q, \qquad D(Q^{-1}) = -Q^{-1}\delta Q Q^{-1}. \tag{277}$$

Congruence of the δP equation by $G_B^{1/2}$ gives the Sylvester equation in C. Both coefficients T, Q are positive definite, so both linear equations have unique symmetric solutions. □

Corollary 2 (First-order balance is a tangent projection)**.** *At $G_B = G_A = H$,*

$$P = Q = I, \qquad H\delta Q + \delta Q H = -\operatorname{sym} C_p(\delta p), \tag{278}$$

$$D\mathcal{B}_p[\delta p] = \delta p + V_p(\delta Q), \qquad D\mathcal{B}_p = \operatorname{Proj}_{T_p\mathcal{S}}, \tag{279}$$

$$\|D\mathcal{B}_p\| = 1, \qquad \delta p \in T_p\mathcal{S} \Longrightarrow \delta Q = 0. \tag{280}$$

Proof. Since $\delta G_A - \delta G_B = -2 \operatorname{sym} C_p(\delta p)$ and $\delta P = 2\delta Q$, the Sylvester equation follows. For symmetric δQ, $V_p(\delta Q)$ is normal to $T_p\mathcal{S}$. Furthermore,

$$\operatorname{sym} C_p(\delta p + V_p(\delta Q)) = \operatorname{sym} C_p(\delta p) + H\delta Q + \delta Q H = 0. \tag{281}$$

The tangential component is unchanged and the normal component is removed. □

Lemma 3 (Differential of the horizontal projector)**.** *For $V = V_p$, $N = V^*V$, $V^\dagger = N^{-1}V^*$, and $\Pi = I - VV^\dagger$,*

$$\delta V(\Omega) = (\delta B \Omega, -\Omega \delta A), \tag{282}$$

$$D\Pi_p[\delta p] = -\Pi(\delta V)V^\dagger - (V^\dagger)^*(\delta V)^*\Pi. \tag{283}$$

If $G_B, G_A \succeq \alpha I$, then

$$\|D\Pi_p[\delta p]\|_{\mathrm{op}} \le \sqrt{2/\alpha}\, \|\delta p\|. \tag{284}$$

Proof. Differentiate $\Pi = I - VN^{-1}V^*$ and use $\delta N = (\delta V)^*V + V^*(\delta V)$:

$$\delta\Pi = -(\delta V)N^{-1}V^* - VN^{-1}(\delta V)^* + VN^{-1}(\delta N)N^{-1}V^* \tag{285}$$

$$= -\Pi(\delta V)V^\dagger - (V^\dagger)^*(\delta V)^*\Pi, \tag{286}$$

$$\|\delta V\|_{\mathrm{op}} \le \|\delta p\|, \qquad \sigma_{\min}(V)^2 = \lambda_{\min}(N) \ge 2\alpha. \tag{287}$$

The norm bound follows from $\|V^\dagger\| \le (2\alpha)^{-1/2}$. □

Proposition 13 (Finite uniform constants on a Gram band)**.** *Suppose $\alpha I \preceq G_A, G_B \preceq \gamma I$ and put*

$$h = (\gamma/\alpha)^{1/4}, \qquad C_\Pi = \sqrt{2/\alpha}, \tag{288}$$

$$C_Q = \frac{\gamma^{3/2}}{2\alpha^2}(\gamma/\alpha)^{1/4}\sqrt{1+(\gamma/\alpha)^2}, \qquad C_{\mathcal{B}} = h + \sqrt{\gamma}\sqrt{1+h^4}\,C_Q. \tag{289}$$

Then

$$\|Q\|_2, \|Q^{-1}\|_2 \le h, \qquad \|DQ_p[\delta p]\|_F \le C_Q\|\delta p\|, \tag{290}$$

$$\|D\mathcal{B}_p[\delta p]\| \le C_{\mathcal{B}}\|\delta p\|, \tag{291}$$

$$\|D(T_{Q(p)}m)[\delta p, \delta m]\| \le h\|\delta m\| + h^2\|m\|C_Q\|\delta p\|, \tag{292}$$

where $T_Q m = (m_B Q, Q^{-1}m_A)$. For $\|G_z\|_F \le G_f$ and $\|\nabla^2 f_z\|_{F\to F} \le L_f$, the fresh factor gradient obeys

$$g_z(p) = (G_z A^\top, B^\top G_z), \tag{293}$$

$$\|g_z(p)\| \le \sqrt{2\gamma}G_f =: G_{\mathrm{fac}}, \qquad \|Dg_z(p)\|_{\mathrm{op}} \le 2\gamma L_f + G_f =: L_{\mathrm{fac}}. \tag{294}$$

Proof.

$$\alpha P^2 \preceq PG_BP = G_A \preceq \gamma I \Longrightarrow \|P\|_2 \le \sqrt{\gamma/\alpha}, \tag{295}$$

$$\gamma P^2 \succeq G_A \succeq \alpha I \Longrightarrow \lambda_{\min}(P) \ge \sqrt{\alpha/\gamma}, \tag{296}$$

$$\alpha I \preceq T \preceq \gamma I, \qquad \|\delta P\|_F \le \frac{\gamma}{2\alpha^2}\|\delta G_A - P(\delta G_B)P\|_F, \tag{297}$$

$$\|\delta G_A - P(\delta G_B)P\|_F \le 2\sqrt{\gamma}\sqrt{1+(\gamma/\alpha)^2}\|\delta p\|, \tag{298}$$

$$\|\delta Q\|_F \le \frac{\|\delta P\|_F}{2\lambda_{\min}(Q)}. \tag{299}$$

Apply the exact differentials to obtain $C_Q, C_{\mathcal{B}}$. For $L_p = D\pi_p$,

$$Dg_z(p)[\delta p] = L_p^*\nabla^2 f_z[L_p\delta p] + (G_z\delta A^\top, \delta B^\top G_z), \tag{300}$$

$$\|L_p\|_{\mathrm{op}} \le \sqrt{2\gamma}, \qquad \|(G_z\delta A^\top, \delta B^\top G_z)\| \le G_f\|\delta p\|. \tag{301}$$

These imply the factor-gradient bounds. □

Theorem 18 (Jacobian of the exact balanced momentum state)**.** *For an exact, undamped connection, use a state (p, m) with $p \in \mathcal{S}$ and $\Pi_p m = m$:*

$$u = \beta m + (1-\beta)g_z(p), \qquad v = \Pi_p u, \qquad z_p = p - \eta v, \tag{302}$$

$$Q = Q(z_p), \qquad p^+ = \mathcal{B}(z_p), \qquad w = T_Q v, \qquad m^+ = \Pi_{p^+} w. \tag{303}$$

This is the first-moment iteration expressed with the next balance and state projection performed at the end of the current step. Its exact differential is

$$\delta v = (D\Pi_p[\delta p])u + \Pi_p\{\beta\delta m + (1-\beta)Dg_z(p)[\delta p]\}, \tag{304}$$

$$\delta z_p = \delta p - \eta\delta v, \qquad \delta Q = DQ_{z_p}[\delta z_p], \tag{305}$$

$$\delta p^+ = D\mathcal{B}_{z_p}[\delta z_p], \tag{306}$$

$$\delta w_B = \delta v_B Q + v_B\delta Q, \qquad \delta w_A = Q^{-1}\delta v_A - Q^{-1}\delta Q Q^{-1}v_A, \tag{307}$$

$$\delta m^+ = (D\Pi_{p^+}[\delta p^+])w + \Pi_{p^+}\delta w. \tag{308}$$

Proof. Projection is linear in its vector argument and $g_z(p)$ is horizontal:

$$\Pi_p\{\beta m + (1-\beta)g_z(p)\} = \beta\Pi_p m + (1-\beta)g_z(p). \tag{309}$$

At the next iteration, projecting transported v before or after forming the new first moment gives the same result. This proves the state equivalence. The differential follows successively from the product rule, Eq. (273), and Eq. (283). □

Corollary 3 (An explicit state sensitivity matrix)**.** *Assume the preceding Gram band holds at p, z_p, p^+ and along all connecting state and oracle interpolation paths. Let $\|m\| \le M$ and put*

$$U = \beta M + (1-\beta)G_{\mathrm{fac}}, \tag{310}$$

$$A_v = C_\Pi U + (1-\beta)L_{\mathrm{fac}}, \qquad C_m = U(C_\Pi h C_{\mathcal{B}} + h^2 C_Q). \tag{311}$$

Then the componentwise Jacobian bound is

$$\begin{pmatrix} \|\delta p^+\| \\ \|\delta m^+\| \end{pmatrix} \le \underbrace{\begin{pmatrix} C_{\mathcal{B}}(1+\eta A_v) & C_{\mathcal{B}}\eta\beta \\ hA_v + C_m(1+\eta A_v) & h\beta + C_m\eta\beta \end{pmatrix}}_{\mathsf{M}} \begin{pmatrix} \|\delta p\| \\ \|\delta m\| \end{pmatrix}. \tag{312}$$

For $d_\lambda = \|p - p'\| + \lambda\|m - m'\|$, $\lambda > 0$,

$$d_\lambda(U_z s, U_z s') \le \Lambda_\lambda d_\lambda(s, s'), \tag{313}$$

$$\Lambda_\lambda = \max\{\mathsf{M}_{11} + \lambda\mathsf{M}_{21}, \mathsf{M}_{22} + \mathsf{M}_{12}/\lambda\}, \tag{314}$$

$$d_\lambda(U_z s, U_{z'} s) \le 2G_{\mathrm{fac}}(1-\beta)\{\eta C_{\mathcal{B}} + \lambda(h + \eta C_m)\}. \tag{315}$$

Proof. The exact Jacobian gives

$$\|\delta v\| \le A_v\|\delta p\| + \beta\|\delta m\|, \tag{316}$$

$$\|\delta z_p\| \le \|\delta p\| + \eta\|\delta v\|, \tag{317}$$

$$\|\delta p^+\| \le C_{\mathcal{B}}\|\delta z_p\|, \tag{318}$$

$$\|\delta m^+\| \le h\|\delta v\| + C_m\|\delta z_p\|. \tag{319}$$

Substitution yields M. Multiplying by $(1, \lambda)$ proves the common-example bound by integration along the prescribed path. For a changed example at a fixed state, $\|\delta v\| \le (1-\beta)\|g_z - g_{z'}\| \le 2(1-\beta)G_{\mathrm{fac}}$ and $\|\delta z_p\| \le \eta\|\delta v\|$. The same last two inequalities give the data perturbation bound. □

For $0 < \beta < 1$ and $\lambda = \eta\beta/(1-\beta)$, its injection constant divided by 2η is

$$C_U = G_{\mathrm{fac}}\{(1-\beta)C_{\mathcal{B}} + \beta(h + \eta C_m)\}. \tag{320}$$

The coarse matrix M supplies finite bounds; it does not establish $\Lambda_\lambda = 1 + O(\eta)$ with a step-independent constant. Such a sharper statement must preserve the cancellations in the exact Jacobian. In particular, no contraction follows by deleting its state blocks.

D. Joint Deep-Chain Geometry: Exact Spectra and Depth Dependence

D.1. Joint group action and the normal operator

Definition 6. For $L \ge 2$,

$$W_1 \in \mathbb{R}_*^{r\times d}, \qquad W_i \in \mathrm{GL}(r)\ (1 < i < L), \qquad W_L \in \mathbb{R}_*^{m\times r}, \tag{321}$$

$$\pi_L(W) = W_L \cdots W_1, \qquad \mathcal{G}_L = \mathrm{GL}(r)^{L-1}, \tag{322}$$

$$W_1' = Q_1^{-1}W_1, \qquad W_i' = Q_i^{-1}W_i Q_{i-1}, \qquad W_L' = W_L Q_{L-1}. \tag{323}$$

Let $\Omega = (\Omega_1, \dots, \Omega_{L-1})$ and

$$(\mathcal{V}_W\Omega)_1 = -\Omega_1 W_1, \tag{324}$$
$$(\mathcal{V}_W\Omega)_i = W_i\Omega_{i-1} - \Omega_i W_i, \quad 1 < i < L, \tag{325}$$
$$(\mathcal{V}_W\Omega)_L = W_L\Omega_{L-1}, \tag{326}$$
$$(\mathcal{V}_W^* u)_i = W_{i+1}^\top u_{i+1} - u_i W_i^\top. \tag{327}$$

Proposition 14 (Joint projection, not pairwise projection)**.**

$$D\pi_L(W)[u] = \sum_{i=1}^{L} L_i u_i R_i, \tag{328}$$
$$L_i = W_L \cdots W_{i+1}, \qquad R_i = W_{i-1} \cdots W_1, \tag{329}$$
$$L_L = I_m, \qquad R_1 = I_d, \tag{330}$$
$$\ker D\pi_L(W) = \operatorname{range} \mathcal{V}_W, \tag{331}$$
$$\Omega_u = (\mathcal{V}_W^*\mathcal{V}_W)^{-1}\mathcal{V}_W^* u, \qquad \Pi_W u = u - \mathcal{V}_W\Omega_u. \tag{332}$$

For $\mathcal{N}_W = \mathcal{V}_W^*\mathcal{V}_W$,

$$(\mathcal{N}_W\Omega)_i = W_{i+1}^\top W_{i+1}\Omega_i + \Omega_i W_i W_i^\top - W_i\Omega_{i-1}W_i^\top - W_{i+1}^\top\Omega_{i+1}W_{i+1}. \tag{333}$$

Terms involving Ω_0, Ω_L are omitted.

Proof.

$$D\pi_L[\mathcal{V}_W\Omega] = -L_1\Omega_1 W_1 + \sum_{i=2}^{L-1} L_i(W_i\Omega_{i-1} - \Omega_i W_i)R_i + W_L\Omega_{L-1}R_L = 0, \tag{334}$$

$$\sum_i \langle u_i, (\mathcal{V}_W\Omega)_i\rangle_F = \sum_{i=1}^{L-1} \left\langle W_{i+1}^\top u_{i+1} - u_i W_i^\top, \Omega_i \right\rangle_F. \tag{335}$$

If $\mathcal{V}_W\Omega = 0$,

$$\Omega_1 W_1 = 0 \Longrightarrow \Omega_1 = 0, \tag{336}$$
$$W_i\Omega_{i-1} - \Omega_i W_i = 0,\ \Omega_{i-1} = 0 \Longrightarrow \Omega_i = 0. \tag{337}$$

Thus $\mathcal{N}_W \succ 0$. The prefix coordinates $T_i = W_i \cdots W_1$ determine all gauge matrices by $Q_i = T_i(T_i')^\dagger$. Hence

$$\{D\pi_L[u] : u_2 = \cdots = u_{L-1} = 0\} \supseteq \{UZ_1 + Z_2V^\top : Z_1 \in \mathbb{R}^{r\times d},\ Z_2 \in \mathbb{R}^{m\times r}\} \tag{338}$$
$$= T_X\mathcal{M}_r. \tag{339}$$

Here the endpoint multipliers have column space $\operatorname{range} U$ and row space $\operatorname{range} V^\top$, respectively; arbitrary endpoint perturbations therefore realize both displayed summands. Surjectivity gives

$$\dim\ker D\pi_L = rd + mr + (L-2)r^2 - r(m+d-r) = (L-1)r^2 \tag{340}$$
$$= \dim\operatorname{range}\mathcal{V}_W. \tag{341}$$

The kernel inclusion and dimension identity prove equality. Expanding $\mathcal{V}_W^*\mathcal{V}_W$ gives Eq. (333). □

D.2. Jointly balanced representatives

Theorem 19 (Canonical chain and residual orthogonal gauges)**.** *For $X = U\Sigma V^\top$, define $D = \Sigma^{1/L}$ and*

$$W_1^0 = DV^\top, \qquad W_i^0 = D\ (1 < i < L), \qquad W_L^0 = UD. \tag{342}$$

Then

$$\pi_L(W^0) = X, \qquad (W^0_{i+1})^\top W^0_{i+1} = W^0_i (W^0_i)^\top = D^2, \tag{343}$$

$$\kappa(W^0_i) = \kappa(X)^{1/L}. \tag{344}$$

All jointly balanced chains in a fixed regular product fiber differ by $\mathcal{G}_L^O = \mathrm{O}(r)^{L-1}$.

Proof. For a jointly balanced chain, put $H_i = W_i W_i^\top = W_{i+1}^\top W_{i+1}$. Then

$$H_i W_i = W_i H_{i-1} \quad (1 < i < L), \tag{345}$$

$$T_i T_i^\top = H_i^i, \qquad T_i = W_i \cdots W_1, \tag{346}$$

$$XX^\top = (W_L W_L^\top)^L, \qquad X^\top X = (W_1^\top W_1)^L. \tag{347}$$

The identities follow inductively from

$$W_i H_{i-1}^{i-1} W_i^\top = H_i^{i-1} W_i W_i^\top = H_i^i. \tag{348}$$

If W, W' have the same product, $T_i' = Q_i^{-1} T_i$, and both are balanced,

$$W_1'^\top W_1' = (X^\top X)^{1/L} = W_1^\top W_1, \tag{349}$$

$$W_1^\top (Q_1^{-\top} Q_1^{-1} - I) W_1 = 0 \Longrightarrow Q_1 Q_1^\top = I. \tag{350}$$

Using $W_i' = Q_i^{-1} W_i Q_{i-1}$ and $Q_{i-1} \in \mathrm{O}(r)$,

$$W_i'^\top W_i' = W_{i-1}' (W_{i-1}')^\top = Q_{i-1}^\top W_i^\top W_i Q_{i-1} \tag{351}$$

$$\Longrightarrow W_i^\top (Q_i^{-\top} Q_i^{-1} - I) W_i = 0 \Longrightarrow Q_i Q_i^\top = I. \tag{352}$$

This proves the claim by induction. □

D.3. Exact normal-equation spectrum and a fast spectral solve

Theorem 20 (Discrete-sine diagonalization)**.** *At Eq.* (342)*, put $d_a = \sigma_a^{1/L}$ and $\theta_k = k\pi/L$. The complete spectrum of $\mathcal{N}_W$ is*

$$\lambda_{abk} = d_a^2 + d_b^2 - 2 d_a d_b \cos\theta_k, \quad a, b \le r, \quad 1 \le k < L. \tag{353}$$

Consequently

$$\lambda_{\min}(\mathcal{N}_W) = 2\sigma_r^{2/L}(1 - \cos(\pi/L)), \tag{354}$$

$$\lambda_{\max}(\mathcal{N}_W) = 2\sigma_1^{2/L}(1 + \cos(\pi/L)), \tag{355}$$

$$\kappa(\mathcal{N}_W) = \kappa(X)^{2/L} \cot^2\Big(\frac{\pi}{2L}\Big). \tag{356}$$

Proof. For each matrix entry (a, b), write $\omega_i = (\Omega_i)_{ab}$. Equation (333) becomes

$$(\mathcal{N}_W \omega)_i = (d_a^2 + d_b^2)\omega_i - d_a d_b(\omega_{i-1} + \omega_{i+1}), \tag{357}$$

$$\omega_0 = \omega_L = 0, \tag{358}$$

$$s_i^{(k)} = \sqrt{2/L} \sin(ik\pi/L), \tag{359}$$

$$s_{i-1}^{(k)} + s_{i+1}^{(k)} = 2\cos(k\pi/L) s_i^{(k)}, \qquad \sum_{i=1}^{L-1} s_i^{(k)} s_i^{(\ell)} = \delta_{k\ell}. \tag{360}$$

Hence $\mathcal{N}_W s^{(k)} = \lambda_{abk} s^{(k)}$. Moreover

$$\lambda_{abk} = (d_a - d_b)^2 + 2 d_a d_b (1 - \cos\theta_k) \tag{361}$$

$$\ge 2\sigma_r^{2/L}(1 - \cos(\pi/L)), \tag{362}$$

$$\lambda_{abk} \le d_a^2 + d_b^2 + 2 d_a d_b \cos(\pi/L) \tag{363}$$

$$\le 2\sigma_1^{2/L}(1 + \cos(\pi/L)). \tag{364}$$

Equality is attained at $(a, b, k) = (r, r, 1)$ and $(1, 1, L-1)$. □

Corollary 4 (Damped inverse and residual-to-projection error)**.** *For $F_i = (\mathcal{V}_W^* u)_i$ and $\varepsilon \geq 0$,*

$$\widehat{F}_{abk} = \sqrt{2/L} \sum_{i=1}^{L-1} \sin(ik\pi/L)(F_i)_{ab}, \tag{365}$$

$$\widehat{\Omega}_{abk} = \frac{\widehat{F}_{abk}}{\lambda_{abk} + 2\varepsilon}, \tag{366}$$

$$(\Omega_i)_{ab} = \sqrt{2/L} \sum_{k=1}^{L-1} \sin(ik\pi/L)\widehat{\Omega}_{abk}. \tag{367}$$

The canonical solve takes $O(Lr^2 \log L)$ operations using sine transforms. If $\|(\mathcal{N}_W + 2\varepsilon I)\widetilde{\Omega} - F\|_F \leq \tau$,

$$\left\|\widetilde{\Omega} - \Omega_\varepsilon\right\|_F \leq \frac{\tau}{\lambda_{\min} + 2\varepsilon}, \tag{368}$$

$$\left\|\mathcal{V}_W(\widetilde{\Omega} - \Omega_\varepsilon)\right\|_F \leq \frac{\sqrt{\lambda_{\max}}\tau}{\lambda_{\min} + 2\varepsilon}, \tag{369}$$

$$\|\mathcal{V}_W(\Omega_\varepsilon - \Omega_0)\|_F \leq \frac{2\varepsilon\sqrt{\lambda_{\max}}}{\lambda_{\min}(\lambda_{\min} + 2\varepsilon)} \|F\|_F . \tag{370}$$

Proof. Use Eq. (353) entrywise and

$$\|\mathcal{V}_W\|^2 = \lambda_{\max}(\mathcal{N}_W), \qquad \left|\frac{1}{\lambda + 2\varepsilon} - \frac{1}{\lambda}\right| = \frac{2\varepsilon}{\lambda(\lambda + 2\varepsilon)}. \tag{371}$$

□

Corollary 5 (Depth asymptotics and CG complexity)**.**

$$1 - \cos(\pi/L) = \frac{\pi^2}{2L^2} + O(L^{-4}), \tag{372}$$

$$\lambda_{\min}(\mathcal{N}_W) = \frac{\pi^2 \sigma_r^{2/L}}{L^2}(1 + O(L^{-2})), \tag{373}$$

$$\kappa(\mathcal{N}_W) = \frac{4L^2}{\pi^2}\kappa(X)^{2/L}(1 + O(L^{-2})). \tag{374}$$

For exact arithmetic conjugate gradients,

$$\|\Omega^{(k)} - \Omega_*\|_{\mathcal{N}} \leq 2\left(\frac{\sqrt{\kappa(\mathcal{N})} - 1}{\sqrt{\kappa(\mathcal{N})} + 1}\right)^k \|\Omega^{(0)} - \Omega_*\|_{\mathcal{N}}. \tag{375}$$

These are normal-solve conditioning statements, not task-speedup guarantees.

D.4. Product metric and optimization spectrum of a depth-L chain

Theorem 21 (Depth-dependent product preconditioner)**.** *At a jointly balanced representative,*

$$K_X^{(L)}(G) = \sum_{i=1}^{L} L_i L_i^\top G R_i^\top R_i \tag{376}$$

$$= \sum_{i=1}^{L} (XX^\top)^{(L-i)/L} G (X^\top X)^{(i-1)/L}, \tag{377}$$

$$g_X^{(L)}(Z, Y) = \left\langle Z, (K_X^{(L)})^\dagger Y \right\rangle_F . \tag{378}$$

Zero exponents in this formula denote I_m or I_d. The tangent eigenvalues are

$$\chi_{ab}^{(L)} = \sum_{i=1}^{L} \sigma_a^{2(L-i)/L} \sigma_b^{2(i-1)/L}, \tag{379}$$

$$\chi_{a,\perp}^{(L)} = \sigma_a^{2-2/L}, \qquad \chi_{\perp,b}^{(L)} = \sigma_b^{2-2/L}, \tag{380}$$

$$\chi_{ab}^{(L)} = \begin{cases} \dfrac{\sigma_a^2 - \sigma_b^2}{\sigma_a^{2/L} - \sigma_b^{2/L}}, & \sigma_a \neq \sigma_b, \\ L\sigma_a^{2-2/L}, & \sigma_a = \sigma_b. \end{cases} \tag{381}$$

In particular,

$$a^{2-2/L}\|P_X G\|_F^2 \le \|\operatorname{grad}_{g^{(L)}} f\|_{g^{(L)}}^2 \le L b^{2-2/L}\|P_X G\|_F^2. \tag{382}$$

Proof. For $J_W = D\pi_L(W)$,

$$(J_W^* G)_i = L_i^\top G R_i^\top, \qquad J_W J_W^* G = \sum_i L_i L_i^\top G R_i^\top R_i. \tag{383}$$

At the canonical chain, the top-left block acts entrywise by $\chi_{ab}^{(L)}$; the two off-diagonal tangent blocks receive only the first and last summands, respectively. Finally,

$$\sum_{j=0}^{L-1} u^{L-1-j} v^j = \frac{u^L - v^L}{u - v}, \qquad \lim_{v \to u} \frac{u^L - v^L}{u - v} = L u^{L-1}. \tag{384}$$

□

D.5. Aligned spectral dynamics and a finite-horizon obstruction

Proposition 15 (Exact aligned dynamics)**.** *Assume fixed singular bases and $G = U \operatorname{diag}(\gamma_i) V^\top$. Then*

$$\dot{X} = -K_X^{(L)} G, \qquad \dot{\sigma}_i = -L\sigma_i^{2-2/L}\gamma_i. \tag{385}$$

For $f(X) = \frac{1}{2}\|X - X_*\|_F^2$ with $X_* = U \operatorname{diag}(s_i) V^\top$,

$$\dot{\sigma}_i = L\sigma_i^{2-2/L}(s_i - \sigma_i), \tag{386}$$

$$\sigma_i^+ = \sigma_i \left[1 + \eta(s_i - \sigma_i)\sigma_i^{(L-2)/L}\right]^L \tag{387}$$

provided the factor update preserves positive singular coordinates.

Proof. For diagonal balanced factors $w_i = \sigma_i^{1/L}$,

$$\partial_{w_i} f = (\sigma_i - s_i)\sigma_i^{(L-1)/L}, \tag{388}$$

$$w_i^+ = \sigma_i^{1/L} - \eta(\sigma_i - s_i)\sigma_i^{(L-1)/L}, \qquad \sigma_i^+ = (w_i^+)^L. \tag{389}$$

□

Corollary 6 (Small-mode hitting time)**.** *Let $0 < \sigma_0 < \theta s < s$, $0 < \theta < 1$. For $L = 2$,*

$$\sigma(t) = \frac{s}{1 + (s/\sigma_0 - 1)e^{-2st}}. \tag{390}$$

For $L > 2$, $p = 2 - 2/L > 1$, the time τ_θ to reach θs satisfies

$$\tau_\theta = \int_{\sigma_0}^{\theta s} \frac{du}{L u^p (s - u)}, \tag{391}$$

$$\frac{\sigma_0^{1-p} - (\theta s)^{1-p}}{Ls(p-1)} \le \tau_\theta \le \frac{\sigma_0^{1-p} - (\theta s)^{1-p}}{L(1-\theta)s(p-1)}. \tag{392}$$

Proof. On $u \in [\sigma_0, \theta s]$, $(1-\theta)s \le s - u \le s$. Integrate $u^{-p}/(Ls)$ and $u^{-p}/(L(1-\theta)s)$. □

D.6. Global orbit energy of a deep factorization

Theorem 22 (Schatten variational identity and factor conditioning)**.** *Let $p = 2/L$. Over all compatible full-rank chains with product X,*

$$\inf_{\pi_L(W)=X} \frac{1}{L}\sum_{i=1}^{L}\|W_i\|_F^2 = \sum_{a=1}^{r}\sigma_a(X)^{2/L} = \|X\|_{S_p}^p, \tag{393}$$

$$\inf_{\pi_L(W)=X} \max_i \kappa(W_i) = \kappa(X)^{1/L}. \tag{394}$$

The canonical jointly balanced chain attains both infima.

Proof. Multiplicative singular-value majorization and scalar Hölder imply

$$\sum_a \sigma_a(X)^{2/L} \le \sum_a \prod_{i=1}^{L}\sigma_a(W_i)^{2/L} \tag{395}$$

$$\le \prod_{i=1}^{L}\left(\sum_a \sigma_a(W_i)^2\right)^{1/L} \tag{396}$$

$$= \prod_{i=1}^{L}\|W_i\|_F^{2/L} \le \frac{1}{L}\sum_i \|W_i\|_F^2\,, \tag{397}$$

$$\kappa(X) \le \prod_i \kappa(W_i) \le \Big(\max_i \kappa(W_i)\Big)^L. \tag{398}$$

At Eq. (342),

$$\left\|W_i^0\right\|_F^2 = \sum_a \sigma_a^{2/L}, \qquad \kappa(W_i^0) = \kappa(X)^{1/L}. \tag{399}$$

□

E. Nonasymptotic Convergence and Explicit Local Landscape Constants

E.1. Retraction smoothness from ambient derivatives

Assumption 1.

$$X_t \in \mathcal{K}_{a,b}, \qquad f(X) \ge f_* > -\infty, \tag{400}$$

$$\|\nabla f(X)\|_F \le G_f, \qquad \|\nabla^2 f(X)\|_{F\to F} \le L_f. \tag{401}$$

The derivative bounds hold along the used ambient product segments. Each horizontal step has lift norm at most ρ and preserves rank r.

Lemma 4 (Explicit pullback smoothness)**.** *For*

$$L_R = G_f + L_f(\sqrt{2b} + \rho/2)^2, \tag{402}$$

$$f(R_X(\xi)) \le f(X) + g_X(\mathrm{grad}_g\, f, \xi) + \frac{L_R}{2}\|\xi\|_g^2\,. \tag{403}$$

Proof. For the horizontal lift, set $Z = L_p\xi$, $Q = \xi_B\xi_A$:

$$R_X(\xi) - X = Z + Q, \tag{404}$$

$$\|Z\|_F \le \sqrt{2b}\,\|\xi\|_g\,, \tag{405}$$

$$\|Q\|_F \le \tfrac{1}{2}\|\xi\|_g^2\,, \tag{406}$$

$$f(X+Z+Q) \le f(X) + \langle\nabla f, Z\rangle_F + \langle\nabla f, Q\rangle_F + \tfrac{L_f}{2}\|Z+Q\|_F^2 \tag{407}$$

$$\le f(X) + g_X(\mathrm{grad}_g\, f, \xi) + \tfrac{1}{2}L_R\|\xi\|_g^2\,. \tag{408}$$

□

E.2. Deterministic stationarity and the PL regime

Theorem 23 (Finite-time stationarity)**.** *Let $g_t = \operatorname{grad}_g f(X_t)$ and $X_{t+1} = R_{X_t}(-\eta_t g_t)$. For $0 < \eta_t \le 1/L_R$,*

$$f(X_{t+1}) \le f(X_t) - \tfrac{\eta_t}{2} \|g_t\|_g^2, \tag{409}$$

$$\sum_{t<T} \eta_t \|g_t\|_g^2 \le 2\Delta_0, \qquad \Delta_0 = f(X_0) - f_*, \tag{410}$$

$$\min_{t<T} \|g_t\|_g^2 \le \frac{2\Delta_0}{\sum_{t<T} \eta_t}, \tag{411}$$

$$\min_{t<T} \|P_{X_t} \nabla f(X_t)\|_F^2 \le \frac{2\Delta_0}{a \sum_{t<T} \eta_t}. \tag{412}$$

Proof.

$$f(X_{t+1}) - f(X_t) \le -\eta_t(1 - L_R\eta_t/2) \|g_t\|_g^2 \le -\tfrac{1}{2}\eta_t \|g_t\|_g^2. \tag{413}$$

Telescope and apply Eq. (119). □

Assumption 2 (Quotient PL)**.** On an invariant optimization region,

$$\tfrac{1}{2} \left\|\operatorname{grad}_g f(X)\right\|_g^2 \ge \mu_g(f(X) - f_*), \qquad \mu_g > 0. \tag{414}$$

Theorem 24 (Last-iterate contraction)**.** *For $0 < \eta \le 1/L_R$ and $0 < \mu_g\eta \le 1$,*

$$f(X_T) - f_* \le (1 - \mu_g\eta)^T \Delta_0, \tag{415}$$

$$\dot{X} = -K_X \nabla f(X) \Longrightarrow \frac{d}{dt}(f(X) - f_*) \le -2\mu_g(f(X) - f_*), \tag{416}$$

$$f(X(t)) - f_* \le e^{-2\mu_g t}\Delta_0. \tag{417}$$

If $\frac{1}{2}\|P_X \nabla f\|_F^2 \ge \mu_E(f - f_*)$, one may take $\mu_g = a\mu_E$.

Proof.

$$\Delta_{t+1} \le \Delta_t - \tfrac{\eta}{2} \|g_t\|_g^2 \le (1 - \eta\mu_g)\Delta_t, \tag{418}$$

$$\tfrac{1}{2} \|g_t\|_g^2 \ge \tfrac{a}{2}\|P_X \nabla f\|_F^2 \ge a\mu_E(f - f_*). \tag{419}$$

□

E.3. Biased stochastic gradients and error floors

Assumption 3. With $\mathcal{F}_t$ denoting the pre-sampling filtration,

$$\widehat{g}_t = g_t + b_t + \zeta_t, \qquad \mathbb{E}[\zeta_t \mid \mathcal{F}_t] = 0, \tag{420}$$

$$\mathbb{E}[\|\zeta_t\|_g^2 \mid \mathcal{F}_t] \le \nu_t^2, \qquad b_t \in T_{X_t}\mathcal{M}_r \text{ is } \mathcal{F}_t\text{-measurable.} \tag{421}$$

All iterates and sampled retraction steps satisfy Assumption 1 almost surely.

Theorem 25 (Biased stochastic stationarity)**.** *For $X_{t+1} = R_{X_t}(-\eta_t \widehat{g}_t)$, $0 < \eta_t \le (4L_R)^{-1}$, and $S_T = \sum_{t<T} \eta_t$,*

$$\mathbb{E}_t f(X_{t+1}) \le f(X_t) - \tfrac{\eta_t}{2} \|g_t\|_g^2 + 2\eta_t \|b_t\|_g^2 + \tfrac{L_R\eta_t^2}{2}\nu_t^2, \tag{422}$$

$$\frac{\sum_{t<T} \eta_t \mathbb{E} \|g_t\|_g^2}{S_T} \le \frac{2\Delta_0}{S_T} + \frac{4\sum_{t<T} \eta_t \mathbb{E} \|b_t\|_g^2}{S_T} + \frac{L_R \sum_{t<T} \eta_t^2 \nu_t^2}{S_T}. \tag{423}$$

For $\Pr(J = t) = \eta_t/S_T$, the left side equals $\mathbb{E} \|g_J\|_g^2$.

Proof.

$$\mathbb{E}_t f(X_{t+1}) \le f(X_t) - \eta_t \|g_t\|_g^2 - \eta_t g(g_t, b_t) + \tfrac{L_R \eta_t^2}{2}\{\|g_t + b_t\|_g^2 + \nu_t^2\}, \tag{424}$$

$$|g(g_t, b_t)| \le \tfrac{1}{4}\|g_t\|_g^2 + \|b_t\|_g^2, \qquad \|g_t + b_t\|_g^2 \le 2\|g_t\|_g^2 + 2\|b_t\|_g^2, \tag{425}$$

$$-\tfrac{3}{4} + L_R\eta_t \le -\tfrac{1}{2}, \qquad 1 + L_R\eta_t \le \tfrac{5}{4} \le 2. \tag{426}$$

Take total expectations, sum, and use $\mathbb{E} f(X_T) \ge f_*$. □

Corollary 7 (Explicit schedules). *For $\nu_t \le \nu$ and $b_t = 0$,*

$$\eta = cT^{-1/2} \Longrightarrow \frac{1}{T}\sum_{t<T} \mathbb{E}\|g_t\|_g^2 \le \frac{2\Delta_0/c + L_R c \nu^2}{\sqrt{T}}, \tag{427}$$

$$c_* = \sqrt{\frac{2\Delta_0}{L_R \nu^2}} \quad (\nu > 0), \tag{428}$$

$$T \ge 16 L_R^2 c^2 \Longrightarrow \eta \le (4L_R)^{-1}. \tag{429}$$

For $\eta_t = c(t+1)^{-\alpha}$, $1/2 < \alpha < 1$,

$$S_T \ge \frac{c}{1-\alpha}\big((T+1)^{1-\alpha} - 1\big), \tag{430}$$

$$\sum_{t<T} \eta_t^2 \le c^2\left(1 + \frac{1}{2\alpha - 1}\right), \tag{431}$$

$$\frac{\sum_t \eta_t \mathbb{E}\|g_t\|_g^2}{S_T} = O(T^{-(1-\alpha)}). \tag{432}$$

A bounded bias adds $4\sup_t \mathbb{E}\|b_t\|_g^2$ to the stationarity bound.

Corollary 8 (Stochastic PL neighborhood). *For constant η, $\|b_t\|_g \le B$, and $\nu_t \le \nu$,*

$$\mathbb{E}\Delta_{t+1} \le (1 - \mu_g \eta)\mathbb{E}\Delta_t + 2\eta B^2 + \tfrac{1}{2} L_R \eta^2 \nu^2, \tag{433}$$

$$\mathbb{E}\Delta_T \le (1 - \mu_g\eta)^T \Delta_0 + \frac{2B^2 + L_R \eta \nu^2/2}{\mu_g}. \tag{434}$$

E.4. Transported momentum: full tracking and absorption

Assumption 4 (Momentum regularity). For $0 \le \beta < 1$,

$$v_t = \beta \mathcal{T}_{t-1,t} v_{t-1} + (1-\beta)(g_t + \zeta_t), \quad v_{-1} = 0, \tag{435}$$

$$X_{t+1} = R_{X_t}(-\eta v_t), \tag{436}$$

$$\|\mathcal{T}_{t-1,t} w\|_g \le (1 + c_T \eta)\|w\|_g, \tag{437}$$

$$\|\mathcal{T}_{t-1,t} g_{t-1} - g_t\|_g \le L_T c_R \eta \|v_{t-1}\|_g, \tag{438}$$

$$\mathbb{E}_t \|\zeta_t\|_g^2 \le \nu^2. \tag{439}$$

The transport is determined before sampling ζ_t.

Lemma 5 (Tracking recursion). *For $e_t = v_t - g_t$, $q_t = \mathbb{E}\|e_t\|_g^2$, $V_t = \mathbb{E}\|v_t\|_g^2$,*

$$q_t \le \beta(1 + c_T\eta)^2 q_{t-1} + \frac{\beta^2 L_T^2 c_R^2 \eta^2}{1-\beta} V_{t-1} + (1-\beta)^2 \nu^2, \tag{440}$$

$$q_0 \le \beta^2 \|g_0\|_g^2 + (1-\beta)^2 \nu^2. \tag{441}$$

Proof. Let $d_t = \mathcal{T} g_{t-1} - g_t$. Then

$$e_t = \beta \mathcal{T} e_{t-1} + \beta d_t + (1-\beta)\zeta_t, \tag{442}$$

$$\|\beta \mathcal{T} e_{t-1} + \beta d_t\|_g^2 \le \beta \|\mathcal{T} e_{t-1}\|_g^2 + \frac{\beta^2}{1-\beta}\|d_t\|_g^2. \tag{443}$$

The noise cross term has conditional expectation zero. □

Theorem 26 (Momentum with non-isometric transport)**.** *Assume*

$$\beta(1 + c_T\eta)^2 \le (1+\beta)/2, \qquad \eta \le \min\left\{\frac{1}{2L_R}, \frac{1-\beta}{2\sqrt{2}\,\beta L_T c_R}\right\}. \tag{444}$$

Restrictions involving β are omitted if $\beta = 0$. Then

$$\frac{1}{T}\sum_{t<T} \mathbb{E}\,\|g_t\|_g^2 \le \frac{2\Delta_0}{\eta T} + \frac{2q_0}{(1-\beta)T} + 2(1-\beta)\nu^2. \tag{445}$$

Proof. For $\rho = (1+\beta)/2$,

$$(1-\rho)\sum_{t<T} q_t \le q_0 + \frac{\beta^2 L_T^2 c_R^2 \eta^2}{1-\beta}\sum_{t<T} V_t + T(1-\beta)^2\nu^2, \tag{446}$$

$$\sum_{t<T} q_t \le \frac{2q_0}{1-\beta} + \frac{2\beta^2 L_T^2 c_R^2 \eta^2}{(1-\beta)^2}\sum_{t<T} V_t + 2T(1-\beta)\nu^2. \tag{447}$$

The polarization identity yields

$$2g(g_t, v_t) = \|g_t\|_g^2 + \|v_t\|_g^2 - \|e_t\|_g^2\,, \tag{448}$$

$$\mathbb{E} f(X_{t+1}) \le \mathbb{E} f(X_t) - \tfrac{\eta}{2}\mathbb{E}\,\|g_t\|_g^2 - \tfrac{\eta}{4}V_t + \tfrac{\eta}{2}q_t. \tag{449}$$

The coefficient $2\beta^2 L_T^2 c_R^2 \eta^2/(1-\beta)^2 \le 1/4$ gives

$$\tfrac{\eta}{2}\sum_{t<T} \mathbb{E}\,\|g_t\|_g^2 + \tfrac{\eta}{8}\sum_{t<T} V_t \le \Delta_0 + \frac{\eta q_0}{1-\beta} + \eta T(1-\beta)\nu^2. \tag{450}$$

Drop the nonnegative velocity sum and divide by $\eta T/2$. □

Corollary 9 (Isometric transport and vanishing-noise schedules)**.** *If $c_T = 0$, the sharper conditions $\eta \le \min\{(2L_R)^{-1}, (1-\beta)/(2\beta L_T c_R)\}$ imply*

$$\frac{1}{T}\sum_{t<T} \mathbb{E}\,\|g_t\|_g^2 \le \frac{2\Delta_0}{\eta T} + \frac{q_0}{(1-\beta)T} + (1-\beta)\nu^2. \tag{451}$$

For $1-\beta = T^{-1/2}$ and $\eta = cT^{-1/2}$, Eq. (445) *is $O(T^{-1/2})$ whenever its uniform regularity and stepsize conditions hold. For $\nu = 0$ and fixed admissible β, η, it is $O(T^{-1})$.*

E.5. Uniform velocity bounds for the implemented local transport

Proposition 16 (A closed invariant velocity ball)**.** *Assume $\|\widehat{g}_t\|_g \le G$ almost surely, and that smooth balance–project transport obeys*

$$\|\mathcal{T} w\|_g \le (1 + C_{\mathrm{bal}}\eta^2 M^2)\|w\|_g \quad \textit{whenever } \|v_{t-1}\|_g \le M. \tag{452}$$

If

$$M = 2G, \qquad \beta C_{\mathrm{bal}}\eta^2 M^2 \le (1-\beta)/2, \tag{453}$$

then $\|v_t\|_g \le M$ for all t.

Proof. For $\|v_{t-1}\|_g \le M$,

$$\|v_t\|_g \le \beta(1 + C_{\mathrm{bal}}\eta^2 M^2)M + (1-\beta)G \tag{454}$$

$$\le \beta M + \tfrac{1}{2}(1-\beta)M + \tfrac{1}{2}(1-\beta)M = M. \tag{455}$$

The induction starts at $v_{-1} = 0$. □

Proposition 17 (Geometry-to-oracle error budget)**.** *Let $E_{\mathrm{syl},t}$ be the right side of the direction-error bound in Eq.* (156)*; let $E_{\mathrm{coord},t}$ and $E_{\mathrm{state},t}$ bound horizontal coordinate and state errors. If the remaining vertical component has coefficient Ω_t and horizontal part h_t, define*

$$Q_t = \sqrt{2b}\,\|h_t\|_F\,\|\Omega_t\|_F + b\,\|\Omega_t\|_F^2\,, \tag{456}$$

$$B_t = E_{\mathrm{syl},t} + E_{\mathrm{coord},t} + E_{\mathrm{state},t} + C_{\mathrm{chart}}\eta_t Q_t. \tag{457}$$

A local inverse of the retraction converts a product defect $\eta_t^2 Q_t$ into a tangent-direction defect bounded by $C_{\mathrm{chart}}\eta_t Q_t$. An explicit admissible constant and the required inverse-domain condition are given below. For a bias bound $\|b_t\|_g \le B_t$, Eq. (423) *has the explicit numerical floor*

$$\frac{4\sum_t \eta_t B_t^2}{\sum_t \eta_t}. \tag{458}$$

Lemma 6 (Quantitative inverse of the factor retraction)**.** *Identify $T_X\mathcal{M}_r$ with its minimum-energy factor lift at a balanced p. For $0 < \rho < \sqrt{a}$ and $\|\xi\|, \|\zeta\| \le \rho$,*

$$\|R_X(\xi) - R_X(\zeta)\|_F \ge (\sqrt{a} - \rho)\|\xi - \zeta\|, \tag{459}$$

$$\|R_X^{-1}(Y) - R_X^{-1}(Y')\| \le \frac{\|Y - Y'\|_F}{\sqrt{a} - \rho}, \quad Y, Y' \in R_X(B_\rho). \tag{460}$$

Consequently $C_{\mathrm{chart}} = (\sqrt{a} - \rho)^{-1}$ is sufficient when both perturbed products belong to this same inverse domain. For $\rho \le \sqrt{a}/2$, $C_{\mathrm{chart}} \le 2/\sqrt{a}$.

Proof. Put $\delta = \xi - \zeta$ and $\bar{\xi} = (\xi + \zeta)/2$. Then

$$R_X(\xi) - R_X(\zeta) = L_p\delta + \bar{\xi}_B\delta_A + \delta_B\bar{\xi}_A, \tag{461}$$

$$\|L_p\delta\|_F^2 \ge a\|\delta\|^2, \tag{462}$$

$$\|\bar{\xi}_B\delta_A + \delta_B\bar{\xi}_A\|_F \le \sqrt{\|\bar{\xi}_B\|_F^2 + \|\bar{\xi}_A\|_F^2}\sqrt{\|\delta_A\|_F^2 + \|\delta_B\|_F^2} \le \rho\|\delta\|. \tag{463}$$

The reverse triangle inequality gives the first claim and injectivity. Since $\rho < \sqrt{a}$, both factors remain full rank. The derivative has smallest singular value at least $\sqrt{a} - \rho > 0$, so its inverse exists locally; injectivity joins these inverses on $R_X(B_\rho)$. The first claim gives the stated finite-difference inverse bound. For $Y = R_X(-\eta h)$ and $Y' = Y + \Delta$ in that image,

$$\left\|-\eta^{-1}R_X^{-1}(Y') - h\right\| \le \frac{\|\Delta\|_F}{\eta(\sqrt{a} - \rho)}. \tag{464}$$

□

E.6. Predictable preconditioning and general angle conditions

Theorem 27 (Predictable positive tangent operator)**.** *Let $H_t = H_t^* \in \mathcal{F}_t$-measurable with $\alpha I \preceq H_t \preceq \gamma I$, $\alpha > 0$, and update $X_{t+1} = R_{X_t}(-\eta H_t(g_t + \zeta_t))$. For $\eta \le \alpha/(L_R\gamma^2)$,*

$$\frac{1}{T}\sum_{t<T}\mathbb{E}\,\|g_t\|_g^2 \le \frac{2\Delta_0}{\alpha\eta T} + \frac{L_R\eta\gamma^2\nu^2}{\alpha}. \tag{465}$$

Proof.

$$\mathbb{E}_t g(g_t, H_t(g_t+\zeta_t)) = g(g_t, H_t g_t) \geq \alpha \left\|g_t\right\|_g^2, \tag{466}$$

$$\mathbb{E}_t \|H_t(g_t+\zeta_t)\|_g^2 \leq \gamma^2(\left\|g_t\right\|_g^2 + \nu^2), \tag{467}$$

$$\mathbb{E}_t f(X_{t+1}) \leq f(X_t) - \tfrac{1}{2}\alpha\eta \left\|g_t\right\|_g^2 + \tfrac{1}{2}L_R\eta^2\gamma^2\nu^2. \tag{468}$$

□

Proposition 18 (Direction-angle certificate)**.** *For an arbitrary horizontal direction d_t, assume*

$$g(g_t, d_t) \geq c_d \left\|g_t\right\|_g^2 - \epsilon_t, \qquad \left\|d_t\right\|_g^2 \leq C_d \left\|g_t\right\|_g^2 + v_t, \qquad c_d > 0. \tag{469}$$

If $\eta_t \leq c_d/(L_R C_d)$, then

$$f(X_{t+1}) - f(X_t) \leq -\tfrac{1}{2}c_d\eta_t \left\|g_t\right\|_g^2 + \eta_t\epsilon_t + \tfrac{1}{2}L_R\eta_t^2 v_t, \tag{470}$$

$$\frac{\sum_t \eta_t \left\|g_t\right\|_g^2}{S_T} \leq \frac{2\Delta_0 + 2\sum_t \eta_t\epsilon_t + L_R\sum_t \eta_t^2 v_t}{c_d S_T}. \tag{471}$$

A current-sample-dependent Adam denominator does not satisfy the predictability premise merely by being positive.

E.7. Rank geometry plus RIP yields an explicit local PL constant

Assumption 5 (Restricted isometry)**.** For a linear map $\mathcal{A} : \mathbb{R}^{m\times d} \to \mathbb{R}^n$,

$$(1-\delta)\left\|Z\right\|_F^2 \leq \left\|\mathcal{A}Z\right\|_2^2 \leq (1+\delta)\left\|Z\right\|_F^2, \quad \operatorname{rank} Z \leq 4r, \quad 0 \leq \delta < \sqrt{35}/6. \tag{472}$$

Let $\operatorname{rank} X_* = \operatorname{rank} X = r$, $E = X - X_*$, and $f(X) = \frac{1}{2}\|\mathcal{A}E\|_2^2$.

Lemma 7 (Normal component is quadratic in the product error)**.** *For $\sigma_r(X) \geq a$ and $\left\|E\right\|_F < a$,*

$$\left\|(I - P_X)E\right\|_F \leq \frac{\left\|E\right\|_F^2}{2(a - \left\|E\right\|_F)}. \tag{473}$$

If $\left\|E\right\|_F \leq a/4$, then

$$\left\|P_X E\right\|_F \geq c_0 \left\|E\right\|_F, \qquad c_0 = \sqrt{35}/6. \tag{474}$$

Proof. In the X singular bases,

$$X_* \equiv \begin{pmatrix} \Sigma - E_{11} & -E_{12} \\ -E_{21} & -E_{22} \end{pmatrix}, \tag{475}$$

$$\operatorname{rank} X_* = r \Longrightarrow E_{22} = -E_{21}(\Sigma - E_{11})^{-1}E_{12}, \tag{476}$$

$$\left\|E_{22}\right\|_F \leq \frac{\left\|E_{21}\right\|_F \left\|E_{12}\right\|_F}{a - \left\|E_{11}\right\|_2} \leq \frac{\left\|E\right\|_F^2}{2(a - \left\|E\right\|_F)}, \tag{477}$$

$$\frac{\left\|E_{22}\right\|_F}{\left\|E\right\|_F} \leq \frac{\left\|E\right\|_F}{2(a - \left\|E\right\|_F)} \leq \frac{1}{6}, \tag{478}$$

$$\left\|P_X E\right\|_F^2 = \left\|E\right\|_F^2 - \left\|E_{22}\right\|_F^2 \geq \tfrac{35}{36}\left\|E\right\|_F^2. \tag{479}$$

□

Theorem 28 (Explicit RIP-to-PL transfer)**.** *Under Eq.* (472)*, $\sigma_r(X) \geq a$, and $\left\|E\right\|_F \leq a/4$,*

$$\left\|P_X \nabla f(X)\right\|_F \geq (c_0 - \delta)\left\|E\right\|_F, \tag{480}$$

$$\tfrac{1}{2}\|\operatorname{grad}_g f(X)\|_g^2 \geq \mu_{\mathrm{RIP}} f(X), \qquad \mu_{\mathrm{RIP}} = \frac{a(c_0-\delta)^2}{1+\delta}. \tag{481}$$

Proof. RIP polarization for $\operatorname{rank} Z, \operatorname{rank} Y \le 2r$ gives

$$|\langle \mathcal{A}Z, \mathcal{A}Y\rangle - \langle Z, Y\rangle_F| \le \delta \|Z\|_F \|Y\|_F . \tag{482}$$

Choose $Z = P_X E$, $\operatorname{rank} Z \le 2r$. Then

$$\langle \nabla f(X), Z\rangle_F = \langle \mathcal{A}E, \mathcal{A}Z\rangle \tag{483}$$
$$\ge \|Z\|_F^2 - \delta \|E\|_F \|Z\|_F , \tag{484}$$
$$\|P_X \nabla f(X)\|_F \ge \|Z\|_F - \delta \|E\|_F \ge (c_0 - \delta)\|E\|_F , \tag{485}$$
$$\left\|\operatorname{grad}_g f\right\|_g^2 \ge a(c_0-\delta)^2 \|E\|_F^2 , \tag{486}$$
$$f(X) \le \tfrac{1}{2}(1+\delta)\|E\|_F^2 . \tag{487}$$

□

Theorem 29 (A sharp pullback RIP Hessian enclosure)**.** *Under Eq.* (472)*, let $E = X - X_*$, let $k_\pm(X)$ be the extreme tangent eigenvalues of K_X, and put*

$$\omega_E = \|E\|_2 + \delta\|E\|_F. \tag{488}$$

Then

$$\{(1-\delta)k_-(X) - \omega_E\} I \preceq \operatorname{Hess}_g f(X) \preceq \{(1+\delta)k_+(X) + \omega_E\} I. \tag{489}$$

On $\mathcal{K}_{a,b}$, the sufficient radius

$$\|E\|_F \le \frac{a(1-\delta)}{2(1+\delta)} \tag{490}$$

gives $\operatorname{Hess}_g f(X) \succeq a(1-\delta)I/2$.

Proof. For a unit rank-one matrix $uv^\top$, restricted polarization gives

$$|\langle \mathcal{A}E, \mathcal{A}(uv^\top)\rangle - \langle E, uv^\top\rangle| \le \delta\|E\|_F, \tag{491}$$
$$\|\mathcal{A}^*\mathcal{A}E - E\|_2 \le \delta\|E\|_F, \tag{492}$$
$$\|G\|_2 = \|\mathcal{A}^*\mathcal{A}E\|_2 \le \omega_E. \tag{493}$$

The needed combined rank is at most $2r+1 \le 3r$. For a unit horizontal lift ξ, $\operatorname{rank}(L_p\xi) \le 2r$, so RIP and Eq. (226) give

$$(1-\delta)\|L_p\xi\|_F^2 - \omega_E \le \operatorname{Hess}_g f[L_p\xi, L_p\xi] \tag{494}$$
$$\le (1+\delta)\|L_p\xi\|_F^2 + \omega_E. \tag{495}$$

Use $k_- \le \|L_p\xi\|_F^2 \le k_+$ and $\omega_E \le (1+\delta)\|E\|_F$. For $\delta = 0$, scalar reconstruction has Hessian $2x + (x - x_*)$; positive and negative errors attain the respective enclosure endpoints. □

Corollary 10 (A matching local geodesic PL constant)**.** *Let $\mathcal{U}$ be a geodesically convex neighborhood containing X_*, with $k_-(X) \ge a$ and $\omega_E \le \omega < a(1-\delta)$ throughout $\mathcal{U}$. Then*

$$\tfrac{1}{2}\|\operatorname{grad}_g f(X)\|_g^2 \ge \{a(1-\delta) - \omega\}\{f(X) - f(X_*)\}, \qquad X \in \mathcal{U}. \tag{496}$$

Proof. Write $\mu = a(1-\delta) - \omega$. Integrate the Hessian lower bound along a minimizing geodesic to X_*:

$$f(X_*) \ge f(X) + g_X(\operatorname{grad}_g f(X), \operatorname{Log}_X X_*) + \tfrac{\mu}{2} d_g(X, X_*)^2, \tag{497}$$
$$f(X) - f(X_*) \le \|\operatorname{grad}_g f(X)\|_g d_g(X, X_*) - \tfrac{\mu}{2} d_g(X, X_*)^2 \le \frac{\|\operatorname{grad}_g f(X)\|_g^2}{2\mu}. \tag{498}$$

This local Hessian-based constant approaches $a(1-\delta)$ as the neighborhood shrinks. The earlier chord-based PL result does not require this additional geodesic-convexity premise. □

Corollary 11 (A sufficient invariant local basin)**.** *Let $s_* = \sigma_r(X_*)$, $a = s_*/2$, and*

$$\|X_0 - X_*\|_F \le \frac{s_*}{8}\sqrt{\frac{1-\delta}{1+\delta}}. \tag{499}$$

For rank-preserving steps satisfying pullback descent,

$$f(X_t) \le f(X_0), \tag{500}$$

$$\|X_t - X_*\|_F^2 \le \frac{2f(X_t)}{1-\delta} \le \frac{1+\delta}{1-\delta}\|X_0 - X_*\|_F^2 \le s_*^2/64, \tag{501}$$

$$\sigma_r(X_t) \ge s_* - \|X_t - X_*\|_F \ge 7s_*/8 \ge a, \tag{502}$$

$$\|X_t - X_*\|_F \le s_*/8 = a/4, \tag{503}$$

$$f(X_T) \le (1 - \eta\mu_{\mathrm{RIP}})^T f(X_0). \tag{504}$$

The step-radius conditions must hold on this sublevel set.

Corollary 12 (Noise and approximate stationarity)**.** *For $y = \mathcal{A}X_* + \xi$ and*

$$\nu_{\mathcal{A}} = \sup_{\operatorname{rank} Z \le 2r,\ \|Z\|_F = 1} |\langle \xi, \mathcal{A}Z\rangle|, \tag{505}$$

inside the same regular basin,

$$\|P_X \nabla f(X)\|_F \ge (c_0 - \delta)\|X - X_*\|_F - \nu_{\mathcal{A}}, \tag{506}$$

$$\left\|\operatorname{grad}_g f(X)\right\|_g \le \epsilon \Longrightarrow \|X - X_*\|_F \le \frac{\nu_{\mathcal{A}} + \epsilon/\sqrt{a}}{c_0 - \delta}. \tag{507}$$

E.8. Dense-Gaussian sample size: an explicit sufficient condition

Lemma 8 (Rank-k Frobenius net)**.** *For $\mathcal{T}_k = \{Z : \operatorname{rank} Z \le k,\ \|Z\|_F = 1\}$,*

$$N(\mathcal{T}_k, \|\cdot\|_F, \varepsilon) \le (1 + 6\sqrt{k}/\varepsilon)^{k(m+d+1)}. \tag{508}$$

Proof. Write $Z = U\operatorname{diag}(s)V^\top$, $\|s\|_2 = 1$. Take $\varepsilon/3$ nets of U, V, s in balls of radii $\sqrt{k}, \sqrt{k}, 1$. Their product cardinality is bounded by

$$(1 + 6\sqrt{k}/\varepsilon)^{mk}(1 + 6\sqrt{k}/\varepsilon)^{dk}(1 + 6/\varepsilon)^k. \tag{509}$$

With net centers constrained to the corresponding Stiefel/sphere sets,

$$\|U\operatorname{diag}(s)V^\top - \widehat{U}\operatorname{diag}(\widehat{s})\widehat{V}^\top\|_F \le \|U - \widehat{U}\|_F + \|s - \widehat{s}\|_2 + \|V - \widehat{V}\|_F \tag{510}$$

$$\le \varepsilon. \tag{511}$$

□

Theorem 30 (Dense Gaussian RIP with stated constants)**.** *Let $(\mathcal{A}Z)_i = n^{-1/2}\langle G_i, Z\rangle$ with independent standard Gaussian matrix entries in G_i. Set $k = \min(4r, m, d)$, $0 < \delta \le 1/2$, and $0 < q < 1$. A sufficient condition for Eq.* (472) *with probability at least $1 - q$ is*

$$n \ge \frac{32}{\delta^2}\left[k(m + d + 1)\log\left(1 + \frac{96\sqrt{2k}}{\delta}\right) + \log\frac{2}{q}\right]. \tag{512}$$

Proof. For fixed $\|Z\|_F = 1$,

$$\Pr\{|\|\mathcal{A}Z\|_2^2 - 1| > u\} \le 2e^{-nu^2/8}, \qquad 0 < u < 1. \tag{513}$$

Use $u = \delta/2$, $\varepsilon = \delta/(16\sqrt{2})$ in Eq. (508) and a union bound. Let $Q_k = \sup_{\mathcal{T}_k} \|\mathcal{A}Z\|_2$. Splitting a rank-$2k$ SVD into two blocks gives $Q_{\min(2k,m,d)} \le \sqrt{2}Q_k$. Hence

$$Q_k \le \sqrt{1+\delta/2} + \sqrt{2}\,\varepsilon Q_k, \tag{514}$$

$$Q_k \le \frac{\sqrt{1+\delta/2}}{1-\delta/16} \le \sqrt{1+\delta}, \tag{515}$$

$$\inf_{\mathcal{T}_k} \|\mathcal{A}Z\|_2 \ge \sqrt{1-\delta/2} - \frac{\delta}{16}\sqrt{1+\delta} \ge \sqrt{1-\delta}. \tag{516}$$

The two last scalar inequalities hold for $0 < \delta \le 1/2$. □

The experiments use rank-one Gaussian measurements, not independent Gaussian entries in each measurement matrix; this sufficient condition is not asserted for that different design.

E.9. A high-probability stationarity inequality

Theorem 31 (Bounded martingale noise)**.** *Let $b_t = 0$, $\|g_t\|_g \le G$, $\|\widehat{g}_t\|_g \le H$, and $\|\zeta_t\|_g \le V$ almost surely. For deterministic steps and $S_T = \sum_t \eta_t$, with probability at least $1-q$,*

$$\frac{\sum_{t<T} \eta_t \|g_t\|_g^2}{S_T} \le \frac{\Delta_0}{S_T} + \frac{L_R H^2 \sum_t \eta_t^2}{2S_T} + \frac{GV\sqrt{2\log(1/q)\sum_t \eta_t^2}}{S_T}. \tag{517}$$

Proof.

$$f(X_{t+1}) - f(X_t) \le -\eta_t \|g_t\|_g^2 - \eta_t g(g_t, \zeta_t) + \tfrac{1}{2} L_R \eta_t^2 H^2, \tag{518}$$

$$M_T = -\sum_{t<T} \eta_t g(g_t, \zeta_t), \tag{519}$$

$$\mathbb{E}_t[M_{t+1} - M_t] = 0, \tag{520}$$

$$|M_{t+1} - M_t| \le \eta_t GV, \tag{521}$$

$$\tag{522}$$

$$\Pr\left\{ M_T > GV\sqrt{2\log(1/q)\sum_t \eta_t^2} \right\} \le q. \tag{523}$$

Telescope the first inequality and apply the displayed martingale bound. □

E.10. Trajectory perturbations and an explicit contraction counterexample

Theorem 32 (Discrete perturbation resolvent)**.** *Let $d_0 = 0$ and*

$$d_{t+1} \le (1 + L_U\eta_t)d_t + \eta_t e_t + \eta_t^2 q_t. \tag{524}$$

Then

$$d_T \le \sum_{t<T} (\eta_t e_t + \eta_t^2 q_t) \prod_{k>t} (1 + L_U \eta_k). \tag{525}$$

If the actual common state map contracts by $1-\mu_U\eta$, $0 \le 1-\mu_U\eta < 1$, $e_t \le e$, $q_t \le q$, then

$$d_T \le (1-\mu_U\eta)^T d_0 + \frac{e+\eta q}{\mu_U}\{1 - (1-\mu_U\eta)^T\}. \tag{526}$$

Proof. Repeated substitution gives the weighted sum. In the contractive case use $\sum_{j=0}^{T-1}(1-\mu_U\eta)^j = [1-(1-\mu_U\eta)^T]/(\mu_U\eta)$. □

Proposition 19 (Ambient strong convexity is insufficient)**.** *For $f(x) = \frac{1}{2}(x-c)^2$, $x, c > 0$, balanced scalar factors $B = A = \sqrt{x}$ give*

$$\nabla^2 f(x) = 1, \qquad U_\eta(x) = x[1-\eta(x-c)]^2, \tag{527}$$

$$U'_\eta(x) = [1-\eta(x-c)][1-\eta(3x-c)], \qquad \lim_{x\downarrow 0} U'_\eta(x) = (1+\eta c)^2 > 1. \tag{528}$$

Thus ambient strong convexity does not imply product-map contraction.

Theorem 33 (Exact scalar geodesics, Hessian, and a contractive local map)**.** *For $m = d = r = 1$ and $x > 0$,*

$$K_x = 2x, \qquad g_x(z,y) = \frac{zy}{2x}, \qquad q = \sqrt{2x}, \tag{529}$$

$$ds_g^2 = \frac{dx^2}{2x} = dq^2, \qquad d_g(x,y) = \sqrt{2}\,|\sqrt{x} - \sqrt{y}|, \tag{530}$$

$$\gamma_{x,y}(t) = \tfrac{1}{2}[(1-t)\sqrt{2x} + t\sqrt{2y}]^2, \qquad \Gamma^x_{xx} = -\frac{1}{2x}, \tag{531}$$

$$\operatorname{grad}_g f(x) = 2xf'(x), \qquad \operatorname{Hess}_g f(x)[z] = (2xf''(x) + f'(x))z. \tag{532}$$

For $f(x) = \frac{1}{2}(x-c)^2$, $c > 0$,

$$\lambda(\operatorname{Hess}_g f(x)) = 3x - c, \qquad \tfrac{1}{2}\|\operatorname{grad}_g f(x)\|_g^2 = 2xf(x). \tag{533}$$

Thus the quotient Hessian is negative for $0 < x < c/3$ despite $f'' = 1$. On $\mathcal{I}_c = [3c/4, 5c/4]$, the exact balanced factor update

$$U_\eta(x) = x[1-\eta(x-c)]^2, \qquad 0 < \eta \le 1/(4c) \tag{534}$$

preserves $\mathcal{I}_c$ and satisfies

$$|U_\eta(x) - U_\eta(y)| \le (1-c\eta)|x-y|, \tag{535}$$

$$d_g(U_\eta(x), U_\eta(y)) \le (1 - 5c\eta/4) d_g(x,y). \tag{536}$$

Proof. The change of coordinates $q = \sqrt{2x}$ gives

$$\frac{dq}{dx} = (2x)^{-1/2}, \qquad \frac{d^2 f(q^2/2)}{dq^2} = 2xf''(x) + f'(x). \tag{537}$$

For $x \in \mathcal{I}_c$ and $\tau = c\eta \le 1/4$,

$$U'_\eta(x) = [1-\eta(x-c)][1-\eta(3x-c)], \tag{538}$$

$$0 < (1-\tau/4)(1-11\tau/4) \le U'_\eta(x) \tag{539}$$

$$\le (1+\tau/4)(1-5\tau/4) = 1 - \tau - 5\tau^2/16 \le 1-\tau. \tag{540}$$

Since $U_\eta(c) = c$, the mean-value theorem implies that each point moves toward c and remains in $\mathcal{I}_c$. The induced update in the isometric coordinate is

$$\widetilde{U}_\eta(q) = q[1-\eta(q^2/2 - c)], \tag{541}$$

$$\widetilde{U}'_\eta(q) = 1 - \eta(3x-c) \in [1-11\tau/4, 1-5\tau/4] \subset (0,1). \tag{542}$$

Apply the mean-value theorem in the q interval to obtain the second contraction bound. Both constants concern the actual scalar update, with no optimizer state suppressed. □

Proposition 20 (Coordinatewise adaptive scaling is not rotation-covariant)**.** *Let $g = (1,2)^\top$ and $R = 2^{-1/2}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$.*

Then

$$R^\top g = 2^{-1/2}(3,1)^\top, \qquad \operatorname{sign}(R^\top g) = (1,1)^\top, \tag{543}$$

$$R^\top \operatorname{sign}(g) = (\sqrt{2}, 0)^\top \ne \operatorname{sign}(R^\top g). \tag{544}$$

Even residual orthogonal gauge consistency of coordinatewise normalized directions requires an additional argument.

F. Exact Local Momentum Dynamics, Damping, and a Certified Nonlinear Basin

F.1. The quotient state near a nondegenerate critical point

Definition 7. Let $X_\star \in \mathcal{M}_r$ be a critical point of $f|_{\mathcal{M}_r}$, $D = r(m+d-r)$, and let f be C^3 locally. Choose a normal chart $\psi : B_R(0) \subset \mathbb{R}^D \to \mathcal{M}_r$ and a smooth tangent frame $E(x)$ such that

$$\psi(0) = X_\star, \qquad g_{ij}(0) = \delta_{ij}, \qquad \Gamma_{ij}^k(0) = 0, \tag{545}$$

$$E(0)^\top g_\star E(0) = I_D, \qquad \mathsf{H} = [\mathrm{Hess}_g\, f(X_\star)]_{E(0)} = \mathsf{H}^\top. \tag{546}$$

The reduced state is $s = (x, \omega) \in \mathbb{R}^{2D}$, where ω contains the incoming first moment in that frame. Set $\alpha = \eta(1-\beta)$.

Theorem 34 (Exact Jacobian on the quotient tangent bundle)**.** *For the exact first-moment iteration in Eq.* (303)*, its reduced state map $\mathscr{U}$ satisfies*

$$\mathscr{U}(0,0) = (0,0), \tag{547}$$

$$D\mathscr{U}(0,0) = \mathsf{J} = \begin{pmatrix} I_D - \alpha\mathsf{H} & -\eta\beta I_D \\ (1-\beta)\mathsf{H} & \beta I_D \end{pmatrix}. \tag{548}$$

For $\mathsf{H}e_i = \lambda_i e_i$, the two-dimensional invariant mode (x_i, ω_i) has matrix

$$\mathsf{J}_{\lambda_i} = \begin{pmatrix} 1-\alpha\lambda_i & -\eta\beta \\ (1-\beta)\lambda_i & \beta \end{pmatrix}, \tag{549}$$

$$\chi_{\lambda_i}(z) = z^2 - (1+\beta-\alpha\lambda_i)z + \beta, \tag{550}$$

$$z_\pm(\lambda_i) = \frac{1+\beta-\alpha\lambda_i \pm \sqrt{(1+\beta-\alpha\lambda_i)^2 - 4\beta}}{2}. \tag{551}$$

Proof. In the chosen chart and frame,

$$[\mathrm{grad}_g\, f(\psi(x))]_{E(x)} = \mathsf{H}x + O(\|x\|^2), \tag{552}$$

$$[\mathcal{T}_{X,Y}]_{E(X),E(Y)} = I_D + O(\|x\| + \|y\|), \tag{553}$$

$$[\mathrm{Retr}_{\psi(x)}(-\eta v)]_\psi = x - \eta[v]_{E(x)} + O((\|x\| + \|v\|)^2). \tag{554}$$

At $(x,\omega) = (0,0)$, every derivative of transport or of the horizontal projector that multiplies the current gradient or moment vanishes. Hence

$$\delta v = (1-\beta)\mathsf{H}\,\delta x + \beta\,\delta\omega, \tag{555}$$

$$\delta x^+ = \delta x - \eta\delta v, \qquad \delta\omega^+ = \delta v. \tag{556}$$

These are precisely Eq. (548). The characteristic polynomial follows from

$$\det(zI - \mathsf{J}_\lambda) = (z - 1 + \alpha\lambda)(z-\beta) + \eta\beta(1-\beta)\lambda. \tag{557}$$

The cancellations use the actual transported state and the quotient Hessian; the ambient Hessian is not substituted for H. □

F.2. An exact stability interval and optimal quadratic tuning

The following modal calculation is the local heavy-ball analysis in the present normalized-momentum convention (Lessard et al., 2016).

Theorem 35 (Necessary and sufficient Schur condition)**.** *Suppose $0 \le \beta < 1$ and $0 < \mu \le \lambda_i \le \Lambda$. Then*

$$\rho(\mathsf{J}) < 1 \quad \Longleftrightarrow \quad 0 < \alpha\Lambda < 2(1+\beta). \tag{558}$$

For an individual positive mode, the upper bound is $\alpha\lambda_i < 2(1+\beta)$.

Proof. Put $t_\lambda = 1 + \beta - \alpha\lambda$ and $z = (1+w)/(1-w)$. Direct expansion gives

$$(1-w)^2\chi_\lambda((1+w)/(1-w)) = \alpha\lambda + 2(1-\beta)w \tag{559}$$

$$+ \{2(1+\beta) - \alpha\lambda\}w^2. \tag{560}$$

The unit disk in z corresponds to $\operatorname{Re} w < 0$. A real quadratic $aw^2 + bw + c$, $b, c > 0$, has both roots with negative real parts exactly when $a > 0$; this follows either from their sum and product or from the conjugate-root case. The strict endpoint conditions also exclude $z = \pm 1$. Apply this criterion to each λ_i. □

Theorem 36 (Optimal interval-wide local spectral factor)**.** *Let $0 < \mu < \Lambda$ and $\kappa_H = \Lambda/\mu$. Over constant $\eta > 0$ and $0 \le \beta < 1$,*

$$\inf_{\eta,\beta} \sup_{\lambda\in[\mu,\Lambda]} \rho(\mathsf{J}_\lambda) = q_\star = \frac{\sqrt{\Lambda} - \sqrt{\mu}}{\sqrt{\Lambda} + \sqrt{\mu}}, \tag{561}$$

$$\beta_\star = q_\star^2, \qquad \alpha_\star = \frac{4}{(\sqrt{\Lambda} + \sqrt{\mu})^2}, \qquad \eta_\star = \frac{1}{\sqrt{\mu\Lambda}}. \tag{562}$$

The infimum concerns the linearized quadratic family.

Proof. Suppose all modal roots satisfy $|z_\pm| \le q < 1$. Since $\chi_\lambda(1) = \alpha\lambda$,

$$(1-q)^2 \le |(1 - z_+(\lambda))(1 - z_-(\lambda))| = \alpha\lambda \le (1+q)^2, \tag{563}$$

$$\frac{\Lambda}{\mu} \le \left(\frac{1+q}{1-q}\right)^2 \Longrightarrow q \ge q_\star. \tag{564}$$

For the displayed parameters,

$$1 + q_\star^2 - \alpha_\star\mu = 2q_\star, \qquad 1 + q_\star^2 - \alpha_\star\Lambda = -2q_\star, \tag{565}$$

$$-2q_\star \le 1 + q_\star^2 - \alpha_\star\lambda \le 2q_\star. \tag{566}$$

Thus every mode has either a conjugate pair of modulus $q_\star$ or a double root $\pm q_\star$. Finally,

$$1 - \beta_\star = \frac{4\sqrt{\mu\Lambda}}{(\sqrt{\Lambda} + \sqrt{\mu})^2} \tag{567}$$

gives the normalized step $\eta_\star$. □

Proposition 21 (Defective endpoints and finite-time powers)**.** *At $\lambda = \mu, \Lambda$ with the optimal parameters, J_λ has a nontrivial size-two Jordan block. For $z_0 = \pm q_\star$,*

$$\mathsf{J}_\lambda^k = z_0^k I + k z_0^{k-1}(\mathsf{J}_\lambda - z_0 I). \tag{568}$$

Consequently $\|\mathsf{J}^k\| = O((k+1)q_\star^k)$, and for every $\widetilde{q} \in (q_\star, 1)$ there is a finite $C_{\widetilde{q}}$ with $\|\mathsf{J}^k\| \le C_{\widetilde{q}}\widetilde{q}^k$. A one-step Euclidean contraction factor $q_\star$ does not follow.

Proof. The off-diagonal entry $-\eta_\star\beta_\star$ is nonzero, so a double-root mode is not $z_0 I$. Cayley–Hamilton gives $(\mathsf{J}_\lambda - z_0 I)^2 = 0$, and the binomial formula gives the identity. The final bound follows from $\sup_{k\ge 0}(k+1)(q_\star/\widetilde{q})^k < \infty$. If $\mu = \Lambda$, the choice $\beta = 0, \eta = 1/\mu$ instead makes $\mathsf{J}^2 = 0$; the position error vanishes in one linearized step. □

Proposition 22 (Negative quotient curvature produces an unstable mode)**.** *If $\eta > 0$, $0 \le \beta < 1$, and $\lambda < 0$, then*

$$z_+(\lambda) > 1, \qquad 0 \le z_-(\lambda) = \beta/z_+(\lambda) < 1. \tag{569}$$

For $\lambda = 0$, the roots are $1, \beta$. Thus the number of eigenvalues outside the unit circle contributed by negative-curvature modes is exactly the negative index of H.

Proof. $\chi_\lambda(1) = \alpha\lambda < 0$ and $\chi_\lambda(z) \to +\infty$ as $z \to +\infty$ imply a real root larger than one. Their product is β. For a zero mode, $\chi_0(z) = (z-1)(z-\beta)$. □

F.3. Damping and projection ablations at first order

Proposition 23 (Transverse-memory spectrum of a damped connection)**.** *At a balanced critical representative $p_\star$ with $H_\star = \mathrm{diag}(\sigma_1, \dots, \sigma_r)$, allow the incoming moment to have both horizontal and full vertical parts. For the update as ordered in Eq.* (29)*, the linearization on quotient position, horizontal memory, and vertical memory is block diagonal:*

$$D\mathscr{U}_\varepsilon(0) \simeq \mathsf{J} \oplus \beta\, 2\varepsilon(\mathscr{L}_\star + 2\varepsilon I)^{-1}, \tag{570}$$

$$\mathscr{L}_\star(\Omega) = H_\star\Omega + \Omega H_\star, \tag{571}$$

$$\mathrm{spec}_{\mathrm{vert}}(D\mathscr{U}_\varepsilon(0)) = \left\{ \frac{2\beta\varepsilon}{\sigma_i + \sigma_j + 2\varepsilon} \right\}_{i,j\le r}. \tag{572}$$

The exact projection has zero transverse-memory eigenvalues. Balance-only has transverse-memory eigenvalue β. The joint depth-L formula replaces $\sigma_i + \sigma_j$ by λ_{abk} in Eq. (353)*.*

Proof. At $p_\star$ write the pre-projection direction as $h + V_\star\Omega$. Then

$$C_\star(h) = 0, \qquad C_\star(V_\star\Omega) = \mathscr{L}_\star\Omega, \tag{573}$$

$$\Pi_\varepsilon(h + V_\star\Omega) = h + V_\star\{2\varepsilon(\mathscr{L}_\star + 2\varepsilon I)^{-1}\Omega\}, \tag{574}$$

$$D\pi_\star V_\star\Omega = 0. \tag{575}$$

The fresh-gradient differential at criticality is horizontal:

$$D(C_p g(p))[\delta p] = 0, \quad g(p_\star) = 0 \Longrightarrow C_\star Dg(p_\star)[\delta p] = 0. \tag{576}$$

Every coordinate-transport derivative multiplies the zero base moment. Horizontal position and memory therefore give J, whereas only β times the displayed damped projector acts on vertical memory. Balanced representative rotations disappear after passing to quotient position. □

Corollary 13 (Equality of the linearized spectral radii)**.** *If $\mathsf{H} \succ 0$ and the Schur condition holds, then*

$$\rho(D\mathscr{U}_\varepsilon(0)) = \rho(\mathsf{J}) = \rho(D\mathscr{U}_{\text{balance-only}}(0)). \tag{577}$$

Proof. Each horizontal mode has determinant β, hence its spectral radius is at least $\sqrt{\beta}$. Every transverse-memory eigenvalue lies in $[0, \beta]$, and $\beta \le \sqrt{\beta}$. This is a first-order statement at a regular critical point. The mixed finite-step terms in Eq. (145) remain. □

F.4. From Schur stability to a quantified nonlinear basin

Theorem 37 (An explicit Lyapunov contraction neighborhood)**.** *Assume $\rho(\mathsf{J}) < 1$, $\mathscr{U} \in C^2(B_R)$, $\mathscr{U}(0) = 0$, and*

$$\sup_{\|s\|_2 \le R} \|D^2\mathscr{U}(s)\|_{\mathrm{op}} \le K_2. \tag{578}$$

Define

$$\mathsf{P} = \sum_{k=0}^{\infty} (\mathsf{J}^k)^\top \mathsf{J}^k, \qquad \|s\|_{\mathsf{P}} = (s^\top \mathsf{P} s)^{1/2}, \tag{579}$$

$$q_0 = \sqrt{1 - \lambda_{\max}(\mathsf{P})^{-1}}, \qquad C_{\mathsf{P}} = \frac{K_2\sqrt{\lambda_{\max}(\mathsf{P})}}{\lambda_{\min}(\mathsf{P})}, \tag{580}$$

$$r_{\mathsf{P}} \le \min\left\{ R\sqrt{\lambda_{\min}(\mathsf{P})}, \frac{1 - q_0}{2C_{\mathsf{P}}} \right\}, \qquad q_{\mathsf{P}} = (1 + q_0)/2. \tag{581}$$

If $C_{\mathsf{P}} = 0$, the second radius restriction is omitted. Then $B_{\mathsf{P}}(r_{\mathsf{P}})$ is invariant and

$$\|\mathscr{U}(s) - \mathscr{U}(s')\|_{\mathsf{P}} \le q_{\mathsf{P}}\|s - s'\|_{\mathsf{P}}, \tag{582}$$

$$\|s_T\|_{\mathsf{P}} \le q_{\mathsf{P}}^T\|s_0\|_{\mathsf{P}}. \tag{583}$$

If $f \circ \psi - f(X_\star) \le \Lambda_{\mathrm{loc}} \|x\|_2^2/2$ there,

$$f(X_T) - f(X_\star) \le \frac{\Lambda_{\mathrm{loc}}}{2\lambda_{\min}(\mathsf{P})} q_{\mathsf{P}}^{2T} \|s_0\|_{\mathsf{P}}^2. \tag{584}$$

Proof. Schur stability makes the series convergent and

$$\mathsf{J}^\top \mathsf{P} \mathsf{J} = \mathsf{P} - I, \qquad \mathsf{P} \succeq I, \tag{585}$$

$$\|\mathsf{J}\|_{\mathsf{P}\to\mathsf{P}}^2 = \sup_{s\neq 0} \frac{s^\top(\mathsf{P}-I)s}{s^\top \mathsf{P} s} = 1 - \lambda_{\max}(\mathsf{P})^{-1}. \tag{586}$$

The derivative-Lipschitz bound gives

$$\|D\mathscr{U}(s) - \mathsf{J}\|_{\mathsf{P}\to\mathsf{P}} \le \sqrt{\frac{\lambda_{\max}(\mathsf{P})}{\lambda_{\min}(\mathsf{P})}} K_2 \|s\|_2 \tag{587}$$

$$\le C_{\mathsf{P}} \|s\|_{\mathsf{P}}, \tag{588}$$

$$\sup_{\|s\|_{\mathsf{P}} \le r_{\mathsf{P}}} \|D\mathscr{U}(s)\|_{\mathsf{P}\to\mathsf{P}} \le q_0 + C_{\mathsf{P}} r_{\mathsf{P}} \le q_{\mathsf{P}} < 1. \tag{589}$$

Integrate along the segment joining s, s' in the convex P ball. Setting $s' = 0$ proves invariance and then the iterate bound. The risk bound follows from the position block. □

Corollary 14 (Robustness inside the certified basin)**.** *For $s_{t+1} = \mathscr{U}(s_t) + e_t$ with $\|e_t\|_{\mathsf{P}} \le \epsilon \le (1-q_{\mathsf{P}}) r_{\mathsf{P}}$ and $\|s_0\|_{\mathsf{P}} \le r_{\mathsf{P}}$,*

$$\|s_T\|_{\mathsf{P}} \le q_{\mathsf{P}}^T \|s_0\|_{\mathsf{P}} + \frac{\epsilon(1-q_{\mathsf{P}}^T)}{1-q_{\mathsf{P}}}. \tag{590}$$

Proof. The one-step bound is $\|s_{t+1}\|_{\mathsf{P}} \le q_{\mathsf{P}} \|s_t\|_{\mathsf{P}} + \epsilon \le r_{\mathsf{P}}$. Induction and a geometric sum give the result. □

F.5. Eckart–Young critical points and their complete quotient spectrum

Theorem 38 (Exact residual-gap Hessian spectrum)**.** *Let $f(X) = \frac{1}{2}\|X - Y\|_F^2$ and let $X = U\Sigma V^\top$ be a regular rank-r critical point. Put $p = m - r$, $q = d - r$, and $s = \min(p, q)$. Then*

$$Y = U\Sigma V^\top + U_\perp N V_\perp^\top, \tag{591}$$

$$\tau_1 \ge \cdots \ge \tau_s \ge 0 \quad \textit{are the singular values of } N. \tag{592}$$

The complete quotient Hessian spectrum, with multiplicities, is

$$\{\sigma_i + \sigma_j\}_{i,j\le r} \;\cup\; \{\sigma_i + \tau_k, \sigma_i - \tau_k\}_{i\le r, k\le s} \;\cup\; \{\sigma_i\}_{i\le r}^{|p-q|}. \tag{593}$$

Its negative index is exactly $\sum_{i,k} \mathbf{1}_{\{\tau_k > \sigma_i\}}$ and its nullity is $\sum_{i,k} \mathbf{1}_{\{\tau_k = \sigma_i\}}$.

Proof. Stationarity is $P_X(X - Y) = 0$, which gives the displayed block form of Y. At $B = U\Sigma^{1/2}$, $A = \Sigma^{1/2}V^\top$, write a horizontal lift as

$$\xi_B = UC_B + U_\perp D_B, \qquad \xi_A = C_A V^\top + E_A V_\perp^\top, \tag{594}$$

$$\Sigma^{1/2} C_B = C_A \Sigma^{1/2}. \tag{595}$$

The core subspace is orthogonal to the off-core subspace in factor norm. On the core, $L_p^* L_p$ has eigenvalues $\sigma_i + \sigma_j$. For column d_i of D_B and transposed row e_i of E_A, the pullback identity gives the remaining quadratic forms

$$\mathrm{Hess}_g f[L_p\xi, L_p\xi] = \|L_p \xi_{\mathrm{core}}\|_F^2 + \sum_{i=1}^{r} \begin{pmatrix} d_i \\ e_i \end{pmatrix}^\top \begin{pmatrix} \sigma_i I_p & -N \\ -N^\top & \sigma_i I_q \end{pmatrix} \begin{pmatrix} d_i \\ e_i \end{pmatrix}. \tag{596}$$

An SVD of N makes each 2×2 coupled block have eigenvalues $\sigma_i \pm \tau_k$; unmatched rectangular coordinates contribute σ_i. The count is $r^2 + r(2s + |p-q|) = r(m + d - r)$. All core and unmatched eigenvalues are positive, proving the two index formulas. □

Corollary 15 (A sharp spectral-gap condition number)**.** *If $m > r$ or $d > r$, X is the best rank-r approximation to Y, and $\sigma_r(Y) > \sigma_{r+1}(Y)$, then*

$$\lambda_{\min}(\mathrm{Hess}_g f(X)) = \sigma_r(Y) - \sigma_{r+1}(Y), \tag{597}$$

$$\lambda_{\max}(\mathrm{Hess}_g f(X)) = 2\sigma_1(Y), \tag{598}$$

$$\kappa(\mathrm{Hess}_g f(X)) = \frac{2\sigma_1(Y)}{\sigma_r(Y) - \sigma_{r+1}(Y)}. \tag{599}$$

Use $\sigma_{r+1}(Y) = 0$ when $r = \min(m, d)$. Every regular critical point is either a global rank-r minimum or has a negative-curvature direction.

Proof. The Eckart–Young theorem (Eckart & Young, 1936) identifies the best approximation with the retained leading singular values. The exact spectrum then has smallest eigenvalue $\sigma_r - \tau_1$ and largest $2\sigma_1$. At an arbitrary critical point, $\tau_1 \le \sigma_r$ makes all retained singular values at least all omitted singular values, so Eckart–Young makes it a global minimum. If $\tau_1 > \sigma_r$, Eq. (593) supplies a strictly negative eigenvalue. Spectral ties account for the stated zero modes. □

F.6. Exact local task conditioning versus connection conditioning

Theorem 39 (Reconstruction spectra at an exact planted solution)**.** *For $f(X) = \frac{1}{2}\|X - X_\star\|_F^2$ at a regular solution, the quotient Hessian is $K_{X_\star}$. For the depth-L balanced metric it is $K_{X_\star}^{(L)}$. Writing $\kappa_\star = \sigma_1(X_\star)/\sigma_r(X_\star)$,*

$$\kappa(\mathrm{Hess}_g f(X_\star)) = \begin{cases} 2\kappa_\star, & m > r \text{ or } d > r, \\ \kappa_\star, & m = d = r, \end{cases} \tag{600}$$

$$\kappa(\mathrm{Hess}_{g^{(L)}} f(X_\star)) = \begin{cases} L\kappa_\star^{2-2/L}, & m > r \text{ or } d > r, \\ \kappa_\star^{2-2/L}, & m = d = r. \end{cases} \tag{601}$$

In contrast, the joint connection normal equation has

$$\kappa(\mathcal{N}_W) = \kappa_\star^{2/L} \cot^2(\pi/(2L)). \tag{602}$$

Proof. At $X_\star$, the ambient gradient vanishes and the ambient Hessian is the identity. Thus Eq. (236) reduces to $\mathrm{Hess}_g f = K_{X_\star}$. Its core eigenvalues are $\sigma_i + \sigma_j$; any nonempty off-diagonal block supplies eigenvalues σ_i. For depth L, each core eigenvalue is the sum of L positive monomials, so

$$L\sigma_r^{2-2/L} \le \chi_{ab}^{(L)} \le L\sigma_1^{2-2/L}, \tag{603}$$

$$\min \chi_{ab}^{(L)} = L\sigma_r^{2-2/L}, \qquad \max \chi_{ab}^{(L)} = L\sigma_1^{2-2/L}. \tag{604}$$

If present, off-diagonal eigenvalues are $\sigma_i^{2-2/L}$. Take the exact ratios and apply Eq. (356). The conditioning of the Sylvester solve and the conditioning of the task Hessian can therefore change in different directions with depth. □

G. Noise Covariance, Averaging, and Intrinsic Statistical Efficiency

G.1. An exactly solvable stochastic tangent model

Definition 8. In a $g_\star$-orthonormal tangent frame, let $\mathsf{H} \succ 0$ and consider the quadratic model

$$F(x) = \tfrac{1}{2} x^\top \mathsf{H} x, \tag{605}$$

$$m_{t+1} = \beta m_t + (1-\beta)(\mathsf{H}x_t + \zeta_t), \tag{606}$$

$$x_{t+1} = x_t - \eta m_{t+1}, \qquad \mathbb{E}\zeta_t = 0, \qquad \mathbb{E}\zeta_t\zeta_t^\top = \mathsf{C}. \tag{607}$$

The noise is IID and independent of the initial state, with finite second moment. Here η, β are constant and satisfy Eq. (558). This model is the linearization of the stochastic quotient iteration, not an exact global representation of a nonlinear network loss.

Theorem 40 (Discrete Lyapunov covariance)**.** *With $\alpha = \eta(1-\beta)$ and*

$$\mathsf{N} = \begin{pmatrix} -\alpha I_D \\ (1-\beta) I_D \end{pmatrix}, \tag{608}$$

the centered state covariance converges to the unique positive semidefinite solution

$$\mathsf{S} = \mathsf{J}\mathsf{S}\mathsf{J}^\top + \mathsf{N}\mathsf{C}\mathsf{N}^\top, \tag{609}$$

$$\mathsf{S} = \sum_{k=0}^{\infty} \mathsf{J}^k \mathsf{N}\mathsf{C}\mathsf{N}^\top (\mathsf{J}^k)^\top, \tag{610}$$

$$\operatorname{vec} \mathsf{S} = (I - \mathsf{J} \otimes \mathsf{J})^{-1} \operatorname{vec}(\mathsf{N}\mathsf{C}\mathsf{N}^\top). \tag{611}$$

Proof. For $s_t = (x_t, m_t)$,

$$s_T = \mathsf{J}^T s_0 + \sum_{t=0}^{T-1} \mathsf{J}^{T-1-t} \mathsf{N} \zeta_t, \tag{612}$$

$$\operatorname{Cov}(s_T) = \mathsf{J}^T \operatorname{Cov}(s_0) (\mathsf{J}^T)^\top + \sum_{k=0}^{T-1} \mathsf{J}^k \mathsf{N}\mathsf{C}\mathsf{N}^\top (\mathsf{J}^k)^\top. \tag{613}$$

The initial term vanishes because $\rho(\mathsf{J}) < 1$. Any difference of two stationary solutions satisfies $\Delta = \mathsf{J}^k \Delta (\mathsf{J}^k)^\top$ for all k, hence $\Delta = 0$. Vectorization gives the final equation since $\rho(\mathsf{J} \otimes \mathsf{J}) = \rho(\mathsf{J})^2 < 1$. □

Theorem 41 (Exact modal variance and stationary excess risk)**.** *Diagonalize* H *and put* $\nu_i^2 = e_i^\top \mathsf{C} e_i$*. At stationarity,*

$$\mathbb{E} x_i^2 = \frac{\eta(1+\beta)\nu_i^2}{\lambda_i\{2(1+\beta) - \eta(1-\beta)\lambda_i\}}, \tag{614}$$

$$\mathbb{E} m_i^2 = \frac{2(1-\beta)\nu_i^2}{2(1+\beta) - \eta(1-\beta)\lambda_i}, \tag{615}$$

$$\mathbb{E}[x_i m_i] = -\frac{(1-\beta)\lambda_i}{1+\beta} \mathbb{E} x_i^2, \tag{616}$$

$$\mathbb{E} F(x) = \frac{\eta}{2} \sum_{i=1}^{D} \frac{(1+\beta)\nu_i^2}{2(1+\beta) - \eta(1-\beta)\lambda_i}. \tag{617}$$

The noise covariance need not commute with H*.*

Proof. Eliminate m in a single mode:

$$x_{t+1} = a_i x_t - \beta x_{t-1} - \alpha \zeta_{t,i}, \qquad a_i = 1 + \beta - \alpha\lambda_i. \tag{618}$$

Write $s_i = \mathbb{E} x_{t,i}^2$ and $c_i = \mathbb{E}[x_{t,i} x_{t-1,i}]$. Independence of the new noise gives

$$c_i = a_i s_i - \beta c_i \Longrightarrow c_i = \frac{a_i}{1+\beta} s_i, \tag{619}$$

$$s_i = (a_i^2 + \beta^2) s_i - 2 a_i \beta c_i + \alpha^2 \nu_i^2, \tag{620}$$

$$s_i \frac{1-\beta}{1+\beta} \{(1+\beta)^2 - a_i^2\} = \alpha^2 \nu_i^2. \tag{621}$$

Use

$$(1+\beta)^2 - a_i^2 = \alpha\lambda_i\{2(1+\beta) - \alpha\lambda_i\} \tag{622}$$

and $\alpha = \eta(1-\beta)$ to obtain the first identity. Since $m_t = (x_{t-1} - x_t)/\eta$,

$$\mathbb{E} m_{t,i}^2 = 2(s_i - c_i)/\eta^2, \qquad \mathbb{E}[x_{t,i} m_{t,i}] = (c_i - s_i)/\eta. \tag{623}$$

Finally $F = \frac{1}{2}\sum_i \lambda_i x_i^2$ uses only diagonal modal variances, regardless of cross-mode noise correlations. □

Corollary 16 (The rate–variance distinction)**.** *For fixed $\beta < 1$ and $\eta \downarrow 0$,*

$$\mathbb{E}F(x) = \frac{\eta}{4}\operatorname{tr}\mathsf{C} + \frac{\eta^2(1-\beta)}{8(1+\beta)}\operatorname{tr}(\mathsf{HC}) + O(\eta^3). \tag{624}$$

For any mode with $\nu_i^2 > 0$, its variance diverges as $\eta(1-\beta)\lambda_i \uparrow 2(1+\beta)$.

Proof. Expand each denominator in Eq. (617) by $(1-u)^{-1} = 1 + u + O(u^2)$. The strict stability margin makes this expansion uniform over the finitely many modes. The divergence follows directly from the exact positive numerator. □

G.2. Averaging removes the fixed-step modal variance inflation

Theorem 42 (Exact boundary identity for the average)**.** *For $\bar{x}_T = T^{-1}\sum_{t=0}^{T-1} x_t$ in Eq.* (607)*,*

$$\sqrt{T}\,\bar{x}_T = -\mathsf{H}^{-1}\frac{1}{\sqrt{T}}\sum_{t=0}^{T-1}\zeta_t + \frac{\mathsf{H}^{-1}}{\sqrt{T}}\left\{\frac{x_0 - x_T}{\eta} + \frac{\beta}{1-\beta}(m_T - m_0)\right\}. \tag{625}$$

For square-integrable s_0,

$$\sqrt{T}\,\bar{x}_T \xrightarrow{d} N(0, \mathsf{H}^{-1}\mathsf{C}\mathsf{H}^{-1}), \tag{626}$$

$$T\,\mathbb{E}[\bar{x}_T\bar{x}_T^\top] \longrightarrow \mathsf{H}^{-1}\mathsf{C}\mathsf{H}^{-1}, \tag{627}$$

$$T\,\mathbb{E}F(\bar{x}_T) \longrightarrow \tfrac{1}{2}\operatorname{tr}(\mathsf{C}\mathsf{H}^{-1}). \tag{628}$$

Proof. Sum both update equations:

$$\sum_{t<T} m_{t+1} = (x_0 - x_T)/\eta, \tag{629}$$

$$(1-\beta)\mathsf{H}\sum_{t<T} x_t = \sum_{t<T}(m_{t+1} - \beta m_t) - (1-\beta)\sum_{t<T}\zeta_t \tag{630}$$

$$= (1-\beta)\sum_{t<T} m_{t+1} + \beta(m_T - m_0) - (1-\beta)\sum_{t<T}\zeta_t. \tag{631}$$

This proves the exact identity. Schur stability and the covariance series imply $\sup_T \mathbb{E}(\|x_T\|^2 + \|m_T\|^2) < \infty$; the boundary term is therefore $O_{L^2}(T^{-1/2})$. The IID central limit theorem gives the distributional limit. The leading noise average has exactly covariance C; Cauchy–Schwarz bounds its cross term with the boundary by $O(T^{-1/2})$. Taking the quadratic-risk trace proves the final claim. This is an averaging statement, not a claim that the unaveraged stationary covariance is independent of momentum. □

G.3. A nonlinear quotient averaging theorem with declared moment stability

The next result gives the retraction and transport remainder calculation underlying stochastic averaging (Polyak & Juditsky, 1992).

Assumption 6 (Local stochastic approximation regime)**.** In one fixed regular chart around a population minimizer, let $\mathsf{H} = \nabla^2(R\circ\psi)(0) \succ 0$, $0 \le \beta < 1$ be fixed, and $\eta_t = c(t+1)^{-\gamma}$ with $1/2 < \gamma < 1$. The coordinate state satisfies

$$m_{t+1} = \beta m_t + (1-\beta)(\mathsf{H}x_t + \zeta_{t+1}) + r_{t+1}^m, \tag{632}$$

$$x_{t+1} = x_t - \eta_t m_{t+1} + r_{t+1}^x, \tag{633}$$

$$\mathbb{E}_t\zeta_{t+1} = 0, \qquad \frac{1}{T}\sum_{t<T}\mathbb{E}_t[\zeta_{t+1}\zeta_{t+1}^\top] \xrightarrow{p} \mathsf{C}, \tag{634}$$

$$\sup_t \mathbb{E}\|\zeta_{t+1}\|^{2+\delta} < \infty \quad \text{for some } \delta > 0, \tag{635}$$

$$\mathbb{E}\|x_t\|^2 \le C\eta_t, \qquad \sup_t \mathbb{E}\|m_t\|^2 \le C, \tag{636}$$

$$\mathbb{E}\|r_{t+1}^m\| \le C\eta_t, \qquad \mathbb{E}\|r_{t+1}^x\| \le C\eta_t^2. \tag{637}$$

Iterates stay in the regular chart, or a localization event with probability tending to one is used. The displayed last-iterate moment stability is a premise; it is stronger than an averaged stationarity estimate.

Proposition 24 (Where the geometric remainders enter)**.** *For the exact quotient iteration, the last two remainder bounds in Eq.* (637) *follow locally from the displayed moment bounds if the gradient has a quadratic Taylor remainder, retraction and transport have bounded local derivatives, and the pre-transport direction has uniformly bounded second moment.*

Proof. For constants uniform on the chart,

$$g(x) = \mathsf{H}x + \mathcal{R}_g(x), \qquad \|\mathcal{R}_g(x)\| \le C_g\|x\|^2, \tag{638}$$

$$[\mathrm{Retr}_X(-\eta v)]_\psi = x - \eta v + \mathcal{R}_R(x, v, \eta), \qquad \|\mathcal{R}_R\| \le C_R\eta^2\|v\|^2, \tag{639}$$

$$[\mathcal{T}_{X,X^+}]_{E(X),E(X^+)} = I + \mathcal{R}_T, \qquad \|\mathcal{R}_T\|_{\mathrm{op}} \le C_T\eta\|v\|. \tag{640}$$

With $\widetilde{m} = \beta m + (1-\beta)(g(x) + \zeta)$ and $m^+ = \mathcal{T}\widetilde{m}$,

$$r^m = (1-\beta)\mathcal{R}_g(x) + \mathcal{R}_T\widetilde{m}, \tag{641}$$

$$r^x = \mathcal{R}_R + \eta(m^+ - \widetilde{m}). \tag{642}$$

Cauchy–Schwarz, $\mathbb{E}\|x\|^2 = O(\eta)$, and the second-moment bounds give $\mathbb{E}\|r^m\| = O(\eta)$ and $\mathbb{E}\|r^x\| = O(\eta^2)$. All frame and transport corrections are retained in these remainders. □

Theorem 43 (Asymptotic linearity of averaged quotient iterates)**.** *Under Assumption 6,*

$$\sqrt{T}\,\bar{x}_T = -\mathsf{H}^{-1}\frac{1}{\sqrt{T}}\sum_{t<T}\zeta_{t+1} + o_p(1), \tag{643}$$

$$\sqrt{T}\,\bar{x}_T \xrightarrow{d} N(0, \mathsf{H}^{-1}\mathsf{C}\mathsf{H}^{-1}). \tag{644}$$

If, in addition, $\sup_T \mathbb{E}\|\sqrt{T}\bar{x}_T\|^{2+\delta} < \infty$ *and the population Taylor remainder is* $O(\|x\|^{2+\delta})$ *for some* $0 < \delta \le 1$*, then*

$$T\,\mathbb{E}[R(\psi(\bar{x}_T)) - R(X_\star)] \longrightarrow \tfrac{1}{2}\,\mathrm{tr}(\mathsf{C}\mathsf{H}^{-1}). \tag{645}$$

Proof. Eliminate m_{t+1} in the two recursions and sum:

$$\begin{aligned}\mathsf{H}\sum_{t<T}x_t = &-\sum_{t<T}\zeta_{t+1} + \sum_{t<T}\frac{x_t - x_{t+1}}{\eta_t} + \frac{\beta}{1-\beta}(m_T - m_0)\\ &+ \sum_{t<T}\frac{r^x_{t+1}}{\eta_t} - \frac{1}{1-\beta}\sum_{t<T}r^m_{t+1}.\end{aligned} \tag{646}$$

Abel summation gives the exact boundary expansion

$$\sum_{t<T}\frac{x_t - x_{t+1}}{\eta_t} = \frac{x_0}{\eta_0} - \frac{x_T}{\eta_{T-1}} + \sum_{t=1}^{T-1}x_t\left(\frac{1}{\eta_t} - \frac{1}{\eta_{t-1}}\right), \tag{647}$$

$$\mathbb{E}\left\|\sum_{t<T}\frac{x_t - x_{t+1}}{\eta_t}\right\| \le C(1 + T^{\gamma/2}), \tag{648}$$

$$\mathbb{E}\sum_{t<T}\left(\|r^x_{t+1}\|/\eta_t + \|r^m_{t+1}\|\right) \le C\sum_{t<T}\eta_t = O(T^{1-\gamma}). \tag{649}$$

After division by $\sqrt{T}$, these errors are $O(T^{(\gamma-1)/2} + T^{1/2-\gamma}) = o(1)$. The moment bound also makes the momentum endpoint term $o_{L^1}(1)$. Uniform $(2+\delta)$ moments give the conditional Lindeberg condition:

$$\frac{1}{T}\sum_{t<T}\mathbb{E}_t\left[\|\zeta_{t+1}\|^2\mathbf{1}_{\{\|\zeta_{t+1}\| > \epsilon\sqrt{T}\}}\right] \xrightarrow{p} 0. \tag{650}$$

The martingale central limit theorem and the conditional covariance premise prove the Gaussian limit. For the final claim, uniform $(2+\delta)$ moments allow passage of quadratic expectations, while

$$T\,\mathbb{E}\|\bar{x}_T\|^{2+\delta} = T^{-\delta/2}\mathbb{E}\|\sqrt{T}\bar{x}_T\|^{2+\delta} \to 0. \tag{651}$$

Expand the population risk to second order. □

Corollary 17 (An implementable rank-preserving average)**.** *Let $\bar{Y}_T = T^{-1}\sum_{t<T} X_t$ and let $\Pi_r(\bar{Y}_T)$ be its best rank-r Frobenius approximation. Under the preceding assumptions,*

$$\sqrt{T}\{\Pi_r(\bar{Y}_T) - \psi(\bar{x}_T)\} \xrightarrow{p} 0. \tag{652}$$

Thus its first-order product-space Gaussian limit is the same.

Proof. Near a regular $X_\star$, the spectral gap $\sigma_r(X_\star) > 0 = \sigma_{r+1}(X_\star)$ gives a smooth local nearest-rank projection, with

$$\Pi_r(X_\star + E) = X_\star + P_{X_\star}E + O(\|E\|_F^2). \tag{653}$$

No distinctness of the positive singular values is required. For $J = D\psi(0)$,

$$X_t = X_\star + Jx_t + O(\|x_t\|^2), \tag{654}$$

$$\bar{Y}_T = X_\star + J\bar{x}_T + O_p\left(T^{-1}\sum_{t<T}\eta_t\right) = X_\star + J\bar{x}_T + O_p(T^{-\gamma}). \tag{655}$$

Since $P_{X_\star}J = J$, the difference in the claim is $O_p(T^{-\gamma} + \|\bar{x}_T\|^2)$, which is $o_p(T^{-1/2})$. □

Proposition 25 (Coordinate and gauge invariance of statistical efficiency)**.** *For $y = Tx + O(\|x\|^2)$ with invertible T,*

$$\mathsf{H}_y = T^{-\top}\mathsf{H}_x T^{-1}, \qquad \mathsf{C}_y = T^{-\top}\mathsf{C}_x T^{-1}, \tag{656}$$

$$\mathsf{H}_y^{-1}\mathsf{C}_y\mathsf{H}_y^{-1} = T(\mathsf{H}_x^{-1}\mathsf{C}_x\mathsf{H}_x^{-1})T^\top, \tag{657}$$

$$\operatorname{tr}(\mathsf{C}_y\mathsf{H}_y^{-1}) = \operatorname{tr}(\mathsf{C}_x\mathsf{H}_x^{-1}). \tag{658}$$

Proof. At a stationary population point, Hessians transform by congruence; per-example gradients transform as covectors. Apply the chain rule and the cyclic trace identity. Gauge representatives induce the same quotient estimator and therefore the same scalar efficiency constant. □

H. Statistical Generalization: State Stability and Intrinsic Capacity

H.1. Statistical model and product sensitivity

Assumption 7.

$$S = (z_1, \dots, z_n) \sim \mathcal{D}^n, \qquad R(X) = \mathbb{E}_{z\sim\mathcal{D}}\ell(X; z), \tag{659}$$

$$R_S(X) = n^{-1}\sum_i \ell(X; z_i), \qquad 0 \le \ell(X; z) \le 1, \tag{660}$$

$$|\ell(X; z) - \ell(Y; z)| \le L_\ell \|X - Y\|_F. \tag{661}$$

All connecting quotient paths used below lie in the stated regular spectral band.

Lemma 9.

$$\|X - Y\|_F \le \sqrt{2b}\, d_g(X, Y), \tag{662}$$

$$|\ell(X; z) - \ell(Y; z)| \le C_\ell d_g(X, Y), \qquad C_\ell = L_\ell\sqrt{2b}. \tag{663}$$

Proof. For every admissible connecting path γ,

$$\|X - Y\|_F \le \int_0^1 \|\dot{\gamma}(t)\|_F\, dt \le \sqrt{2b}\int_0^1 \|\dot{\gamma}(t)\|_g\, dt. \tag{664}$$

Take the infimum over paths. □

H.2. Uniform replace-one stability for the full optimizer state

Definition 9. Let $S^{(j)} = (z_1, \dots, z'_j, \dots, z_n)$, with independent z'_j. The two runs use the same initialization and sampled indices. Let $s_t = (X_t, \mathsf{m}_t, \mathsf{v}_t, \dots)$ contain every persistent optimizer variable, and let $d_*(s, s') \geq d_g(X, X')$.

Assumption 8 (Actual state-map regularity)**.** For the implemented one-example map $U_{t,z}$,

$$d_*(U_{t,z}s, U_{t,z}s') \leq (1 + L_U\eta_t)d_*(s, s'), \tag{665}$$

$$d_*(U_{t,z}s, U_{t,z'}s) \leq 2C_U\eta_t, \tag{666}$$

$$|\ell(X(s); z) - \ell(X(s'); z)| \leq C_\ell d_*(s, s'). \tag{667}$$

Theorem 44 (Stability with all expansion factors retained)**.** *For $I_t \sim \mathrm{Unif}\{1, \dots, n\}$,*

$$\mathbb{E}d_*(s_T(S), s_T(S^{(j)})) \leq \frac{2C_U}{n} \sum_{t<T} \eta_t \prod_{k=t+1}^{T-1} (1 + L_U\eta_k), \tag{668}$$

$$\epsilon_{\mathrm{stab}} \leq \frac{2C_\ell C_U}{n} \sum_{t<T} \eta_t \prod_{k=t+1}^{T-1} (1 + L_U\eta_k), \tag{669}$$

$$|\mathbb{E}[R(X_T) - R_S(X_T)]| \leq \epsilon_{\mathrm{stab}}. \tag{670}$$

Proof. Let $\Delta_t = d_*(s_t, s'_t)$, $a_t = 1 + L_U\eta_t$.

$$I_t \neq j \Longrightarrow \Delta_{t+1} \leq a_t\Delta_t, \tag{671}$$

$$I_t = j \Longrightarrow \Delta_{t+1} \leq d_*(U_{t,z_j}s_t, U_{t,z_j}s'_t) + d_*(U_{t,z_j}s'_t, U_{t,z'_j}s'_t) \tag{672}$$

$$\leq a_t\Delta_t + 2C_U\eta_t, \tag{673}$$

$$\mathbb{E}_t\Delta_{t+1} \leq a_t\Delta_t + 2C_U\eta_t/n, \tag{674}$$

$$\mathbb{E}\Delta_T \leq \left(\prod_{k<T} a_k\right)\Delta_0 + \frac{2C_U}{n} \sum_{t<T} \eta_t \prod_{k>t} a_k. \tag{675}$$

Since $\Delta_0 = 0$, this proves the first two claims. Exchangeability gives

$$\mathbb{E}[R(X_T(S)) - R_S(X_T(S))] = \frac{1}{n} \sum_j \mathbb{E}[\ell(X_T(S^{(j)}); z_j) - \ell(X_T(S); z_j)]. \tag{676}$$

Bound each summand in absolute value by ϵ_{stab}. □

Corollary 18 (Closed forms for nonexpansive, expansive, and contractive maps)**.** *For constant η,*

$$L_U = 0 \Longrightarrow \epsilon_{\mathrm{stab}} \leq \frac{2C_\ell C_U \eta T}{n}, \tag{677}$$

$$L_U > 0 \Longrightarrow \epsilon_{\mathrm{stab}} \leq \frac{2C_\ell C_U}{nL_U}\{(1 + L_U\eta)^T - 1\}, \tag{678}$$

$$d_*(U_z s, U_z s') \leq (1 - \mu_U\eta)d_*(s, s') \Longrightarrow \epsilon_{\mathrm{stab}} \leq \frac{2C_\ell C_U}{n\mu_U}\{1 - (1 - \mu_U\eta)^T\}. \tag{679}$$

In the last line, $0 \leq 1 - \mu_U\eta < 1$ is required.

Proof. Use the exact sums

$$\eta \sum_{j=0}^{T-1} (1 + L_U\eta)^j = \frac{(1 + L_U\eta)^T - 1}{L_U}, \quad \eta \sum_{j=0}^{T-1} (1 - \mu_U\eta)^j = \frac{1 - (1 - \mu_U\eta)^T}{\mu_U}. \tag{680}$$

□

Corollary 19 (Stability constants from the implemented factor-state Jacobian)**.** *Suppose the domain and interpolation assumptions of Eq.* (315) *hold uniformly for the coupled runs, with constant* $\eta, \lambda, \Lambda_\lambda, C_U$*. For* $0 \le \ell \le 1$ *with product Lipschitz constant* L_ℓ*,*

$$C_{\ell,\mathrm{fac}} = L_\ell \sqrt{2\gamma}, \tag{681}$$

$$\epsilon_{\mathrm{stab}} \le \frac{2C_{\ell,\mathrm{fac}} C_U \eta}{n} \sum_{j=0}^{T-1} \Lambda_\lambda^j \tag{682}$$

$$= \frac{2C_{\ell,\mathrm{fac}} C_U \eta}{n} \begin{cases} T, & \Lambda_\lambda = 1, \\ (\Lambda_\lambda^T - 1)/(\Lambda_\lambda - 1), & \Lambda_\lambda \neq 1. \end{cases} \tag{683}$$

Proof. For two balanced factor states in the Gram band,

$$\|BA - B'A'\|_F \le \|(B - B')A\|_F + \|B'(A - A')\|_F \tag{684}$$

$$\le \sqrt{\gamma}(\|B - B'\|_F + \|A - A'\|_F) \le \sqrt{2\gamma}\|p - p'\|. \tag{685}$$

Thus the loss is $C_{\ell,\mathrm{fac}}$-Lipschitz in d_λ. Eq. (315) gives the coupled recursion

$$\mathbb{E}\Delta_{t+1} \le \Lambda_\lambda \mathbb{E}\Delta_t + 2C_U\eta/n, \qquad \Delta_0 = 0. \tag{686}$$

Unroll it and use the exact geometric sum. This argument uses the full factor-state Jacobian; neither C_U nor its momentum contribution is assigned zero. □

H.3. An explicit momentum-state sensitivity calculation

Proposition 26 (Weighted state metric in a flat chart)**.** *For*

$$v^+ = \beta v + (1 - \beta)\nabla \ell_z(x), \qquad x^+ = x - \eta v^+, \tag{687}$$

$$\|\nabla \ell_z(x)\| \le G, \qquad \|\nabla \ell_z(x) - \nabla \ell_z(x')\| \le L\|x - x'\|, \tag{688}$$

set $\delta_x = \|x - x'\|$*,* $\delta_v = \|v - v'\|$*. Then*

$$\begin{pmatrix} \delta_x^+ \\ \delta_v^+ \end{pmatrix} \le \begin{pmatrix} 1 + \eta(1-\beta)L & \eta\beta \\ (1-\beta)L & \beta \end{pmatrix} \begin{pmatrix} \delta_x \\ \delta_v \end{pmatrix} + 2(1-\beta)G \begin{pmatrix} \eta \\ 1 \end{pmatrix} \mathbf{1}_{\{I_t = j\}}. \tag{689}$$

For $0 < \beta < 1$ *and*

$$D_\eta = \delta_x + \lambda_\eta \delta_v, \qquad \lambda_\eta = \frac{\eta\beta}{1-\beta}, \tag{690}$$

$$D_\eta^+ \le (1 + \eta L) D_\eta + 2\eta G \mathbf{1}_{\{I_t = j\}}. \tag{691}$$

Proof. The row-vector identity is

$$(1, \lambda_\eta) \begin{pmatrix} 1 + \eta(1-\beta)L & \eta\beta \\ (1-\beta)L & \beta \end{pmatrix} = (1 + \eta L, \lambda_\eta), \tag{692}$$

$$(1, \lambda_\eta)\, 2(1-\beta)G \begin{pmatrix} \eta \\ 1 \end{pmatrix} = 2\eta G. \tag{693}$$

□

On the quotient, Eq. (667) must additionally include retraction and transport derivatives. Equations (197)–(237) provide finite local constants on the regular band; they are not set to zero. For varying steps,

$$D_{\eta_{t+1}}(s_{t+1}, s'_{t+1}) \le \max\{1, \lambda_{\eta_{t+1}}/\lambda_{\eta_t}\} D_{\eta_t}(s_{t+1}, s'_{t+1}). \tag{694}$$

H.4. Symmetrization and a nuclear-norm capacity bound

Definition 10. For $\mathcal{X}_\tau = \{X : \|X\|_* \le \tau\}$ and $h_X(z) = h_0(z) + \langle X, Z(z)\rangle$,

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_\tau) = \frac{1}{n}\mathbb{E}_\epsilon \sup_{X\in\mathcal{X}_\tau} \sum_i \epsilon_i h_X(z_i), \tag{695}$$

$$\Pr(\epsilon_i = 1) = \Pr(\epsilon_i = -1) = \tfrac{1}{2}. \tag{696}$$

Assume $\|Z(z)\|_F \le R_Z$.

Theorem 45 (Exact dual-norm reduction)**.**

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_\tau) = \frac{\tau}{n}\mathbb{E}_\epsilon \left\| \sum_i \epsilon_i Z(z_i) \right\|_2 \tag{697}$$

$$\le \frac{\tau R_Z}{\sqrt{n}}. \tag{698}$$

For $0 \le \ell(h; z) \le 1$ and scalar Lipschitz constant L_h, with probability at least $1 - q$,

$$\sup_{\|X\|_*\le\tau} |R(X) - R_S(X)| \le \frac{2L_h\tau R_Z}{\sqrt{n}} + 3\sqrt{\frac{\log(4/q)}{2n}}. \tag{699}$$

For balanced factors, $\|X\|_* = \frac{1}{2}(\|B\|_F^2 + \|A\|_F^2)$; if $\operatorname{rank} X \le r$ and $\|X\|_2 \le b$, one may take $\tau = rb$.

Proof. The fixed offset has zero sign expectation. Nuclear–spectral duality gives

$$\sup_{\|X\|_*\le\tau} \left\langle X, \sum_i \epsilon_i Z_i \right\rangle_F = \tau \left\| \sum_i \epsilon_i Z_i \right\|_2, \tag{700}$$

$$\mathbb{E}_\epsilon \left\| \sum_i \epsilon_i Z_i \right\|_2 \le \left(\mathbb{E}_\epsilon \left\| \sum_i \epsilon_i Z_i \right\|_F^2 \right)^{1/2} \tag{701}$$

$$= \left(\sum_i \|Z_i\|_F^2 \right)^{1/2} \le R_Z\sqrt{n}. \tag{702}$$

For a one-Lipschitz scalar map ϕ and arbitrary a_h, t_h,

$$\tfrac{1}{2}\sup_h(a_h + \phi(t_h)) + \tfrac{1}{2}\sup_h(a_h - \phi(t_h))$$

$$= \tfrac{1}{2}\sup_{h,h'}[a_h + a_{h'} + \phi(t_h) - \phi(t_{h'})] \tag{703}$$

$$\le \tfrac{1}{2}\sup_{h,h'}[a_h + a_{h'} + |t_h - t_{h'}|] \tag{704}$$

$$= \tfrac{1}{2}\sup_h(a_h + t_h) + \tfrac{1}{2}\sup_h(a_h - t_h). \tag{705}$$

Iterating over coordinates gives $\widehat{\mathfrak{R}}_S(\ell \circ \mathcal{H}_\tau) \le L_h\widehat{\mathfrak{R}}_S(\mathcal{H}_\tau)$. For an independent ghost sample S',

$$\mathbb{E}_S \sup_X(R(X) - R_S(X)) \le \mathbb{E}_{S,S'} \sup_X \frac{1}{n}\sum_i [\ell(X; z_i') - \ell(X; z_i)] \tag{706}$$

$$= \mathbb{E}_{S,S',\epsilon} \sup_X \frac{1}{n}\sum_i \epsilon_i[\ell(X; z_i') - \ell(X; z_i)] \tag{707}$$

$$\le 2\mathbb{E}_S\widehat{\mathfrak{R}}_S(\ell \circ \mathcal{H}_\tau). \tag{708}$$

Both the supremum gap and empirical loss complexity change by at most $1/n$ when one observation is replaced. Bounded differences and a union bound give

$$\sup_X (R - R_S) \le 2\mathbb{E}_S \widehat{\mathfrak{R}}_S + \sqrt{\log(2/q)/(2n)}, \tag{709}$$

$$\mathbb{E}_S \widehat{\mathfrak{R}}_S \le \widehat{\mathfrak{R}}_S + \sqrt{\log(2/q)/(2n)}. \tag{710}$$

Apply these one-sided bounds to both ℓ and $-\ell$, each with failure probability $q/2$. The loss ranges have length one, and their centered empirical complexities obey the same contraction bound. Substitution and a union bound prove Eq. (699). □

H.5. Explicit intrinsic covers from Schur coordinates

Theorem 46 (A Lipschitz constant for the rank chart)**.** *On a Schur-chart domain with*

$$\left\|C^{-1}\right\|_2, \left\|(C')^{-1}\right\|_2 \le s^{-1}, \qquad \|D\|_F, \|D'\|_F, \|E\|_F, \|E'\|_F \le h, \tag{711}$$

the map in Eq. (71) *obeys*

$$\|\Psi(C,D,E) - \Psi(C',D',E')\|_F \le \left(1 + \frac{h^2}{s^2}\right) \sqrt{\|C-C'\|_F^2 + \|D-D'\|_F^2 + \|E-E'\|_F^2}. \tag{712}$$

Proof. With $F = EC^{-1}D$,

$$F - F' = (E-E')C^{-1}D + E'C^{-1}(D-D') + E'(C^{-1} - (C')^{-1})D', \tag{713}$$

$$C^{-1} - (C')^{-1} = -C^{-1}(C-C')(C')^{-1}, \tag{714}$$

$$\|F-F'\|_F \le (h/s)\|E-E'\|_F + (h/s)\|D-D'\|_F + (h^2/s^2)\|C-C'\|_F, \tag{715}$$

$$\|F-F'\|_F^2 \le (2h^2/s^2 + h^4/s^4)(\|C-C'\|_F^2 + \|D-D'\|_F^2 + \|E-E'\|_F^2). \tag{716}$$

The squared norm of the full chart difference adds the three coordinate block norms, and $1 + 2h^2/s^2 + h^4/s^4 = (1 + h^2/s^2)^2$. □

Theorem 47 (A fully explicit global regular-band cover)**.** *Put*

$$D_r = r(m+d-r), \qquad J_r = \binom{m}{r}\binom{d}{r}, \tag{717}$$

$$s_0 = \frac{a^r}{b^{r-1}\sqrt{J_r}}, \qquad C_0 = 1 + \frac{rb^2}{s_0^2}, \qquad R_0 = \sqrt{r}\, b. \tag{718}$$

Then

$$N(\mathcal{K}_{a,b}, \|\cdot\|_F, \varepsilon) \le J_r(1 + 2C_0R_0/\varepsilon)^{D_r}. \tag{719}$$

Proof. For $X = U\Sigma V^\top$, Cauchy–Binet gives

$$\sum_{|I|=r} \det(U_I)^2 = \det(U^\top U) = 1, \qquad \sum_{|J|=r} \det(V_J)^2 = 1. \tag{720}$$

Hence there are I, J with

$$|\det X_{I,J}| = |\det U_I|\, \det\Sigma\, |\det V_J| \ge a^r/\sqrt{J_r}, \tag{721}$$

$$\sigma_r(X_{I,J}) \ge \frac{|\det X_{I,J}|}{\sigma_1(X_{I,J})^{r-1}} \ge s_0. \tag{722}$$

After row and column permutations, X lies in a Schur chart whose coordinate norm is at most $\|X\|_F \le R_0$ and whose two off-diagonal block norms are at most $h = \sqrt{r}\, b$. The preceding theorem gives constant C_0. A maximal ε/C_0-separated subset of the admissible coordinate domain is a net with centers in that domain. Disjoint radius-$\varepsilon/(2C_0)$ balls and the ambient radius-R_0 bound give at most $(1 + 2C_0R_0/\varepsilon)^{D_r}$ centers. Summing over all J_r charts proves the claim. □

Theorem 48 (Intrinsic-dimension uniform generalization)**.** *Under Eq.* (661)*, with probability at least* $1-q$*,*

$$\sup_{X\in\mathcal{K}_{a,b}} |R(X)-R_S(X)| \le 2L_\ell\varepsilon + \sqrt{\frac{\log(2J_r/q)+D_r\log(1+2C_0R_0/\varepsilon)}{2n}}. \tag{723}$$

Proof. For a finite ε net $\mathcal{N}$,

$$\Pr\left\{\max_{Y\in\mathcal{N}}|R(Y)-R_S(Y)|>t\right\} \le 2|\mathcal{N}|e^{-2nt^2}, \tag{724}$$

$$|R(X)-R_S(X)| \le |R(Y)-R_S(Y)| + |R(X)-R(Y)| + |R_S(X)-R_S(Y)| \tag{725}$$

$$\le |R(Y)-R_S(Y)| + 2L_\ell\varepsilon. \tag{726}$$

Insert Eq. (719) and choose $t=\sqrt{\log(2|\mathcal{N}|/q)/(2n)}$. □

Corollary 20 (A rank-bound cover without a lower singular-value margin)**.** *For* $\mathcal{C}_{r,R}=\{X:\operatorname{rank}X\le r,\ \|X\|_F\le R\}$*,*

$$N(\mathcal{C}_{r,R},\|\cdot\|_F,\varepsilon) \le (1+6\sqrt{r}\,R/\varepsilon)^{r(m+d+1)}, \tag{727}$$

$$\sup_{\mathcal{C}_{r,R}}|R-R_S| \le 2L_\ell\varepsilon + \sqrt{\frac{r(m+d+1)\log(1+6\sqrt{r}\,R/\varepsilon)+\log(2/q)}{2n}} \tag{728}$$

with probability at least $1-q$*. This bound has no* a^{-1} *constant, but uses a larger exponent than* D_r*.*

H.6. PAC–Bayes on quotient coordinates and volume cancellation

The change-of-measure step below is the standard exponential PAC–Bayes construction (Seldin et al., 2012); the chart, volume, and gauge cancellations are made explicit for $\mathcal{M}_r$.

Definition 11. For probability measures $Q\ll P$,

$$\mathrm{KL}(Q\|P)=\int\log\left(\frac{dQ}{dP}\right)dQ, \tag{729}$$

$$R(Q)=\int R(X)\,dQ(X), \qquad R_S(Q)=\int R_S(X)\,dQ(X). \tag{730}$$

A prior is independent of S. A posterior may be any measurable function of S and of the optimization randomness.

Lemma 10 (Variational change of measure)**.** *For measurable* F *with* $\int e^F\,dP<\infty$*,*

$$\int F\,dQ \le \mathrm{KL}(Q\|P)+\log\int e^F\,dP. \tag{731}$$

Proof. Writing $q=dQ/dP$, Jensen's inequality gives

$$\int F\,dQ-\mathrm{KL}(Q\|P)=\int\log(e^F/q)\,dQ \tag{732}$$

$$\le \log\int_{\{q>0\}} e^F q^{-1}\,dQ \le \log\int e^F\,dP. \tag{733}$$

Truncation proves the extended-valued form whenever the right side is finite. □

Theorem 49 (Simultaneous posterior and temperature bound)**.** *Let* $0\le\ell\le 1$*,* $S\sim\mathcal{D}^n$*, and let* P *be a fixed prior on a measurable predictor space. Fix* $\lambda_0>0$ *independently of* S*, and put*

$$\lambda_j=2^j\lambda_0, \qquad w_j=\frac{6}{\pi^2(j+1)^2}, \qquad j=0,1,\dots. \tag{734}$$

With probability at least $1-q$*, simultaneously for every* j *and every posterior* $Q\ll P$*,*

$$R(Q)-R_S(Q)\le\frac{\mathrm{KL}(Q\|P)+\log\frac{\pi^2(j+1)^2}{6q}}{\lambda_j n}+\frac{\lambda_j}{8}. \tag{735}$$

Proof. For fixed X, independence and the range-one exponential bound imply

$$\mathbb{E}_S \exp\{\lambda n(R(X) - R_S(X))\} = \prod_{i=1}^{n} \mathbb{E}_{z_i} \exp\{\lambda(R(X) - \ell(X; z_i))\} \tag{736}$$

$$\leq \exp(n\lambda^2/8). \tag{737}$$

Tonelli's theorem and prior independence yield

$$Z_j(S) = \int \exp\{\lambda_j n(R(X) - R_S(X)) - n\lambda_j^2/8\}\, dP(X), \tag{738}$$

$$\mathbb{E}_S Z_j(S) \leq 1, \qquad \Pr\{Z_j > (qw_j)^{-1}\} \leq qw_j. \tag{739}$$

Since $\sum_{j\geq 0} w_j = 1$, with probability at least $1 - q$,

$$\log Z_j(S) \leq \log(1/(qw_j)) \quad \text{for all } j. \tag{740}$$

On this single event, apply Eq. (731) to $F = \lambda_j n(R - R_S) - n\lambda_j^2/8$ for arbitrary Q. No union bound over posteriors is needed. □

Proposition 27 (Pushforward KL and Riemannian volume)**.** *Let $\Psi : \Theta \subset \mathbb{R}^{D_r} \to \mathcal{M}_r$ be an injective regular Schur chart. For Lebesgue densities p, q and coordinate metric $g_{ij}(\theta)$, set*

$$J_g(\theta) = \sqrt{\det(g_{ij}(\theta))}, \qquad d\mathrm{vol}_g(\Psi(\theta)) = J_g(\theta)\, d\theta. \tag{741}$$

Then

$$\frac{d\Psi_{\#}P}{d\mathrm{vol}_g}(\Psi(\theta)) = \frac{p(\theta)}{J_g(\theta)}, \qquad \frac{d\Psi_{\#}Q}{d\mathrm{vol}_g}(\Psi(\theta)) = \frac{q(\theta)}{J_g(\theta)}, \tag{742}$$

$$\mathrm{KL}(\Psi_{\#}Q \| \Psi_{\#}P) = \mathrm{KL}(Q\|P). \tag{743}$$

Proof. The coordinate volume formula and the Radon–Nikodym chain rule give

$$\mathrm{KL}(\Psi_{\#}Q \| \Psi_{\#}P) = \int_\Theta \log\left(\frac{q(\theta)/J_g(\theta)}{p(\theta)/J_g(\theta)}\right) \frac{q(\theta)}{J_g(\theta)} J_g(\theta)\, d\theta \tag{744}$$

$$= \int_\Theta q(\theta) \log(q(\theta)/p(\theta))\, d\theta. \tag{745}$$

In particular, a Jacobian factor cannot be dropped from only one of the two densities. □

Theorem 50 (Local quotient dimension in a deterministic-predictor bound)**.** *Fix finitely many charts Ψ_a and coordinate balls $B(\theta_{0,a}, R_a) \subset \Theta_a$, independently of S. Let prior chart weights $\pi_a > 0$ satisfy $\sum_a \pi_a = 1$. On the disjoint union of the labeled balls, take*

$$P(a, d\theta) = \pi_a \frac{\mathbf{1}_{\{\|\theta - \theta_{0,a}\| \leq R_a\}}}{v_{D_r} R_a^{D_r}}\, d\theta, \tag{746}$$

$$v_D = \frac{\pi^{D/2}}{\Gamma(D/2 + 1)}. \tag{747}$$

Suppose Ψ_a is C_a-Lipschitz and ℓ is L_ℓ-Lipschitz in product Frobenius distance. On the event in Eq. (735)*, for every possibly data-dependent a, θ, ρ with $\|\theta - \theta_{0,a}\| + \rho \leq R_a$ and $\rho > 0$,*

$$R(\Psi_a(\theta)) - R_S(\Psi_a(\theta)) \leq 2L_\ell C_a \rho + \frac{\lambda_j}{8} + \frac{D_r \log(R_a/\rho) + \log(1/\pi_a) + \log\frac{\pi^2(j+1)^2}{6q}}{\lambda_j n}, \qquad j \geq 0. \tag{748}$$

The dimensional term is $D_r = r(m + d - r)$.

Proof. Take the posterior concentrated on label a and uniform on $B(\theta,\rho)$. Its density ratio on its support is constant:

$$\frac{dQ}{dP}=\frac{v_{D_r}R_a^{D_r}}{\pi_a v_{D_r}\rho^{D_r}}, \qquad \mathrm{KL}(Q\|P)=D_r\log(R_a/\rho)+\log(1/\pi_a). \tag{749}$$

For $\vartheta\sim Q$,

$$\|\Psi_a(\vartheta)-\Psi_a(\theta)\|_F\le C_a\rho, \tag{750}$$

$$|R(Q)-R(\Psi_a(\theta))|\le L_\ell C_a\rho, \qquad |R_S(Q)-R_S(\Psi_a(\theta))|\le L_\ell C_a\rho. \tag{751}$$

Add the two approximation terms to Eq. (735). Overlapping chart images cause no problem: the labeled union is the hypothesis space, and the loss depends on its product image. □

Corollary 21 (Radius minimization and gauge variables)**.** *For fixed a,θ,j and $L_\ell C_a>0$, the optimal radius in Eq.* (748) *is*

$$\rho_*=\min\left\{R_a-\|\theta-\theta_{0,a}\|,\frac{D_r}{2L_\ell C_a\lambda_j n}\right\}, \tag{752}$$

provided the first argument is positive. If the second argument is feasible, the radius-dependent terms equal

$$\frac{D_r}{\lambda_j n}\left[1+\log\left(\frac{2L_\ell C_a R_a\lambda_j n}{D_r}\right)\right]. \tag{753}$$

For any fixed probability law ν on auxiliary gauge coordinates,

$$\mathrm{KL}(Q\otimes\nu\|P\otimes\nu)=\mathrm{KL}(Q\|P). \tag{754}$$

Proof. For $F(\rho)=2L_\ell C_a\rho+D_r\log(R_a/\rho)/(\lambda_j n)$,

$$F'(\rho)=2L_\ell C_a-\frac{D_r}{\lambda_j n\rho}, \qquad F''(\rho)=\frac{D_r}{\lambda_j n\rho^2}>0. \tag{755}$$

The density ratio for the product measures is $(dQ/dP)(\theta)$; integrating over ν proves Eq. (754). Gauge cancellation compares priors and posteriors with the same gauge law. It does not authorize replacing a data-dependent prior by a fixed prior without a correction. □

H.7. Random reshuffling and full-batch variants

Corollary 22 (Random reshuffling without conditional independence)**.** *The bound in Eq.* (670) *also holds for a uniformly random permutation in each epoch, with deterministic step lengths and uniform state-map constants.*

Proof. The pathwise recursion is

$$\Delta_{t+1}\le a_t\Delta_t+2C_U\eta_t\mathbf{1}_{\{I_t=j\}}, \tag{756}$$

$$\Delta_T\le 2C_U\sum_{t<T}\eta_t\mathbf{1}_{\{I_t=j\}}\prod_{k>t}a_k. \tag{757}$$

Each permutation position has marginal probability $\Pr(I_t=j)=1/n$. Since the weights $\eta_t\prod_{k>t}a_k$ are deterministic, taking expectations yields the same bound. No conditional probability $\Pr(I_t=j\mid\mathcal{F}_t)=1/n$ is asserted. □

Corollary 23 (Full-batch sample perturbation)**.** *Suppose a full-batch state update has the common-data Lipschitz constant in Eq.* (667) *and*

$$d_*(U_{t,S}s,U_{t,S^{(j)}}s)\le 2C_U\eta_t/n. \tag{758}$$

Then Eq. (670) *holds without sampling randomness. For an averaged gradient, a sufficient oracle condition is*

$$\|\nabla R_S(x)-\nabla R_{S^{(j)}}(x)\|\le\frac{1}{n}\|\nabla\ell(x;z_j)-\nabla\ell(x;z_j')\|\le 2G/n \tag{759}$$

together with an update-map sensitivity bound to that oracle.

H.8. Multilayer spectral propagation and margin risk

Assumption 9. For $1 \le l \le H$,

$$W_l = W_{0,l} + c_l X_l, \qquad \operatorname{rank} X_l = r_l, \qquad \|X_l\|_2 \le b_l, \tag{760}$$

$$h_l = \phi_l(W_l h_{l-1}), \qquad h_0 = x, \qquad \|x\|_2 \le R_x, \tag{761}$$

$$\|\phi_l(u) - \phi_l(v)\|_2 \le \rho_l \|u - v\|_2, \qquad \phi_l(0) = 0, \tag{762}$$

$$s_l = \|W_{0,l}\|_2 + |c_l| b_l, \qquad A_l = R_x |c_l| \left(\prod_{j=1}^{H} \rho_j \right) \prod_{j \ne l} s_j. \tag{763}$$

Biases and other trainable parameters are fixed or covered separately.

Lemma 11 (Telescoping output sensitivity)**.** *For two networks in this class,*

$$\|h_X(x) - h_Y(x)\|_2 \le \sum_{l=1}^{H} A_l \|X_l - Y_l\|_2. \tag{764}$$

Proof. Let $h^{(l)}$ replace the first l layers of one network by those of the other. Then

$$h_X - h_Y = \sum_{l=1}^{H} (h^{(l)} - h^{(l-1)}), \tag{765}$$

$$\|h^{(l)} - h^{(l-1)}\|_2 \le \left(\prod_{j>l} \rho_j s_j \right) \rho_l |c_l| \|X_l - Y_l\|_2 \left(R_x \prod_{j<l} \rho_j s_j \right) \tag{766}$$

$$= A_l \|X_l - Y_l\|_2. \tag{767}$$

□

Theorem 51 (Network generalization with all rank coordinates counted)**.** *Let the layer-l cover have*

$$\log N_l(\varepsilon_l) \le \log J_l + D_l \log(1 + 2C_l R_l/\varepsilon_l), \quad D_l = r_l(m_l + d_l - r_l). \tag{768}$$

For an output-L_{out}-Lipschitz loss in $[0,1]$, with probability at least $1 - q$,

$$\sup_X |R(X) - R_S(X)| \le 2L_{\mathrm{out}} \sum_l A_l \varepsilon_l + \sqrt{\frac{\log(2/q) + \sum_l \log J_l + \sum_l D_l \log(1 + 2C_l R_l/\varepsilon_l)}{2n}}. \tag{769}$$

Proof.

$$|\mathcal{N}| \le \prod_l N_l(\varepsilon_l), \qquad \log |\mathcal{N}| \le \sum_l \log N_l(\varepsilon_l), \tag{770}$$

$$\|h_X - h_{\widehat{X}}\|_2 \le \sum_l A_l \varepsilon_l. \tag{771}$$

Apply a finite-net Hoeffding union bound and pay the approximation cost in population and empirical risk. □

Corollary 24 (Explicit dependence on non-adapter trainable weights)**.** *If other weights range over a P-dimensional Euclidean ball of radius R_{oth} and their output sensitivity is L_{oth}, the preceding bound additionally contains*

$$2L_{\mathrm{out}} L_{\mathrm{oth}} \varepsilon_{\mathrm{oth}} \quad \textit{and} \quad P \log(1 + 2R_{\mathrm{oth}}/\varepsilon_{\mathrm{oth}}) \tag{772}$$

in the approximation term and the logarithm, respectively. Trainable embeddings, dense backbones, biases, and attention parameters are not omitted from P.

Corollary 25 (Classification margin)**.** *Let*

$$m(h,z) = h_y - \max_{k\neq y} h_k, \qquad \ell_\gamma(h,z) = \min\{1, \max\{0, 1 - m(h,z)/\gamma\}\}. \tag{773}$$

Then

$$\mathbf{1}_{\{m(h,z)\leq 0\}} \leq \ell_\gamma(h,z), \tag{774}$$

$$|\ell_\gamma(h,z) - \ell_\gamma(h',z)| \leq \frac{\sqrt{2}}{\gamma}\|h - h'\|_2. \tag{775}$$

Writing $U_n(\gamma)$ for Eq. (769) *with $L_{\text{out}} = \sqrt{2}/\gamma$,*

$$\Pr_{z\sim\mathcal{D}}\{m(h_X,z) \leq 0\} \leq \frac{1}{n}\sum_i \ell_\gamma(h_X, z_i) + U_n(\gamma). \tag{776}$$

Proposition 28 (Fixed graph propagation)**.** *For graph-valued layers $H_l = \phi_l(P_l H_{l-1} W_l^\top)$, take $\|H_0\|_F \leq R_x$ and Frobenius output distance. With fixed P_l, the exact sensitivity coefficient is*

$$A_l^{\text{graph}} = R_x|c_l|\left(\prod_{j=1}^{H}\rho_j\|P_j\|_2\right)\prod_{j\neq l} s_j = A_l\prod_{j=1}^{H}\|P_j\|_2. \tag{777}$$

In particular, the propagation at the perturbed layer is included. For symmetric normalized adjacency with self-loops,

$$P = D^{-1/2}(A+I)D^{-1/2}, \qquad P \sim D^{-1}(A+I), \tag{778}$$

$$P = P^\top, \qquad \|P\|_2 \leq 1. \tag{779}$$

This operator bound does not make transductive nodes IID. Data-dependent attention needs separate Lipschitz constants.

H.9. Validation selection and union classes

Theorem 52 (Selection among trained candidates)**.** *Let $h_1, \ldots, h_M$ be independent of an IID validation sample of size n_v, conditionally on the training data and training randomness. If $\widehat{h} = \arg\min_j \widehat{R}_v(h_j)$, then with probability at least $1-q$,*

$$R(\widehat{h}) \leq \min_j R(h_j) + 2\sqrt{\frac{\log(2M/q)}{2n_v}}. \tag{780}$$

Proof. For $\alpha = \sqrt{\log(2M/q)/(2n_v)}$,

$$\Pr\{\max_j |R(h_j) - \widehat{R}_v(h_j)| > \alpha\} \leq 2Me^{-2n_v\alpha^2} = q, \tag{781}$$

$$R(\widehat{h}) \leq \widehat{R}_v(\widehat{h}) + \alpha \leq \widehat{R}_v(h_*) + \alpha \leq R(h_*) + 2\alpha. \tag{782}$$

□

Proposition 29 (Checkpoints, hyperparameters, and execution aliases)**.** *For M_{cfg} configurations and K_j candidate checkpoints,*

$$M_{\text{eff}} \leq \sum_{j=1}^{M_{\text{cfg}}} K_j, \qquad \log M_{\text{eff}} \leq \log M_{\text{cfg}} + \log(\max_j K_j), \tag{783}$$

$$N\left(\bigcup_{j=1}^{M}\mathcal{C}_j, \varepsilon\right) \leq \sum_{j=1}^{M} N(\mathcal{C}_j,\varepsilon) \leq M\max_j N(\mathcal{C}_j,\varepsilon). \tag{784}$$

Duplicate executable configurations do not create independent candidates. The validation theorem presupposes that the candidate family is fixed without adaptive use of that validation sample.

H.10. End-to-end excess risk and the stationarity distinction

Theorem 53 (Optimization plus uniform generalization)**.** *For a fixed class $\mathcal{C}$, suppose*

$$\sup_{X\in\mathcal{C}} |R(X) - R_S(X)| \le U_n, \tag{785}$$

$$R_S(X_T) - \inf_{X\in\mathcal{C}} R_S(X) \le \epsilon_{\mathrm{opt}}. \tag{786}$$

Then

$$R(X_T) - \inf_{X\in\mathcal{C}} R(X) \le \epsilon_{\mathrm{opt}} + 2U_n. \tag{787}$$

If instead expected uniform stability is ϵ_{stab} and expected empirical suboptimality is at most ϵ_{opt},

$$\mathbb{E}R(X_T) - \inf_{X\in\mathcal{C}} R(X) \le \epsilon_{\mathrm{opt}} + \epsilon_{\mathrm{stab}}. \tag{788}$$

Proof. For any fixed $X \in \mathcal{C}$,

$$R(X_T) - R(X) = [R(X_T) - R_S(X_T)] + [R_S(X_T) - R_S(X)] + [R_S(X) - R(X)] \tag{789}$$

$$\le 2U_n + \epsilon_{\mathrm{opt}}. \tag{790}$$

Take an infimizing sequence. In expectation, the first bracket is bounded by stability, the second by empirical suboptimality, and $\mathbb{E}[R_S(X) - R(X)] = 0$. □

Corollary 26 (PL population bound and low-rank approximation error)**.** *If the empirical PL minimum is the class minimum,*

$$\epsilon_{\mathrm{opt}}(T) \le (1-\mu_g\eta)^T\Delta_0 \quad \textit{in the deterministic case}, \tag{791}$$

$$\mathbb{E}\epsilon_{\mathrm{opt}}(T) \le (1-\mu_g\eta)^T\Delta_0 + \frac{2B^2 + L_R\eta\nu^2/2}{\mu_g} \quad \textit{in the stochastic case}. \tag{792}$$

For an unrestricted comparison class $\mathcal{H}$,

$$R(X_T) - \inf_{h\in\mathcal{H}} R(h) = \left[R(X_T) - \inf_{X\in\mathcal{C}_r} R(X)\right] + \underbrace{\inf_{X\in\mathcal{C}_r} R(X) - \inf_{h\in\mathcal{H}} R(h)}_{\epsilon_{\mathrm{approx}}(r)}. \tag{793}$$

Proposition 30 (Zero gradient alone does not bound excess risk)**.** *For $c > 1$ on the positive rank-one stratum,*

$$f(x) = ((x-c)^2 - 1)^2, \qquad x \in [c-1, c+1], \tag{794}$$

$$\mathrm{grad}_g f(c) = 0, \qquad f(c) - \min_x f(x) = 1. \tag{795}$$

Hence nonconvex stationarity cannot replace ϵ_{opt} in Eq. (788) *without an additional landscape inequality.*

Proposition 31 (Exact invariance of risk under balancing)**.** *For every invertible Q and product-only predictor,*

$$R(BQ, Q^{-1}A) = R(B, A), \qquad R_S(BQ, Q^{-1}A) = R_S(B, A), \tag{796}$$

$$\|(BQ)(Q^{-1}A)\|_* = \|BA\|_*, \qquad \mathrm{Gap}(BQ, Q^{-1}A) = \mathrm{Gap}(B, A). \tag{797}$$

Thus orbit balancing alone does not shrink the represented predictor class. All IID theorems above require the declared sampling model; they are not automatically applied to correlated language tokens or nodes of a single graph.

I. Localized Fast Rates, Influence Functions, and Gauge-Singular Fisher Information

I.1. Whitened Newton decrement and a leading-constant sharp risk bound

Write $F(\theta) = R(\psi(\theta))$, $F_n(\theta) = R_S(\psi(\theta))$ in a fixed D-dimensional regular chart with $\nabla F(0) = 0$. For a positive definite reference matrix H define $\|\theta\|_H^2 = \theta^\top H\theta$ and $d_n^2 = \nabla F_n(0)^\top H^{-1}\nabla F_n(0)$. The reference H is independent of the sample in the concentration statement, or is fixed after conditioning on a noise-independent design.

Theorem 54 (Whitened local risk certificate)**.** *Suppose on $\|\theta\|_H \le r$ that*

$$(1-\delta_H)H \preceq \nabla^2 F_n(\theta), \qquad \nabla^2 F(\theta) \preceq (1+\epsilon_H)H, \qquad 0 \le \delta_H < 1. \tag{798}$$

If $d_n < (1-\delta_H)r$, the constrained empirical minimum is unique and interior, and

$$\|\widehat{\theta}_n\|_H \le \frac{d_n}{1-\delta_H}, \tag{799}$$

$$F(\widehat{\theta}_n) - F(0) \le \frac{1+\epsilon_H}{2(1-\delta_H)^2} d_n^2. \tag{800}$$

For an empirical ϵ_T-minimizer in the same ball,

$$F(\theta_T) - F(0) \le \frac{1+\epsilon_H}{2}\left\{\frac{d_n}{1-\delta_H} + \sqrt{\frac{2\epsilon_T}{1-\delta_H}}\right\}^2. \tag{801}$$

Proof. Use $z = H^{1/2}\theta$. The empirical Hessian is at least $(1-\delta_H)I$ and its center gradient has norm d_n. On $\|z\| = r$ its outward derivative is at least $(1-\delta_H)r^2 - d_n r > 0$. Thus the minimum is interior. Strong monotonicity gives $(1-\delta_H)\|\widehat{z}\| \le d_n$. The population Hessian bound yields $F(\theta) - F(0) \le (1+\epsilon_H)\|\theta\|_H^2/2$. Empirical strong convexity bounds $\|\theta_T - \widehat{\theta}_n\|_H^2 \le 2\epsilon_T/(1-\delta_H)$. Apply the triangle inequality and keep the square unexpanded. For the quadratic pair

$$F(\theta) - F(0) = \tfrac{1}{2}(1+\epsilon_H)\|\theta\|_H^2, \tag{802}$$

$$F_n(\theta) - F_n(0) = s^\top\theta + \tfrac{1}{2}(1-\delta_H)\|\theta\|_H^2, \tag{803}$$

the minimizer is $-(1-\delta_H)^{-1}H^{-1}s$ and Eq. (800) is an equality. Thus both the coefficient $1/2$ and the relative-Hessian factor $(1-\delta_H)^{-2}$ are sharp for these stated sector conditions. □

Lemma 12 (Laurent–Massart and Hsu–Kakade–Zhang score concentration)**.** *Suppose the independent centered whitened scores $w_i = H^{-1/2}\nabla\ell_0(z_i)$ satisfy*

$$\mathbb{E}e^{u^\top w_i} \le e^{u^\top \mathsf{V} u/2}, \qquad u \in \mathbb{R}^D, \quad \mathsf{V} \succeq 0. \tag{804}$$

Then, for every $u_0 > 0$, with probability at least $1 - e^{-u_0}$,

$$d_n^2 \le \frac{1}{n}\left\{\operatorname{tr}\mathsf{V} + 2\sqrt{\operatorname{tr}(\mathsf{V}^2)u_0} + 2\|\mathsf{V}\|_2 u_0\right\}. \tag{805}$$

Proof. This is the Gaussian quadratic-form bound of Laurent–Massart (Laurent & Massart, 2000) and its sub-Gaussian extension (Hsu et al., 2012); the short proof also fixes the constants. Put $W = n^{-1/2}\sum_i w_i$ and let $g \sim N(0, I_D)$ be independent. For $0 < t < (2\|\mathsf{V}\|_2)^{-1}$,

$$\mathbb{E}e^{t\|W\|^2} = \mathbb{E}_g\mathbb{E}_W e^{\sqrt{2t}g^\top W} \le \mathbb{E}_g e^{tg^\top \mathsf{V} g} = \det(I - 2t\mathsf{V})^{-1/2}, \tag{806}$$

$$\log \mathbb{E}e^{t(\|W\|^2 - \operatorname{tr}\mathsf{V})} \le \frac{t^2\operatorname{tr}(\mathsf{V}^2)}{1 - 2t\|\mathsf{V}\|_2}. \tag{807}$$

The scalar inequality used here is $-\log(1-x) - x \le x^2/[2(1-x)]$, $0 \le x < 1$. Chernoff optimization with

$$t = \frac{\sqrt{u_0}}{\sqrt{\operatorname{tr}(\mathsf{V}^2)} + 2\|\mathsf{V}\|_2\sqrt{u_0}} \tag{808}$$

gives the tail threshold in Eq. (805); $W = \sqrt{n}\,H^{-1/2}\nabla F_n(0)$. The zero-covariance case follows by continuity. □

Corollary 27 (A condition-number-free fast rate)**.** *Suppose the Hessian event in Theorem 54 fails with probability at most q_H and its radius condition is satisfied by the score threshold. With probability at least $1 - q_H - e^{-u_0}$,*

$$F(\widehat{\theta}_n) - F(0) \le \frac{1+\epsilon_H}{2n(1-\delta_H)^2}\left\{\operatorname{tr}\mathsf{V} + 2\sqrt{\operatorname{tr}(\mathsf{V}^2)u_0} + 2\|\mathsf{V}\|_2 u_0\right\}. \tag{809}$$

In the isotropic whitened case $\mathsf{V} = \sigma^2 I_D$ this becomes

$$\frac{(1+\epsilon_H)\sigma^2}{2n(1-\delta_H)^2}\{D + 2\sqrt{Du_0} + 2u_0\}. \tag{810}$$

Proof. Intersect the score-tail event with the Hessian event and substitute Eq. (805). No independence of these two events is needed. Whitening retains the covariance–curvature interaction; there is no separate Λ/μ^2 prefactor. □

Proposition 32 (A Newton–Kantorovich localization certificate)**.** *Let $A = \nabla^2 F_n(0) \succ 0$ and $d = \|A^{-1/2}\nabla F_n(0)\|$. Suppose, on an A-ball containing the radius below,*

$$\|A^{-1/2}\{\nabla^2 F_n(\theta) - A\}A^{-1/2}\|_2 \le K\|\theta\|_A. \tag{811}$$

If $2Kd < 1$, define

$$r_- = \frac{1-\sqrt{1-2Kd}}{K} = \frac{2d}{1+\sqrt{1-2Kd}}, \qquad d \le r_- \le 2d. \tag{812}$$

Then the local stationary point is unique in this ball and $\|\widehat{\theta}_n\|_A \le r_-$. If $\nabla^2 F(\theta) \preceq (1+\epsilon_A)A$ there,

$$F(\widehat{\theta}_n) - F(0) \le \tfrac{1}{2}(1+\epsilon_A)r_-^2. \tag{813}$$

For $K = 0$, interpret $r_- = d$.

Proof. This is the affine-invariant Newton–Kantorovich majorant argument (Deuflhard, 2004), proved here through Banach's contraction principle. In whitened coordinates consider the frozen-Jacobian map

$$\mathcal{N}(\theta) = \theta - A^{-1}\nabla F_n(\theta). \tag{814}$$

Taylor's integral remainder and the derivative premise give

$$\|\mathcal{N}(\theta)\|_A \le d + \tfrac{K}{2}\|\theta\|_A^2, \tag{815}$$

$$\|D\mathcal{N}(\theta)\|_{A\to A} \le K\|\theta\|_A. \tag{816}$$

The majorant equation $d + Kr_-^2/2 = r_-$ makes the ball invariant, and $Kr_- = 1 - \sqrt{1-2Kd} < 1$ makes it a contraction. Its fixed point solves $\nabla F_n = 0$ and the Hessian is positive throughout the ball. The population-risk bound follows by integration. The argument is a localization certificate; it does not require changing QuotientOpt into a Newton iteration. □

I.2. An explicit finite-sample event construction

Assumption 10. Let $\psi : \Theta \to \mathcal{M}_r$ be a fixed regular rank chart, $\Theta = \{\theta \in \mathbb{R}^D : \|\theta\|_2 \le R_0\}$, $D = r(m+d-r)$, and set

$$\ell_\theta(z) = \ell(\psi(\theta); z), \qquad F(\theta) = \mathbb{E}\ell_\theta(z), \qquad F_n(\theta) = \frac{1}{n}\sum_{i=1}^n \ell_\theta(z_i), \tag{817}$$

$$\nabla F(0) = 0, \qquad 2\mu I \preceq \nabla^2 F(\theta) \preceq \Lambda I, \quad \theta \in \Theta. \tag{818}$$

The observations z_i are IID. Per-example coordinate Hessians satisfy

$$\|\nabla^2 \ell_\theta(z)\|_2 \le B, \qquad \|\nabla^2 \ell_\theta(z) - \nabla^2 \ell_{\theta'}(z)\|_2 \le M\|\theta - \theta'\|_2. \tag{819}$$

The centered score $s(z) = \nabla \ell_0(z)$ has directional sub-Gaussian constant ν:

$$\mathbb{E}e^{u^\top s(z)} \le e^{\nu^2\|u\|^2/2}, \qquad u \in \mathbb{R}^D. \tag{820}$$

The model uses its stated derivative and moment conditions; a bounded-loss assumption is not needed in this section.

Lemma 13 (A uniform empirical Hessian event with explicit constants)**.** *For $0 < q < 1$, the sample-size condition*

$$n \ge \frac{32B^2}{\mu^2}\left[D\log 9 + D\log\left(1 + \frac{8MR_0}{\mu}\right) + \log\frac{4}{q}\right] \tag{821}$$

implies, with probability at least $1 - q/2$,

$$\sup_{\theta\in\Theta}\|\nabla^2 F_n(\theta) - \nabla^2 F(\theta)\|_2 \le \mu, \qquad \nabla^2 F_n(\theta) \succeq \mu I. \tag{822}$$

Proof. For a symmetric matrix A and a $1/4$ net $\mathcal{V}$ of the unit sphere,

$$\|A\|_2 \le \max_{v\in\mathcal{V}} |v^\top A v| + \tfrac{1}{2}\|A\|_2, \tag{823}$$

$$\|A\|_2 \le 2\max_{v\in\mathcal{V}} |v^\top A v|, \qquad |\mathcal{V}| \le 9^D. \tag{824}$$

Take a $\delta_\theta = \mu/(4M)$ net of Θ of cardinality at most $(1 + 8MR_0/\mu)^D$, with centers in Θ. For $M = 0$, a single coordinate center suffices. For fixed θ, v, the IID variables $v^\top \nabla^2 \ell_\theta(z_i) v$ lie in $[-B, B]$, so

$$\Pr\left\{\left|v^\top(\nabla^2 F_n(\theta) - \nabla^2 F(\theta))v\right| > \mu/4\right\} \le 2e^{-n\mu^2/(32B^2)}. \tag{825}$$

A union bound gives operator error at most $\mu/2$ at all net centers. Off the net, the empirical and population Hessians each have Lipschitz constant M, adding at most $2M\delta_\theta = \mu/2$. The sample-size condition bounds the union probability by $q/2$. □

Lemma 14 (Score concentration in the intrinsic dimension)**.** *With probability at least* $1 - q/2$*,*

$$\|\bar{s}_n\|_2 = \left\|\frac{1}{n}\sum_i s(z_i)\right\|_2 \le t_n := \nu\sqrt{\frac{8(D\log 5 + \log(2/q))}{n}}. \tag{826}$$

Proof. A $1/2$ sphere net $\mathcal{V}$ has at most 5^D elements and $\|a\| \le 2\max_{v\in\mathcal{V}} v^\top a$. By the sub-Gaussian premise and exponential Markov inequality,

$$\Pr\{v^\top \bar{s}_n > u\} \le e^{-nu^2/(2\nu^2)}, \tag{827}$$

$$\Pr\{\|\bar{s}_n\| > t\} \le 5^D e^{-nt^2/(8\nu^2)}. \tag{828}$$

Set $t = t_n$. □

Theorem 55 (A localized $1/n$ excess-risk bound)**.** *Suppose Eq.* (821) *holds and* $t_n \le \mu R_0/2$*. With probability at least* $1 - q$*, the minimum* $\widehat{\theta}_n = \arg\min_{\theta\in\Theta} F_n(\theta)$ *is unique, interior, and satisfies*

$$\nabla F_n(\widehat{\theta}_n) = 0, \qquad \|\widehat{\theta}_n\|_2 \le t_n/\mu, \tag{829}$$

$$F(\widehat{\theta}_n) - F(0) \le \frac{\Lambda t_n^2}{2\mu^2} = \frac{4\Lambda\nu^2}{\mu^2 n}\{D\log 5 + \log(2/q)\}. \tag{830}$$

More generally, every $\theta_T \in \Theta$ *with* $F_n(\theta_T) - F_n(\widehat{\theta}_n) \le \epsilon_T$ *obeys*

$$F(\theta_T) - F(0) \le \frac{\Lambda}{2}\left(\frac{t_n}{\mu} + \sqrt{\frac{2\epsilon_T}{\mu}}\right)^2. \tag{831}$$

Proof. Work on the intersection of the two preceding events. For $\|\theta\| = R_0$,

$$\theta^\top \nabla F_n(\theta) = \theta^\top \bar{s}_n + \int_0^1 \theta^\top \nabla^2 F_n(t\theta)\theta\, dt \tag{832}$$

$$\ge -R_0 t_n + \mu R_0^2 > 0. \tag{833}$$

Thus a boundary point is improved by moving inward. Compactness gives an interior minimizer, and strong convexity gives uniqueness. Strong monotonicity implies

$$\mu\|\widehat{\theta}_n\|^2 \le \langle \nabla F_n(\widehat{\theta}_n) - \nabla F_n(0), \widehat{\theta}_n\rangle \tag{834}$$

$$= -\langle \bar{s}_n, \widehat{\theta}_n\rangle \le t_n\|\widehat{\theta}_n\|. \tag{835}$$

Population smoothness and $\nabla F(0) = 0$ give $F(\theta) - F(0) \le \Lambda\|\theta\|^2/2$. Finally,

$$\tfrac{\mu}{2}\|\theta_T - \widehat{\theta}_n\|^2 \le F_n(\theta_T) - F_n(\widehat{\theta}_n) \le \epsilon_T, \tag{836}$$

$$\|\theta_T\| \le \|\widehat{\theta}_n\| + \|\theta_T - \widehat{\theta}_n\|. \tag{837}$$

Combining the inequalities proves both claims. □

Corollary 28 (Closing the deterministic optimization–statistics chain)**.** *If the empirical quotient iteration is in the certified basin of Theorem 37 around its empirical minimum, one may insert*

$$\epsilon_T \le C_{\mathrm{init}} q_{\mathsf{P}}^{2T} \tag{838}$$

into Eq. (831)*. In particular, $\epsilon_T \le t_n^2/(2\mu)$ is sufficient for*

$$F(\theta_T) - F(0) \le 2\Lambda t_n^2/\mu^2. \tag{839}$$

For $C_{\mathrm{init}} > t_n^2/(2\mu)$ this holds whenever

$$T \ge \frac{\log(2\mu C_{\mathrm{init}}/t_n^2)}{2\log(1/q_{\mathsf{P}})}. \tag{840}$$

Proof. The optimization theorem gives an empirical suboptimality bound. Then $\sqrt{2\epsilon_T/\mu} \le t_n/\mu$ in Eq. (831). Solving the geometric inequality gives the required iteration count. The sample size n counts independent observations. Repeated optimization epochs on those observations affect ϵ_T, not n. □

Proposition 33 (Obtaining the coordinate curvature premise)**.** *Let a normal chart at $X_\star$ have $\mathsf{H} = \nabla^2 F(0) \succeq h_\star I$, and suppose $\nabla^2 F$ is K_F-Lipschitz there. If $R_0 \le h_\star/(2K_F)$, then*

$$\nabla^2 F(\theta) \succeq h_\star I/2 = 2\mu I, \qquad \mu = h_\star/4. \tag{841}$$

At an exact RIP sensing solution one may take $h_\star \ge a(1-\delta)$.

Proof. Weyl's inequality gives $\lambda_{\min}(\nabla^2 F(\theta)) \ge h_\star - K_F\|\theta\|$. At the chart center, coordinate and quotient Hessians agree in the orthonormal normal frame. The RIP Hessian result at $E = 0$ supplies the last bound. A bound on the actual coordinate Hessian variation is still needed to choose R_0. □

I.3. Influence functions and the exact leading generalization constant

The local estimating-equation expansion is the standard influence-function argument for regular M-estimators (van der Vaart, 1998); the quantitative remainder below makes its dependence on the rank chart explicit.

Theorem 56 (A quantitative influence expansion)**.** *Write*

$$\mathsf{H} = \nabla^2 F(0) \succ 0, \qquad \mathsf{H}_n = \nabla^2 F_n(0), \qquad \mathsf{C} = \mathbb{E}[s(z)s(z)^\top]. \tag{842}$$

On an event where the empirical Hessian is at least μI, its Lipschitz constant is at most M, and the local minimum is interior,

$$\widehat{\theta}_n = -\mathsf{H}^{-1}\bar{s}_n + \Delta_n, \tag{843}$$

$$\|\Delta_n\| \le \|\mathsf{H}^{-1}\|_2 \left\{ \frac{\|\mathsf{H}_n - \mathsf{H}\|_2 \|\bar{s}_n\|}{\mu} + \frac{M\|\bar{s}_n\|^2}{2\mu^2} \right\}. \tag{844}$$

If the event has probability tending to one, $\mathsf{H}_n - \mathsf{H} = o_p(1)$, and the score has finite covariance,

$$\sqrt{n}\,\widehat{\theta}_n \xrightarrow{d} N(0, \mathsf{H}^{-1}\mathsf{C}\mathsf{H}^{-1}), \tag{845}$$

$$n\{F(\widehat{\theta}_n) - F(0)\} \xrightarrow{d} \tfrac{1}{2} Z^\top \mathsf{H}^{-1} Z, \qquad Z \sim N(0, \mathsf{C}). \tag{846}$$

Proof. The stationarity equation and a Taylor remainder give

$$0 = \bar{s}_n + \mathsf{H}_n\widehat{\theta}_n + \rho_n, \qquad \|\rho_n\| \le \tfrac{M}{2}\|\widehat{\theta}_n\|^2, \tag{847}$$

$$\widehat{\theta}_n + \mathsf{H}^{-1}\bar{s}_n = \mathsf{H}^{-1}\{(\mathsf{H} - \mathsf{H}_n)\widehat{\theta}_n - \rho_n\}. \tag{848}$$

Insert $\|\widehat{\theta}_n\| \le \|\bar{s}_n\|/\mu$. The IID score central limit theorem gives $\sqrt{n}\,\bar{s}_n \Rightarrow N(0, \mathsf{C})$, and the displayed error is $o_p(n^{-1/2})$. The population Taylor expansion then proves the quadratic limit. □

Theorem 57 (Leading expected risk, training optimism, and generalization)**.** *Under Assumption 10, consider the localized empirical minimum equal to* 0 *when the uniform Hessian and interior-score event fails. Assume* $\mathbb{E}|\ell_0(z)| < \infty$*. For fixed chart and dimension, as* $n \to \infty$*,*

$$\mathbb{E}[F(\widehat{\theta}_n) - F(0)] = \frac{\operatorname{tr}(\mathsf{C}\mathsf{H}^{-1})}{2n} + O(n^{-3/2}), \tag{849}$$

$$\mathbb{E}[F_n(\widehat{\theta}_n) - F_n(0)] = -\frac{\operatorname{tr}(\mathsf{C}\mathsf{H}^{-1})}{2n} + O(n^{-3/2}), \tag{850}$$

$$\mathbb{E}[F(\widehat{\theta}_n) - F_n(\widehat{\theta}_n)] = \frac{\operatorname{tr}(\mathsf{C}\mathsf{H}^{-1})}{n} + O(n^{-3/2}). \tag{851}$$

Proof. The two concentration arguments imply a failure probability $O(e^{-cn})$ for a fixed sufficiently small score threshold and the fixed chart. Independence and sub-Gaussian scores yield

$$\mathbb{E}[\bar{s}_n \bar{s}_n^\top] = \mathsf{C}/n, \qquad \mathbb{E}\|\bar{s}_n\|^4 = O(n^{-2}), \qquad \mathbb{E}\|\mathsf{H}_n - \mathsf{H}\|_2^2 = O(n^{-1}). \tag{852}$$

The last bound follows by bounding operator norm by Frobenius norm and summing the entrywise variances of the empirical Hessian. On the successful event, the influence expansion gives

$$F(\widehat{\theta}_n) - F(0) = \tfrac{1}{2}\bar{s}_n^\top \mathsf{H}^{-1}\bar{s}_n + \mathcal{E}_n, \tag{853}$$

$$F_n(\widehat{\theta}_n) - F_n(0) = -\tfrac{1}{2}\bar{s}_n^\top \mathsf{H}^{-1}\bar{s}_n + \widetilde{\mathcal{E}}_n, \tag{854}$$

$$|\mathcal{E}_n| + |\widetilde{\mathcal{E}}_n| \le C\{\|\mathsf{H}_n - \mathsf{H}\|_2 \|\bar{s}_n\|^2 + \|\bar{s}_n\|^3\}. \tag{855}$$

Hessian norms are bounded, and the influence bound justifies absorbing higher-order terms into this inequality on the successful event. Cauchy–Schwarz gives expectation $O(n^{-3/2})$. Off that event the localized differences are zero; the omitted quadratic score expectation is at most

$$C(\mathbb{E}\|\bar{s}_n\|^4)^{1/2} \Pr(\text{failure})^{1/2} = O(n^{-1}e^{-cn/2}). \tag{856}$$

Taking the trace proves the first two statements. For the third, use the exact decomposition

$$F(\widehat{\theta}_n) - F_n(\widehat{\theta}_n) = [F(\widehat{\theta}_n) - F(0)] - [F_n(\widehat{\theta}_n) - F_n(0)] \tag{857}$$

$$+ F(0) - F_n(0), \tag{858}$$

whose last term has mean zero. □

I.4. Isotropic sensing and a sharp intrinsic dimension constant

Definition 12. Let

$$y = \langle Z, X_\star \rangle + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid Z] = 0, \qquad \mathbb{E}[\varepsilon^2 \mid Z] = \sigma_\varepsilon^2, \tag{859}$$

$$\ell(X; (Z, y)) = \tfrac{1}{2}(\langle Z, X\rangle - y)^2, \qquad \mathbb{E}[\operatorname{vec} Z (\operatorname{vec} Z)^\top] = I_{md}. \tag{860}$$

All required local derivatives and moments are assumed integrable. Bounded-feature conditions for the finite-sample Hessian theorem and finite-moment conditions for asymptotic influence results are separate premises.

Theorem 58 (The intrinsic efficiency constant)**.** *For* $J = D\psi(0)$*, represented as an* $md \times D$ *matrix,*

$$F(\theta) - F(0) = \tfrac{1}{2}\|\psi(\theta) - X_\star\|_F^2, \tag{861}$$

$$s(z) = -\varepsilon J^\top \operatorname{vec} Z, \qquad \mathsf{H} = J^\top J, \qquad \mathsf{C} = \sigma_\varepsilon^2 \mathsf{H}, \tag{862}$$

$$\operatorname{tr}(\mathsf{C}\mathsf{H}^{-1}) = \sigma_\varepsilon^2 D = \sigma_\varepsilon^2 r(m + d - r). \tag{863}$$

Whenever the expectation premises of Theorem 57 or Theorem 43 apply, their leading excess-risk constant is therefore $\sigma_\varepsilon^2 D/2$*.*

Proof. The conditional mean-zero noise removes the population cross term:

$$\mathbb{E}[\ell(X;z) - \ell(X_\star;z)] = \tfrac{1}{2}\mathbb{E}\langle Z, X - X_\star\rangle^2 = \tfrac{1}{2}\|X - X_\star\|_F^2. \tag{864}$$

At the exact solution the population residual gradient vanishes, so the coordinate Hessian is $J^\top J$. The conditional noise variance gives

$$\mathbb{E}[s(z)s(z)^\top] = \mathbb{E}[\varepsilon^2 J^\top \operatorname{vec} Z(\operatorname{vec} Z)^\top J] \tag{865}$$

$$= \sigma_\varepsilon^2 J^\top J. \tag{866}$$

The chart is regular, so J has rank D and H is invertible. The trace identity follows. □

Proposition 34 (Rank-one Gaussian measurements are isotropic)**.** *For independent $a \sim N(0, I_m)$ and $b \sim N(0, I_d)$ with $Z = ab^\top$,*

$$\mathbb{E}[Z_{ij}Z_{k\ell}] = \mathbb{E}[a_i a_k]\mathbb{E}[b_j b_\ell] = \delta_{ik}\delta_{j\ell}. \tag{867}$$

Thus Eq. (860) *applies to these measurements under the stated noise and local moment premises. It does not require independent entries in Z.*

I.5. Fisher degeneracy along gauge fibers

Theorem 59 (Full factor Fisher information and its identifiable image)**.** *Assume $\varepsilon \mid Z \sim N(0, \sigma_\varepsilon^2)$ and isotropic features. At a regular factor pair $p_\star = (B_\star, A_\star)$, let $L_\star = D\pi_{p_\star}$ with the Euclidean factor inner product. The per-observation Fisher information is*

$$\mathsf{I}_{\mathrm{fac}} = \sigma_\varepsilon^{-2} L_\star^* L_\star, \qquad \ker \mathsf{I}_{\mathrm{fac}} = \mathcal{V}_{p_\star}, \qquad \operatorname{rank} \mathsf{I}_{\mathrm{fac}} = D, \tag{868}$$

$$L_\star \mathsf{I}_{\mathrm{fac}}^\dagger L_\star^* = \sigma_\varepsilon^2 P_{X_\star}. \tag{869}$$

Hence the identifiable product-space covariance has trace $\sigma_\varepsilon^2 D$, independent of the r^2 factor gauge coordinates.

Proof. The conditional log-likelihood score in a factor direction ξ is

$$D \log p(y \mid Z, p_\star)[\xi] = \sigma_\varepsilon^{-2}\varepsilon\langle Z, L_\star \xi\rangle. \tag{870}$$

Taking the expected product of two directional scores gives $\sigma_\varepsilon^{-2}\langle L_\star\xi, L_\star\zeta\rangle$. The kernel and rank follow from Theorem 10. For an SVD $L_\star = \mathsf{USV}^\top$ on its nonzero singular subspaces,

$$L_\star (L_\star^* L_\star)^\dagger L_\star^* = \mathsf{US}(\mathsf{S}^2)^{-1}\mathsf{SU}^\top \tag{871}$$

$$= \mathsf{UU}^\top = P_{X_\star}. \tag{872}$$

The factor σ_ε^2 proves the last identity. □

Corollary 29 (Design-adaptive efficiency by Fisher whitening)**.** *Replace isotropy by a feature covariance $\mathsf{M} = \mathbb{E}[\operatorname{vec} Z(\operatorname{vec} Z)^\top]$ that is positive definite on $T_{X_\star}\mathcal{M}_r$. For the same correctly specified homoscedastic model,*

$$\mathsf{H} = J^\top \mathsf{M} J, \qquad \mathsf{C} = \sigma_\varepsilon^2 J^\top \mathsf{M} J = \sigma_\varepsilon^2 \mathsf{H}, \tag{873}$$

$$\operatorname{tr}(\mathsf{C}\mathsf{H}^{-1}) = \sigma_\varepsilon^2 D, \tag{874}$$

$$\mathsf{I}_{\mathrm{fac}} = \sigma_\varepsilon^{-2} L_\star^* \mathsf{M} L_\star, \tag{875}$$

$$L_\star \mathsf{I}_{\mathrm{fac}}^\dagger L_\star^* = \sigma_\varepsilon^2 (P_{X_\star}\mathsf{M}P_{X_\star})^\dagger, \tag{876}$$

$$\operatorname{tr}\{\mathsf{M} L_\star \mathsf{I}_{\mathrm{fac}}^\dagger L_\star^*\} = \sigma_\varepsilon^2 D. \tag{877}$$

Consequently the leading prediction-risk constant $\sigma_\varepsilon^2 D/2$ holds for every such identifiable design under the stated asymptotic expectation premises.

Proof. The conditional variance identity gives $\mathsf{C} = \sigma_\varepsilon^2 \mathsf{H}$ directly. For $L_\star = \mathsf{U}\mathsf{S}\mathsf{V}^\top$ on the nonzero singular subspaces, let $\mathsf{M}_T = \mathsf{U}^\top \mathsf{M} \mathsf{U} \succ 0$. Then

$$(L_\star^* \mathsf{M} L_\star)^\dagger = \mathsf{V}\mathsf{S}^{-1}\mathsf{M}_T^{-1}\mathsf{S}^{-1}\mathsf{V}^\top, \tag{878}$$

$$L_\star (L_\star^* \mathsf{M} L_\star)^\dagger L_\star^* = \mathsf{U}\mathsf{M}_T^{-1}\mathsf{U}^\top = (P_{X_\star} \mathsf{M} P_{X_\star})^\dagger. \tag{879}$$

Taking the M-weighted trace yields D. This cancellation concerns prediction risk. Product Frobenius error instead has leading trace $\sigma_\varepsilon^2 \operatorname{tr}(\mathsf{M}_T^{-1})$, which retains the design conditioning. □

Proposition 35 (A Cramér–Rao bound for locally unbiased chart estimators)**.** *In the preceding Gaussian model, the Cramér–Rao information inequality (van der Vaart, 1998) gives the following result. At a fixed chart parameter let $\widehat{\theta}$ be unbiased in a neighborhood and satisfy the usual differentiation-under-integral conditions. Then*

$$\operatorname{Cov}(\widehat{\theta}) \succeq \frac{\sigma_\varepsilon^2}{n} \mathsf{H}^{-1}, \tag{880}$$

$$\mathbb{E}\left[\tfrac{1}{2}(\widehat{\theta} - \theta)^\top \mathsf{H} (\widehat{\theta} - \theta)\right] \geq \frac{\sigma_\varepsilon^2 D}{2n}. \tag{881}$$

Proof. Let S_n be the likelihood score for n independent observations. At the fixed parameter,

$$\mathbb{E} S_n = 0, \qquad \mathbb{E}[S_n S_n^\top] = n\mathsf{I}, \qquad \mathsf{I} = \sigma_\varepsilon^{-2}\mathsf{H}. \tag{882}$$

Differentiating $\mathbb{E}_\theta \widehat{\theta} = \theta$ gives $\mathbb{E}[(\widehat{\theta} - \theta) S_n^\top] = I_D$. For $A = (n\mathsf{I})^{-1}$,

$$0 \preceq \mathbb{E}[(\widehat{\theta} - \theta - A S_n)(\widehat{\theta} - \theta - A S_n)^\top] \tag{883}$$

$$= \operatorname{Cov}(\widehat{\theta}) - (n\mathsf{I})^{-1}. \tag{884}$$

Take the $\mathsf{H}/2$-weighted trace. This lower bound concerns the stated locally unbiased class and quadratic tangent loss. □

J. Numerical Verification of the Geometric Identities

A separate seed-42 numerical check uses balanced factors of shape 7×3 and 3×6 with nonzero product spectrum $(2, .7, .2)$. It verifies identities rather than treating a small residual as a task-performance result. In double precision the horizontal residual is 8.00×10^{-16}, differential-preservation error 6.16×10^{-16}, full mixed-term finite-step identity error 2.17×10^{-16}, and damped residual identity error 1.06×10^{-15}. The full-observation Hessian at an exact solution has nine gauge zero eigenvalues, matching $r^2 = 9$. Its positive spectrum agrees with the block spectrum of $\mathcal{K}_X$ within 2.22×10^{-15}.

A single-precision forward check of a product-only sine layer changes the output by at most 7.59×10^{-6} after balancing, consistent with floating-point product reconstruction. This checks that balancing preserves the architecture's function; it does not claim an improvement in fitting quality. A fresh backward pass after the section change has horizontal residual 5.49×10^{-10}. The script, shapes, thresholds, and measured values are retained in the geometry-check artifact. All checks pass their declared tolerances. These controlled checks accompany the analytic proofs and are distinct from the complete task-family runs.

The extended calculus has a separate float64 check with seed 918, $(m, d, r) = (5, 4, 2)$ and spectrum $(3.2, .7)$. Central differences use step 10^{-5} along an exactly rank-preserving Schur-chart curve. The measured identities are

$$\frac{\|D_{\mathrm{FD}} K - DK\|}{\|DK\|} = 1.03 \times 10^{-10}, \qquad \frac{\|D_{\mathrm{FD}} K^\dagger - DK^\dagger\|}{\|DK^\dagger\|} = 1.51 \times 10^{-10}, \tag{885}$$

$$\|D_{\mathrm{FD}} P[Z]Y - DP[Z]Y\|_F = 5.96 \times 10^{-11}, \qquad \|\nabla_{\dot\gamma}^g \dot\gamma(0)\|_F = 2.54 \times 10^{-15}, \tag{886}$$

$$|g(\mathrm{Hess}_g\, f[Z], Y) - \mathrm{Hess}_g f[Z, Y]| = 6.63 \times 10^{-15}. \tag{887}$$

For canonical rank-three chains of depths $2, 3, 5, 9$,

$$\max_L \|\operatorname{eig}(\mathcal{N}_W) - \operatorname{sort}(\lambda_{abk})\|_\infty = 7.99 \times 10^{-15}, \tag{888}$$

$$\max_L \frac{\|\mathcal{N}_W \Omega_{\mathrm{DST}} - F\|_F}{\|F\|_F} = 1.24 \times 10^{-15}. \tag{889}$$

These are consistency checks of analytic identities, not substitutes for their proofs or for task-family experiments.

The refined local theory has an additional float64 check with seed 11903. A nonzero ambient normal residual is retained so that the quotient Hessian differs from the zero-residual product preconditioner. The measured errors are

$$\|D_{\mathrm{FD}}\mathscr{U} - D\mathscr{U}\|/\|D\mathscr{U}\| \le 6.12 \times 10^{-11}, \tag{890}$$

$$\|\operatorname{eig}(\operatorname{Hess}_g f) - \operatorname{sort}\{\sigma_i + \sigma_j, \sigma_i \pm \tau_k, \sigma_i\}\| = 2.90 \times 10^{-15}, \tag{891}$$

$$\|\mathsf{J}^\top \mathsf{P}\mathsf{J} - \mathsf{P} + I\|_F = 4.78 \times 10^{-16}. \tag{892}$$

The Jacobian check includes exact projection, positive damping, and balance-only state updates. The stationary modal position and momentum variance formulas agree with the full discrete Lyapunov solve within 1.01×10^{-16}; the exact averaging boundary identity has error 5.07×10^{-15}. The identifiable Fisher covariance has trace 14 under both a balanced factor pair and a nonorthogonal gauge, with product-projector error at most 2.74×10^{-14}. A separate strict-saddle example verifies the complete residual-gap spectrum within 1.38×10^{-15} and has exactly one negative mode, as predicted.

K. Full Proposal Protocol and Reproducibility

The proposal specifies one seed, long training, four methods, seven task families, a fourteen-component ablation, and a hyperparameter grid. The literal search is expanded in the experiment manifest rather than retrospectively narrowed to successful tasks. Learning rates are $\{10^{-5}, 3\cdot10^{-5}, 10^{-4}, 3\cdot10^{-4}, 10^{-3}, 3\cdot10^{-3}, 10^{-2}\}$; momentum coefficients are $\{0, .8, .9, .95, .99\}$; damping is $\{10^{-8}, 10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}\}$; and balance intervals are $\{1, 5, 10, 25\}$. The task-specific rank sets follow the proposal. This gives 700 hyperparameter slots per task/rank/method. Baselines independent of damping and balance interval have aliases for those slots, with identical execution outcomes rather than invented repetitions. The manifest contains 1,447,748 logical slots. Collapsing baseline aliases leaves 797,378 distinct configurations; 41,160 of these have mathematically infeasible classifier rank, leaving 756,218 feasible distinct executions in a literal full-grid run. The exhaustive queue was explicitly paused after 158 completed runs on November 4, 2026; the later focused study is reported separately.

Deep linear. True depth-L matrix chains use $L \in \{2, 4, 8, 16, 32\}$, ranks $\{8, 16, 32\}$, target conditions $\{1, 10, 100, 1000\}$, noise amplitudes $\{0, .01, .05\}$, 128-dimensional input/output, and 2000 epochs. The target singular spectrum and initialization are generated with seed 42. The corrected method projects all interfaces jointly and logs the joint normal-equation residual. These runs have a different implementation identity from the archived pairwise depth scan.

Matrix tasks. Square sizes are $128, 256, 512$ and ranks $4, 8, 16, 32$, with 2000 epochs. Sensing uses independent Gaussian rank-one measurement matrices $a_i b_i^\top$ with $5r(2n - r)$ observations; the measurement model is recorded explicitly and is not called dense Gaussian sensing. Completion observes a seeded 50% entry set, with a separate 10% validation and 40% test entry set. Regression uses independent Gaussian design rows and the same planted low-rank operator. Reconstruction error against the planted operator is retained separately from observed training loss.

Vision. CIFAR-10/100 use their complete official training splits, a seeded 10% validation holdout from training, and the official test splits. Tiny-ImageNet uses the Stanford release with its validation annotations as held-out test data and a training holdout for validation. ImageNet-100 uses the public `https://huggingface.co/datasets/clane9/imagenet-100` release with a recorded fingerprint and label map, a training holdout, and the provided validation split as test data. The models are a three-hidden-layer MLP, a ResNet-18 with an internal low-rank head, and ViT-S with low-rank attention/MLP adapters. No pretrained weights are silently downloaded. The ResNet head contains a 256-dimensional internal low-rank map so that rank 32 is valid even with ten output classes. Image sizes and trainable/frozen parameter counts are logged. All runs use 1000 epochs.

Graphs. Cora, Citeseer, and PubMed use Planetoid's official public splits; ogbn-arxiv uses its official temporal split. GCN, GAT, GraphSAGE, and GCNII are crossed with input, hidden, and classifier module placement and ranks $4, 8, 16$, for 1000 epochs. A requested rank exceeding the classifier output dimension is marked mathematically infeasible for the full-rank quotient. It is not reduced to the number of classes without disclosure. Transductive test metrics are empirical task measurements and are not presented as IID generalization certificates.

INR. The image is an explicitly synthetic 64-by-64 Fourier-pattern target, the SDF target is a sphere on a seeded coordinate sample, and coordinate regression uses a fixed oscillatory function. MLP, SIREN, and Fourier-feature MLP are crossed with ranks $8, 16, 32$ for 2000 epochs. Training, validation, and test coordinate sets are disjoint. The PSNR convention in the replacement suite uses unit reference amplitude, $-10\log_{10}(\mathrm{MSE})$; the target range and this convention must be considered when comparing to the archive's range-normalized PSNR. These are controlled INR tasks, not natural-image benchmark results.

Scientific learning. FNO in the proposal's task list is an architecture, so we cross Poisson, Burgers, and Darcy with PINN, DeepONet, and FNO. The tasks use deterministic manufactured Poisson/Darcy solution families and a Cole–Hopf Burgers reference, with data provenance retained. They do not substitute for a claim about a separately downloaded standard Darcy operator benchmark. The families have 128 training, 16 validation, and 16 test instances on a 32-by-32 grid. PINN losses combine reference supervision and a differentiated PDE residual; the operator models use supervised field fitting. Low-rank hidden or pointwise maps have ranks $8, 16, 32$. Every formal run uses 2000 epochs.

Language models. The decoder configurations use widths/layers/heads $(160, 6, 5)$, $(320, 7, 5)$, and $(512, 8, 8)$ as approximately 10M/25M/50M dense-equivalent architectures. Actual stored, trainable, and frozen-base counts are logged after adapter replacement; these labels are not assertions of identical parameter counts across ranks. Tokenization uses the GPT-2 vocabulary and sequence length 256. WikiText-2 uses its full training text. WikiText-103 and TinyStories use a predeclared first one million token training shard, selected before training. Validation/test budgets are at most 65,536 tokens each. TinyStories has no official test split, so disjoint even/odd rows of its validation release supply validation/test data. One epoch-equivalent evaluation is a complete training-shard pass followed by held-out evaluation; the run must perform 1000 such passes. These are small-model training experiments, not pretrained large-model fine-tuning claims.

Optimizer identity. The corrected QuotientOpt and balance-only arms use normalized first-moment updates $v_t = \beta v_{t-1} + (1-\beta)g_t$, differing by horizontal reprojection. Ordinary SGD-M uses its standard unnormalized buffer, and AdamW uses its ordinary coordinatewise moments. All four receive the same learning-rate search slots; the distinction in momentum convention is retained in interpreting tuned comparisons. Non-factor trainable weights use the same zero-decay AdamW rule in all arms. Balancing precedes differentiation. A smooth Procrustes-aligned balance choice reduces arbitrary residual orthogonal sign changes. Between balance points the implementation uses the exact two-Gram Sylvester equation. No claim is made that the relaxed-balance adaptive state is an invariant quotient algorithm.

Artifacts and selection. Checkpoints are saved at epochs $0, 1, 2, 5, 10, 20, 50, 100, 200, 500, 1000$ and every subsequent 100 epochs, with optimizer and random-generator state. Resume discards log rows after the last complete checkpoint before replaying. Failure, rank infeasibility, and storage exhaustion have separate status. Existing checkpoints are not silently deleted to keep a queue running. Epoch logs retain loss, held-out metrics, geometry diagnostics, time, and memory. Paired gauge disagreement is defined only for dedicated paired gauge experiments; an unavailable measurement is recorded as unavailable rather than zero. Configurations and paper tables are selected using validation information, with all completed results retained and the search size disclosed. Final test endpoints are identified explicitly rather than mixed with best-validation endpoints.

L. Archived Evidence and Implementation Audit

What the archive supports. The archived single-interface objective is $\frac{1}{2}\|BA - X_*\|_F^2$, full-observation matrix reconstruction. It was labeled matrix sensing in the original manuscript, but no nontrivial measurement operator is applied in that code. Under the archived adaptive implementation, AdamW's final relative errors at target conditions 100/1000 are 6.49×10^{-3}/3.87×10^{-3}; balance-only gives 1.44×10^{-3}/3.42×10^{-3} and the adaptive QuotientOpt heuristic gives 1.45×10^{-3}/8.39×10^{-4}. These are case-study observations, not runs of the newly analyzed momentum method. Raw-gradient vertical energy at machine precision agrees with the algebraic identity, while nonzero update energy supports measuring the actual direction.

Archived gauge tests report mean final product disagreements about $1.54/1.44$ for AdamW and about 10^{-3} for balance-based adaptive implementations across the two conditions. These motivate rerunning the proper problem families. They do not prove an update-map contraction condition. Likewise, the reported L4 pairwise-heuristic gain of $6.7\times$ in max-over-gauges disagreement is a result about the old implementation, which reverses at L16. It does not identify an inherent depth threshold

Table 7. **Archived full-observation reconstruction.** Final product disagreement, mean ± population SD over the 11 prescribed gauges, 1000 epochs, LR 10^{-2}. These adaptive-implementation records are distinct from the analyzed first-moment algorithm. This table replaces the archived bar chart without changing the underlying observations.

Historical method	Condition 100 ↓	Condition 1000 ↓
AdamW	1.54 ± 2.22	1.44 ± 2.2
Balance-only	0.000946 ± 0.000356	0.00101 ± 0.000203
QuotientOpt (adaptive archive)	0.00138 ± 0.000253	0.00113 ± 0.000185

Table 8. Fresh completed rank-one sensing gauge stress: 12 prescribed gauges per method, 1000 updates each, seed 42. $\Delta_T = \|X_T^{(Q)} - X_T^{(I)}\|_F / \|X_*\|_F$; the maximum uses completed runs, and failed attempts are counted explicitly. $\bar{\rho}_{\mathrm{upd}}$ averages the measured update-direction vertical-energy ratio. Identity error is the final reconstruction error without a gauge perturbation.

Method	Finished/12	Max. Δ_T	$\bar{\rho}_{\mathrm{upd}}$	Identity error
AdamW	12/12	1.07×10^{0}	1.35×10^{-1}	0.0460
SGD-M	7/12	8.86×10^{-1}	5.65×10^{-5}	0.3797
Balance-only	12/12	5.40×10^{-5}	1.91×10^{-6}	0.9994
QuotientOpt	12/12	5.20×10^{-5}	1.20×10^{-14}	0.9994

for the joint quotient method.

Breadth results requiring replacement. The archived INR-SIREN, GCN-SBM, and PINN-Burgers loops compute gradients before a balance change and then return those gradients as if computed at the new section. A Euclidean gradient is a covector represented using the local Euclidean metric; simply reusing it after a nonorthogonal factor change is incorrect. The adaptive second-moment state also lacks a consistent quotient transformation. The archived breadth table remains in the source archive, but we do not count its claimed cross-family gauge wins as verified evidence for the corrected algorithm. In the SIREN code, B, A occur only through their product before the sine; balance therefore preserves the forward function exactly. The earlier factor-scale expressivity explanation is withdrawn.

M. All Fresh Controlled Comparison Cases

These are complete configuration blocks from the existing proposal: 28 Cora learning-rate runs and 48 sensing gauge attempts. The Cora block has 28 complete 1000-epoch runs. The gauge block has 43 complete 1000-update runs and five confirmed nonfinite-objective failures after the diagnostic replay audit in Appendix N. The recovered AdamW run retains its original diagnostic-stop records and exact optimizer-prefix checks. Configurations, result records, full epoch trajectories, source hashes, paired product trajectories, and selection rules are retained. The task-comparison table uses only validation information to choose the reported learning rates and checkpoints. This is one seed and one fixed momentum coefficient, so no confidence-interval or fully tuned cross-family superiority claim is made. Shared-machine wall times are retained but are not used as a speed benchmark.

N. Stopped-Run Audit and Recovery Controls

Diagnostic stops versus objective failure. The original six stops comprise three diagnostic Gram-eigensolve exceptions, two JSON serialization exceptions, and one nonfinite training objective. A diagnostic exception is not evidence that the optimizer itself diverged. We replayed all six configurations with the same seed-42 CPU initialization, single-thread execution, float32 training, original learning rate, momentum, and 1000-update horizon. Only measurements and an unused pre-update diagnostic were changed. All available overlapping model checkpoints and training-loss observations are bit-identical, and the original stop records are retained. The AdamW dense-$\kappa(Q) = 10^4$ run now completes 1000 updates; its final loss is 0.80224 and reconstruction error is 0.93624. Thus AdamW completes all 12 prescribed gauges. After repairing logging, all five stopped SGD-M cases reach a genuinely nonfinite float32 training objective. The corrected fixed-step comparison has 43 complete runs and five objective failures; it does not attribute the AdamW diagnostic stop to optimization instability.

Table 9. Cora/GCN input adapter, rank 4, seed 42: the complete seven-learning-rate block at $\beta = 0.9$, with 1000 epochs per run. Each run uses its earliest maximum-validation-accuracy checkpoint; each method then selects its learning rate by validation accuracy, breaking ties by smaller learning rate. Test metrics do not enter either selection rule; the final 1000-epoch metric is also shown.

Method	LR	Epoch	Val. (%)	Test sel. (%)	Test 1000 (%)
AdamW	10^{-2}	25	67.2	67.6	66.5
SGD-M	10^{-2}	56	74.2	74.0	73.0
Balance-only	10^{-2}	67	73.0	72.0	69.8
QuotientOpt	10^{-2}	67	73.0	71.9	69.8

Table 10. **All seven learning rates: Cora validation accuracy (%).** Rank-4 input adapter, seed 42, 1000 epochs. Each entry is the maximum validation accuracy of its full run; the corresponding CE trajectories are retained in Figure 3.

Method	$1e-05$	$3e-05$	0.0001	0.0003	0.001	0.003	0.01
AdamW	45.2	56.0	63.4	64.4	65.6	66.2	67.2
SGD-M	36.4	61.6	69.4	70.2	71.8	71.2	74.2
Balance-only	36.4	61.8	69.8	70.6	71.2	71.8	73.0
QuotientOpt	36.6	61.8	69.8	70.6	71.2	71.8	73.0

Measurement repair. For diagnostics only, let $B = U_B \Sigma_B V_B^\top$, $A = U_A \Sigma_A V_A^\top$, and $C = B^\top u_B - u_A A^\top$. The unregularized Sylvester measurement is evaluated in float64 by

$$\Omega = V_B \left[\frac{(V_B^\top C U_A)_{ij}}{s_i(B)^2 + s_j(A)^2} \right]_{ij} U_A^\top, \qquad (h_B, h_A) = (u_B - B\Omega,\ u_A + \Omega A). \tag{893}$$

This avoids forming the diagnostic Gram spectra in float32. Reconstruction errors are also measured using float64 products and norms. Unavailable diagnostics are recorded as unavailable; they do not interrupt a finite optimizer update. Neither the AdamW/SGD-M update nor its gradients or optimizer state use this diagnostic projection.

A separate stabilization control. For each of the five genuinely failed SGD-M cases, we additionally run a disclosed recovery algorithm for all 1000 updates. This is a changed protocol, labeled SGD-M with Armijo backtracking and descent resets, following the sufficient-decrease principle of Armijo (1966). Its results never replace the fixed-step baseline. The original factors, observations, float32 gradient computation, and momentum coefficient $\beta = 0.9$ are retained. Writing $z = (B, A)$, the candidate direction and acceptance rule are

$$\widetilde{v}_t = 0.9 v_{t-1} + g_t, \qquad g_t = \nabla F(z_t), \tag{894}$$

$$v_t = \begin{cases} \widetilde{v}_t, & \langle g_t, \widetilde{v}_t \rangle > 0, \\ g_t, & \text{otherwise}, \end{cases} \tag{895}$$

$$\eta_t^{(0)} = \min\{10^{-3}, 2\eta_{t-1}\}, \quad \eta_0 = 10^{-3}, \quad \eta_t^{(k)} = 2^{-k}\eta_t^{(0)}, \tag{896}$$

$$F(z_t - \eta_t v_t) \le F(z_t) - 10^{-4}\eta_t \langle g_t, v_t \rangle + 10^{-7} \max\{1, |F(z_t)|\}. \tag{897}$$

The last term is the disclosed float32 acceptance tolerance. At most 60 trials are permitted per update. Every accepted step size, rejected trial count, reset, pre-update loss, post-update loss, and acceptance threshold is retained. All 5000 accepted updates satisfy the logged acceptance test and have finite losses. Completion indicates numerical stabilization under this changed protocol; the reconstruction errors below do not establish successful recovery of the planted matrix. The largest conditioning stresses require very small accepted steps, in some cases near numerical stagnation. These controls therefore separate three observations: diagnostic robustness, finite-step stability, and task recovery. They do not turn an unstable fixed-step baseline into a completed run of that same baseline.

O. Fixed Expansion to Four Additional Tasks

This exploratory expansion fixes four tasks before its formal runs: CiteSeer/GCN (Kipf & Welling, 2017) with a rank-4 input adapter; a synthetic sphere signed distance field with a rank-8 SIREN (Sitzmann et al., 2020); Burgers solution-operator

Table 11. Every prescribed sensing gauge. Entries are final Δ_T relative to the same method at the identity gauge. F denotes a recorded numerical failure; no finite metric is substituted. Scalar values are q in $Q = qI$; dense values are $\kappa(Q)$.

Gauge	Value	AdamW	SGD-M	Balance	QO
scalar	10^{-3}	1.07×10^{0}	F	1.28×10^{-5}	1.41×10^{-5}
scalar	10^{-2}	9.30×10^{-1}	F	1.36×10^{-5}	1.50×10^{-5}
scalar	10^{-1}	2.68×10^{-1}	8.71×10^{-1}	1.14×10^{-5}	2.03×10^{-5}
scalar	10^{0}	0	0	0	0
scalar	10^{1}	4.47×10^{-2}	3.63×10^{-1}	1.60×10^{-5}	2.35×10^{-5}
scalar	10^{2}	2.99×10^{-1}	8.86×10^{-1}	1.50×10^{-5}	1.97×10^{-5}
scalar	10^{3}	8.86×10^{-1}	F	1.72×10^{-5}	1.98×10^{-5}
dense	10^{0}	2.82×10^{-2}	5.62×10^{-7}	1.79×10^{-5}	2.14×10^{-5}
dense	10^{1}	3.48×10^{-2}	2.67×10^{-1}	1.46×10^{-5}	1.55×10^{-5}
dense	10^{2}	6.88×10^{-2}	2.92×10^{-1}	2.47×10^{-5}	2.40×10^{-5}
dense	10^{3}	2.47×10^{-1}	F	2.44×10^{-5}	2.88×10^{-5}
dense	10^{4}	9.35×10^{-1}	F	5.40×10^{-5}	5.20×10^{-5}

Table 12. All Cora learning rates in the fixed block. Entries are test accuracies (%) at the checkpoint chosen using that run's validation accuracy, not at a test-selected checkpoint.

LR	AdamW	SGD-M	Balance	QO
10^{-5}	44.9	40.0	40.0	40.0
3×10^{-5}	59.9	62.1	62.2	62.1
10^{-4}	63.0	71.3	71.1	71.3
3×10^{-4}	64.9	71.6	71.4	71.4
10^{-3}	66.5	70.4	70.1	70.1
3×10^{-3}	66.1	71.9	70.3	70.3
10^{-2}	67.6	74.0	72.0	71.9

learning with a rank-8 DeepONet (Lu et al., 2019); and 128×128 rank-8 matrix completion with planted condition number 100. It is an additional task study, not an independently confirmed or completed exhaustive proposal grid. The SDF is synthetic. Burgers uses the documented Cole–Hopf quadrature data and parameter-held-out splits from Appendix K; it is not relabeled as a public benchmark data set.

Each task crosses all four methods with exactly the same nine learning rates

$$\{10^{-5}, 3\times10^{-5}, 10^{-4}, 3\times10^{-4}, 10^{-3}, 3\times10^{-3}, 10^{-2}, 3\times10^{-2}, 10^{-1}\}.$$

We fix $\beta = 0.9$, $\varepsilon = 10^{-6}$, balance interval one, and seed 42. The two larger step sizes extend the proposal's original learning-rate slice equally for every method, to examine the underfitting seen at its central setting. Graph runs have 1000 epochs; the other runs have 2000. The four task blocks contain 36 attempts each, for 144 total. Low-resource execution checks provide timing estimates only and never enter the result tables.

Validation and test separation. Within a run, the earliest minimum-validation-loss checkpoint is selected: cross-entropy for graphs and MSE for the other tasks. Across learning rates, the minimum validation score selects a run, with ties broken by smaller learning rate and then earlier epoch. Test metrics never enter these choices. The additional implementation does not evaluate the test set every epoch: it evaluates the validation-selected checkpoint and final-horizon model after training. The graph split is the official public transductive split; its metrics are not presented as IID population-risk certificates.

Matrix completion uses disjoint fixed masks with 50% training, 10% validation, and 40% test entries. Selection uses only the validation-entry MSE. Full planted-matrix reconstruction error is retained as a separate diagnostic and is not a validation criterion. The other data and model definitions match the documented implementation, including frozen base matrices and the shared optimizer for non-factor parameters.

Complete blocks, stopped trials, and interpretation. A selected initial-slice comparison is published only after all 36 prescribed attempts for that task have terminated. The primary table selects from completed full-horizon runs. If a

Table 13. Secondary loss-based early stopping on the same 28 Cora runs. Select the lowest validation cross-entropy over epochs and learning rates; ties use smaller LR, then earlier epoch. This is distinct from the primary validation-accuracy criterion in Table 9. Test loss is read at the validation-selected checkpoint and is not minimized during selection.

Method	LR	Epoch	Val. CE	Test CE	Test acc. (%)
AdamW	10^{-2}	15	1.1728	1.0525	64.7
SGD-M	10^{-2}	36	0.9079	0.8579	73.1
Balance-only	10^{-2}	37	0.9261	0.8730	72.0
QuotientOpt	10^{-2}	37	0.9261	0.8730	72.0

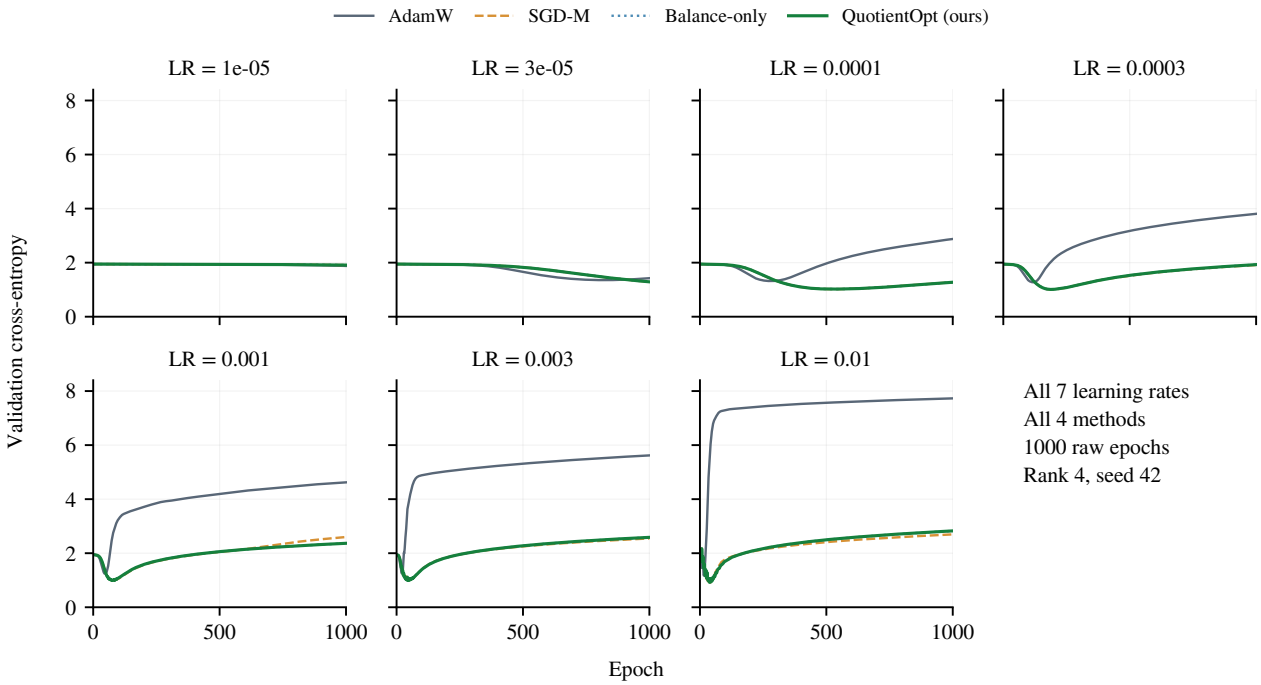


Figure 3. Every learning rate in the complete Cora/GCN input-adapter block: rank 4, seed 42, $\beta = 0.9$. All four methods and all 1000 raw epoch observations are retained, with common axes, no smoothing, and no omitted learning rates. This figure supplies the full learning-rate context for the accuracy-selected core loss curves in Figure 6.

later-stopped trial has a finite best validation checkpoint, its test metric and status are retained in a separate sensitivity analysis. A claim of improvement over all controls must also survive that sensitivity analysis. Stopped runs are not converted into complete-horizon results. Every attempted task, learning rate, method, and failure remains in the full record below; unfavorable outcomes are retained. One fixed seed does not establish seed robustness or a confidence interval. Near-ties with balance-only do not establish a distinct task-accuracy gain from projection.

Sequential factor-step extension. The initial 144-run plan remains unchanged and its complete comparisons are retained separately in Table 16. Motivated by training underfitting and the different factor and non-factor step scales, we add four factor learning rates $\{0.3, 1, 3, 10\}$ for every method and every task, fixing the shared non-factor AdamW learning rate at 10^{-3}. Matrix completion has no non-factor parameters, so its four added settings simply extend the factor learning-rate range. This is a disclosed exploratory extension, not a setting fixed before the initial study. It adds 64 trials. The joint budget has 13 settings per method and 52 attempts per task, for 208 attempts in total. All four methods receive the same two regimes: nine coupled learning rates and four larger factor-only learning rates. Joint selection minimizes validation loss over their union; a joint comparison is reported only after all 52 attempts for that task have terminated. The failed-trial sensitivity rule also applies to this joint selection. Values below a validation MSE scale of 10^{-14} are treated as numerically limited when selecting substantive case-study figures; percentage differences between such values do not establish an optimization advantage.

Table 14. All five SGD-M failure recoveries. The fixed-step column is the update at which the original $\eta = 10^{-3}$ replay first encounters a nonfinite objective. The other columns describe the separate 1000-update backtracking protocol. Final loss is evaluated after the last update; $e_T = \|X_T - X_*\|_F / \|X_*\|_F$.

Gauge	Value	Fixed-step failure	Final loss	Min. η_t	Max. η_t	e_T
scalar	1.00×10^{-3}	5	0.8653	1.22×10^{-7}	3.91×10^{-6}	1.0059
scalar	1.00×10^{-2}	8	0.8653	1.49×10^{-11}	2.50×10^{-4}	1.0059
scalar	1.00×10^{3}	7	0.7944	7.81×10^{-6}	2.50×10^{-4}	0.9611
dense	1.00×10^{3}	9	0.7873	1.91×10^{-9}	5.00×10^{-4}	0.9356
dense	1.00×10^{4}	5	0.9679	3.64×10^{-15}	3.91×10^{-6}	0.9995

Table 15. Additional fixed four-family expansion. Each task has thirteen optimizer settings and four methods, with unchanged full horizons. Counts describe completed runs and recorded failures; a pending attempt is not a task result.

New task	Horizon	Planned	Complete	Failed
CiteSeer / GCN	1000	52	52	0
SDF / SIREN	2000	52	40	12
Burgers / DeepONet	2000	52	45	7
Matrix completion	2000	52	49	3

Table 16. Original nine-learning-rate expansion results before the separate factor-step extension. Every task has all 36 prescribed attempts terminated. Selection uses validation loss; the test loss is reported at its selected checkpoint. These initial outcomes, including unfavorable comparisons, are retained separately from the wider joint search.

Task	AdamW	SGD-M	Balance	QO
CiteSeer / GCN	1.2660	**1.2125**	1.2128	1.2127
SDF / SIREN	6.03×10^{-5}	2.42×10^{-4}	1.10×10^{-5}	$\mathbf{9.36\times10^{-6}}$
Burgers / DeepONet	$\mathbf{8.66\times10^{-5}}$	0.0030	0.0035	0.0042
Matrix completion	$\mathbf{1.06\times10^{-18}}$	5.25×10^{-8}	3.90×10^{-8}	3.89×10^{-8}

Table 17. Validation-selected test losses in completed new-task blocks. All thirteen optimizer-setting attempts per method have finished before inclusion. Selection minimizes validation CE (graph) or validation MSE (other tasks), with ties by smaller LR then earlier epoch. Primary entries use full-horizon runs. Every trial, including failures and finite checkpoints from subsequently stopped trials, is retained in Appendix O. Lower is better.

Task	AdamW	SGD-M	Balance	QO
CiteSeer / GCN	1.3762	1.2508	1.2168	**1.2163**
SDF / SIREN	6.03×10^{-5}	2.42×10^{-4}	1.10×10^{-5}	$\mathbf{9.36\times10^{-6}}$
Burgers / DeepONet	$\mathbf{8.66\times10^{-5}}$	0.0030	5.85×10^{-4}	0.0042
Matrix completion	$\mathbf{1.06\times10^{-18}}$	1.17×10^{-8}	4.14×10^{-17}	4.65×10^{-17}

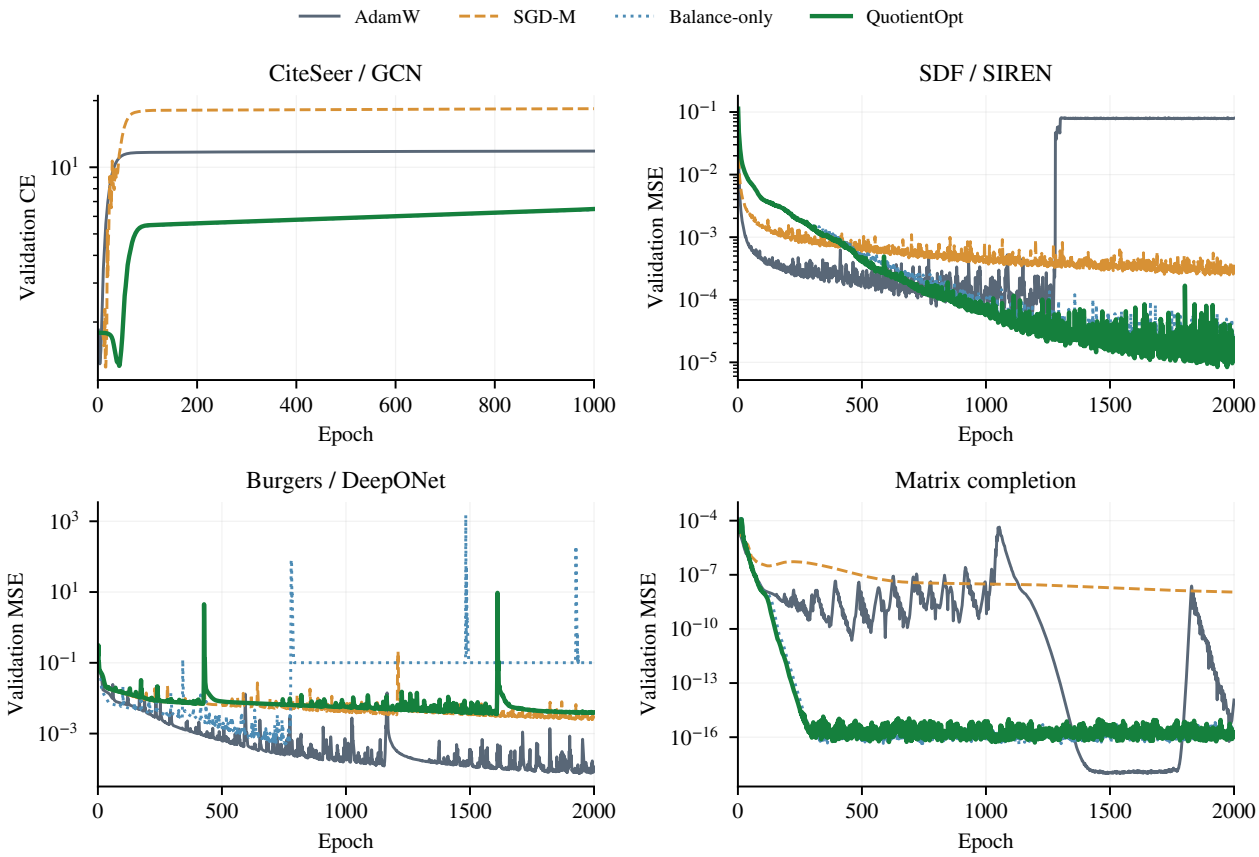


Figure 4. All completed new-task comparison blocks, including unfavorable outcomes. For each method, validation loss selects the learning rate and checkpoint under the fixed rule; the entire selected run is plotted, with no smoothing or interval truncation. The curves are validation measurements, not IID risk certificates.

Descriptive validation iteration profiles. For regression tasks we additionally report the complete family of relative-MSE targets $\{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-6}\}$. With the same initial model and validation data for all methods, define the finite-prefix tuning envelope

$$e_{m,t} = \min_{\lambda \in \Lambda_m} \min_{0 \le s \le \min\{t, T_{m,\lambda}\}} \frac{L_{\mathrm{val}}(z_{m,\lambda,s})}{L_{\mathrm{val}}(z_0)}, \qquad \tau_m(a) = \inf\{t : e_{m,t} \le a\}. \tag{898}$$

Every method has the same 13 optimizer settings. Retained finite prefixes of stopped runs are eligible, so failed configurations do not erase a baseline's earlier finite progress. An unreached target is censored at the horizon. These are descriptive validation tuning profiles, with test metrics excluded, rather than an independently registered confirmation or a single executable optimizer trajectory. They compare iteration budgets; shared-machine wall times are not a controlled speed benchmark. The common initial validation scale is checked against saved initial model states across methods and settings.

Table 18. All descriptive validation-MSE targets in the new regression tasks. Each entry is the first epoch at which the validation tuning profile reaches the stated fraction of the common initial validation MSE. Every method has 13 optimizer settings; finite prefixes of later-stopped runs are eligible. A dash denotes an unreached target by epoch 2000. These are validation-selected iteration profiles, not wall-clock benchmarks or single executable training trajectories.

Task	Relative target	AdamW	SGD-M	Balance	QO
SDF / SIREN	10^{-1}	2	2	2	2
SDF / SIREN	10^{-2}	11	28	16	17
SDF / SIREN	10^{-3}	157	1350	403	403
SDF / SIREN	10^{-4}	—	—	1183	1142
SDF / SIREN	10^{-6}	—	—	—	—
Burgers / DeepONet	10^{-1}	73	292	47	47
Burgers / DeepONet	10^{-2}	297	—	651	584
Burgers / DeepONet	10^{-3}	—	—	—	—
Burgers / DeepONet	10^{-4}	—	—	—	—
Burgers / DeepONet	10^{-6}	—	—	—	—
Matrix completion	10^{-1}	16	25	19	19
Matrix completion	10^{-2}	43	57	48	45
Matrix completion	10^{-3}	67	502	71	69
Matrix completion	10^{-4}	138	—	117	107
Matrix completion	10^{-6}	215	—	162	150

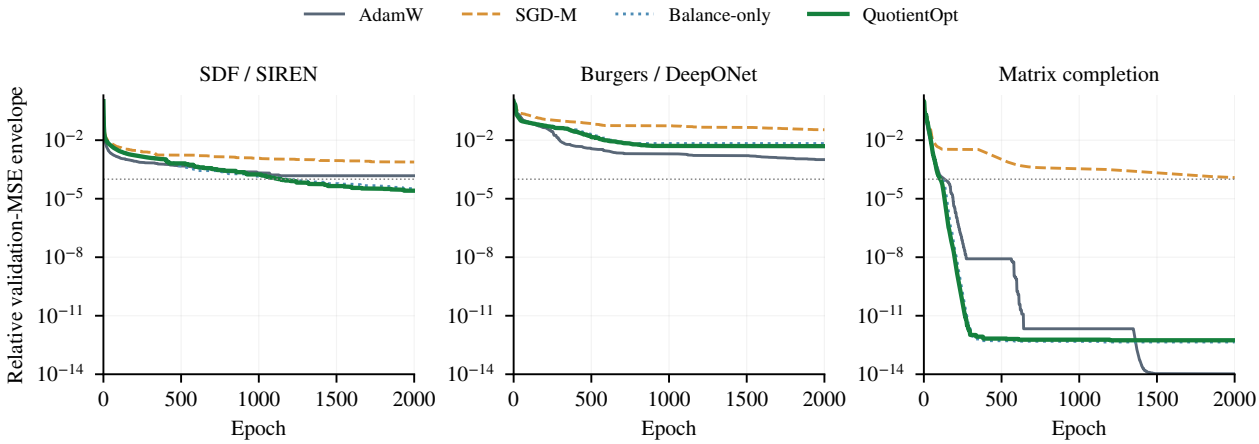


Figure 5. Validation tuning envelopes from every completed regression-task block. At each epoch the envelope takes the best finite validation checkpoint among all 13 settings and earlier epochs, including prefixes of later-stopped trials. It is a descriptive equal-budget tuning profile, not one optimizer trajectory. The horizontal line is the 10^{-4} relative-MSE target; Table 18 retains all five targets. Values below 10^{-14} are drawn at the plotting floor for legibility.

Table 19. All attempted configurations in the new four-family expansion at this snapshot. Entries are validation-selected test losses, not test-minimized checkpoints; LR is the factor learning rate. Non-factor LR equals LR for the original nine settings through 0.1 and is fixed at 10^{-3} for the four added factor LRs 0.3, 1, 3, and 10. E is the selected epoch. C denotes a completed full-horizon run; F denotes a stopped run, whose best finite validation checkpoint is reported when available. Pending and running attempts have no final result and are omitted from this table. Their counts remain in Table 15.

Task / method	LR	E	Val. score	Test loss	Status
CiteSeer / GCN / AdamW	1e-05	1000	1.7272	1.7143	C
CiteSeer / GCN / SGD-M	1e-05	1000	1.7757	1.7693	C
CiteSeer / GCN / Balance-only	1e-05	1000	1.7757	1.7692	C
CiteSeer / GCN / QuotientOpt	1e-05	1000	1.7757	1.7692	C
SDF / SIREN / AdamW	1e-05	2000	0.0012	0.0015	C
SDF / SIREN / SGD-M	1e-05	2000	0.0049	0.0050	C

Task / method	LR	E	Val. score	Test loss	Status
SDF / SIREN / Balance-only	1e-05	2000	0.0072	0.0063	C
SDF / SIREN / QuotientOpt	1e-05	2000	0.0072	0.0063	C
Burgers / DeepONet / AdamW	1e-05	2000	0.0014	0.0016	C
Burgers / DeepONet / SGD-M	1e-05	2000	0.0096	0.0113	C
Burgers / DeepONet / Balance-only	1e-05	2000	0.0096	0.0113	C
Burgers / DeepONet / QuotientOpt	1e-05	2000	0.0096	0.0113	C
Matrix completion / AdamW	1e-05	2000	7.56×10^{-5}	7.48×10^{-5}	C
Matrix completion / SGD-M	1e-05	725	9.10×10^{-5}	9.08×10^{-5}	C
Matrix completion / Balance-only	1e-05	2000	9.11×10^{-5}	9.11×10^{-5}	C
Matrix completion / QuotientOpt	1e-05	2000	9.11×10^{-5}	9.11×10^{-5}	C
CiteSeer / GCN / AdamW	3e-05	695	1.4035	1.3374	C
CiteSeer / GCN / SGD-M	3e-05	1000	1.3855	1.2792	C
CiteSeer / GCN / Balance-only	3e-05	1000	1.3852	1.2795	C
CiteSeer / GCN / QuotientOpt	3e-05	1000	1.3852	1.2795	C
SDF / SIREN / AdamW	3e-05	2000	3.56×10^{-4}	3.74×10^{-4}	C
SDF / SIREN / SGD-M	3e-05	1991	0.0020	0.0021	C
SDF / SIREN / Balance-only	3e-05	2000	0.0034	0.0031	C
SDF / SIREN / QuotientOpt	3e-05	2000	0.0033	0.0031	C
Burgers / DeepONet / AdamW	3e-05	2000	5.20×10^{-4}	5.85×10^{-4}	C
Burgers / DeepONet / SGD-M	3e-05	2000	0.0081	0.0095	C
Burgers / DeepONet / Balance-only	3e-05	2000	0.0082	0.0096	C
Burgers / DeepONet / QuotientOpt	3e-05	2000	0.0081	0.0096	C
Matrix completion / AdamW	3e-05	2000	2.08×10^{-5}	2.20×10^{-5}	C
Matrix completion / SGD-M	3e-05	247	9.10×10^{-5}	9.08×10^{-5}	C
Matrix completion / Balance-only	3e-05	2000	9.10×10^{-5}	9.10×10^{-5}	C
Matrix completion / QuotientOpt	3e-05	2000	9.10×10^{-5}	9.10×10^{-5}	C
CiteSeer / GCN / AdamW	0.0001	231	1.3392	1.2774	C
CiteSeer / GCN / SGD-M	0.0001	355	1.3561	1.2243	C
CiteSeer / GCN / Balance-only	0.0001	356	1.3555	1.2241	C
CiteSeer / GCN / QuotientOpt	0.0001	356	1.3560	1.2244	C
SDF / SIREN / AdamW	0.0001	1989	1.62×10^{-4}	1.64×10^{-4}	C
SDF / SIREN / SGD-M	0.0001	1998	0.0011	0.0011	C
SDF / SIREN / Balance-only	0.0001	2000	8.87×10^{-4}	8.74×10^{-4}	C
SDF / SIREN / QuotientOpt	0.0001	2000	6.94×10^{-4}	7.05×10^{-4}	C
Burgers / DeepONet / AdamW	0.0001	1990	1.34×10^{-4}	1.66×10^{-4}	C
Burgers / DeepONet / SGD-M	0.0001	1988	0.0063	0.0074	C
Burgers / DeepONet / Balance-only	0.0001	1995	0.0064	0.0076	C
Burgers / DeepONet / QuotientOpt	0.0001	1983	0.0064	0.0076	C
Matrix completion / AdamW	0.0001	2000	1.75×10^{-6}	1.77×10^{-6}	C
Matrix completion / SGD-M	0.0001	80	9.10×10^{-5}	9.08×10^{-5}	C
Matrix completion / Balance-only	0.0001	2000	9.08×10^{-5}	9.06×10^{-5}	C
Matrix completion / QuotientOpt	0.0001	2000	9.08×10^{-5}	9.06×10^{-5}	C
CiteSeer / GCN / AdamW	0.0003	96	1.3091	1.2485	C
CiteSeer / GCN / SGD-M	0.0003	139	1.3492	1.2123	C
CiteSeer / GCN / Balance-only	0.0003	139	1.3489	1.2128	C
CiteSeer / GCN / QuotientOpt	0.0003	138	1.3492	1.2152	C
SDF / SIREN / AdamW	0.0003	1994	1.22×10^{-4}	1.21×10^{-4}	C
SDF / SIREN / SGD-M	0.0003	1994	6.41×10^{-4}	6.72×10^{-4}	C
SDF / SIREN / Balance-only	0.0003	1997	6.74×10^{-5}	6.49×10^{-5}	C
SDF / SIREN / QuotientOpt	0.0003	1998	7.49×10^{-5}	8.96×10^{-5}	C
Burgers / DeepONet / AdamW	0.0003	1941	7.40×10^{-5}	8.66×10^{-5}	C
Burgers / DeepONet / SGD-M	0.0003	1977	0.0045	0.0053	C
Burgers / DeepONet / Balance-only	0.0003	1978	0.0047	0.0056	C
Burgers / DeepONet / QuotientOpt	0.0003	1997	0.0046	0.0055	C
Matrix completion / AdamW	0.0003	2000	3.14×10^{-7}	2.71×10^{-7}	C

Task / method	LR	E	Val. score	Test loss	Status
Matrix completion / SGD-M	0.0003	2000	3.99×10^{-5}	3.87×10^{-5}	C
Matrix completion / Balance-only	0.0003	2000	9.02×10^{-5}	8.97×10^{-5}	C
Matrix completion / QuotientOpt	0.0003	2000	9.02×10^{-5}	8.97×10^{-5}	C
CiteSeer / GCN / AdamW	0.001	38	1.3143	1.2552	C
CiteSeer / GCN / SGD-M	0.001	53	1.3352	1.2125	C
CiteSeer / GCN / Balance-only	0.001	53	1.3358	1.2128	C
CiteSeer / GCN / QuotientOpt	0.001	53	1.3357	1.2127	C
SDF / SIREN / AdamW	0.001	1964	5.65×10^{-5}	8.43×10^{-5}	C
SDF / SIREN / SGD-M	0.001	1991	3.49×10^{-4}	3.97×10^{-4}	C
SDF / SIREN / Balance-only	0.001	1929	1.07×10^{-5}	1.10×10^{-5}	C
SDF / SIREN / QuotientOpt	0.001	1972	8.39×10^{-6}	9.36×10^{-6}	C
Burgers / DeepONet / AdamW	0.001	1957	1.45×10^{-4}	1.82×10^{-4}	C
Burgers / DeepONet / SGD-M	0.001	1977	0.0025	0.0030	C
Burgers / DeepONet / Balance-only	0.001	1944	0.0030	0.0036	C
Burgers / DeepONet / QuotientOpt	0.001	1608	0.0036	0.0042	C
Matrix completion / AdamW	0.001	2000	6.64×10^{-8}	7.12×10^{-8}	C
Matrix completion / SGD-M	0.001	2000	5.07×10^{-6}	4.95×10^{-6}	C
Matrix completion / Balance-only	0.001	2000	8.26×10^{-5}	8.12×10^{-5}	C
Matrix completion / QuotientOpt	0.001	2000	8.26×10^{-5}	8.12×10^{-5}	C
CiteSeer / GCN / AdamW	0.003	16	1.3068	1.2660	C
CiteSeer / GCN / SGD-M	0.003	28	1.3386	1.1922	C
CiteSeer / GCN / Balance-only	0.003	28	1.3401	1.1937	C
CiteSeer / GCN / QuotientOpt	0.003	28	1.3401	1.1937	C
SDF / SIREN / AdamW	0.003	1130	4.90×10^{-5}	6.03×10^{-5}	C
SDF / SIREN / SGD-M	0.003	1947	2.47×10^{-4}	2.42×10^{-4}	C
SDF / SIREN / Balance-only	0.003	1993	2.32×10^{-5}	2.63×10^{-5}	C
SDF / SIREN / QuotientOpt	0.003	1943	2.30×10^{-5}	2.40×10^{-5}	C
Burgers / DeepONet / AdamW	0.003	535	2.93×10^{-4}	3.50×10^{-4}	C
Burgers / DeepONet / SGD-M	0.003	585	0.0041	0.0049	C
Burgers / DeepONet / Balance-only	0.003	1978	0.0029	0.0035	C
Burgers / DeepONet / QuotientOpt	0.003	1519	0.0040	0.0047	C
Matrix completion / AdamW	0.003	1999	2.11×10^{-8}	2.20×10^{-8}	C
Matrix completion / SGD-M	0.003	2000	2.01×10^{-6}	1.84×10^{-6}	C
Matrix completion / Balance-only	0.003	2000	1.35×10^{-5}	1.26×10^{-5}	C
Matrix completion / QuotientOpt	0.003	2000	1.35×10^{-5}	1.26×10^{-5}	C
CiteSeer / GCN / AdamW	0.01	7	1.3692	1.3316	C
CiteSeer / GCN / SGD-M	0.01	19	1.3442	1.2166	C
CiteSeer / GCN / Balance-only	0.01	19	1.3505	1.2240	C
CiteSeer / GCN / QuotientOpt	0.01	19	1.3501	1.2234	C
SDF / SIREN / AdamW	0.01	39	0.0780	0.0747	C
SDF / SIREN / SGD-M	0.01	651	5.43×10^{-4}	5.49×10^{-4}	C
SDF / SIREN / Balance-only	0.01	412	2.46×10^{-4}	2.54×10^{-4}	C
SDF / SIREN / QuotientOpt	0.01	485	2.10×10^{-4}	2.13×10^{-4}	C
Burgers / DeepONet / AdamW	0.01	417	0.0040	0.0046	C
Burgers / DeepONet / SGD-M	0.01	41	0.1007	0.1177	C
Burgers / DeepONet / Balance-only	0.01	570	0.0069	0.0081	C
Burgers / DeepONet / QuotientOpt	0.01	770	0.0055	0.0065	F
Matrix completion / AdamW	0.01	2000	3.70×10^{-10}	3.87×10^{-10}	C
Matrix completion / SGD-M	0.01	2000	5.99×10^{-7}	5.46×10^{-7}	C
Matrix completion / Balance-only	0.01	2000	2.20×10^{-6}	2.15×10^{-6}	C
Matrix completion / QuotientOpt	0.01	2000	2.20×10^{-6}	2.15×10^{-6}	C
CiteSeer / GCN / AdamW	0.03	5	1.5255	1.5701	C
CiteSeer / GCN / SGD-M	0.03	27	1.3885	1.3170	C
CiteSeer / GCN / Balance-only	0.03	26	1.5585	1.4382	C
CiteSeer / GCN / QuotientOpt	0.03	26	1.5585	1.4382	C

Task / method	LR	E	Val. score	Test loss	Status
SDF / SIREN / AdamW	0.03	9	0.0797	0.0746	C
SDF / SIREN / SGD-M	0.03	6	0.0786	0.0727	F
SDF / SIREN / Balance-only	0.03	147	0.0015	0.0016	C
SDF / SIREN / QuotientOpt	0.03	145	0.0020	0.0020	C
Burgers / DeepONet / AdamW	0.03	1278	0.0260	0.0242	C
Burgers / DeepONet / SGD-M	0.03	41	0.1007	0.1177	C
Burgers / DeepONet / Balance-only	0.03	62	0.1007	0.1177	C
Burgers / DeepONet / QuotientOpt	0.03	39	0.1007	0.1177	F
Matrix completion / AdamW	0.03	1540	9.61×10^{-19}	1.06×10^{-18}	C
Matrix completion / SGD-M	0.03	2000	2.06×10^{-7}	1.94×10^{-7}	C
Matrix completion / Balance-only	0.03	2000	3.64×10^{-7}	3.23×10^{-7}	C
Matrix completion / QuotientOpt	0.03	2000	3.63×10^{-7}	3.23×10^{-7}	C
CiteSeer / GCN / AdamW	0.1	22	1.6835	1.7497	C
CiteSeer / GCN / SGD-M	0.1	81	1.7301	1.7913	C
CiteSeer / GCN / Balance-only	0.1	100	1.7300	1.8034	C
CiteSeer / GCN / QuotientOpt	0.1	113	1.7375	1.8088	C
SDF / SIREN / AdamW	0.1	5	0.0900	0.0853	C
SDF / SIREN / SGD-M	0.1	1	0.3478	0.3722	F
SDF / SIREN / Balance-only	0.1	11	0.0791	0.0742	F
SDF / SIREN / QuotientOpt	0.1	13	0.0797	0.0755	F
Burgers / DeepONet / AdamW	0.1	1726	0.1006	0.1176	C
Burgers / DeepONet / SGD-M	0.1	170	0.0524	0.0578	C
Burgers / DeepONet / Balance-only	0.1	36	0.1007	0.1177	C
Burgers / DeepONet / QuotientOpt	0.1	–	—	—	F
Matrix completion / AdamW	0.1	641	1.96×10^{-16}	1.88×10^{-16}	C
Matrix completion / SGD-M	0.1	2000	5.11×10^{-8}	5.25×10^{-8}	C
Matrix completion / Balance-only	0.1	2000	4.59×10^{-8}	3.90×10^{-8}	C
Matrix completion / QuotientOpt	0.1	2000	4.57×10^{-8}	3.89×10^{-8}	C
CiteSeer / GCN / AdamW	0.3	4	1.2942	1.3762	C
CiteSeer / GCN / SGD-M	0.3	33	1.3254	1.2699	C
CiteSeer / GCN / Balance-only	0.3	48	1.2749	1.1835	C
CiteSeer / GCN / QuotientOpt	0.3	48	1.2756	1.1842	C
SDF / SIREN / AdamW	0.3	182	0.0774	0.0726	C
SDF / SIREN / SGD-M	0.3	1	0.2931	0.3260	F
SDF / SIREN / Balance-only	0.3	1994	1.50×10^{-4}	1.70×10^{-4}	C
SDF / SIREN / QuotientOpt	0.3	1954	1.37×10^{-4}	1.54×10^{-4}	C
Burgers / DeepONet / AdamW	0.3	61	0.0317	0.0314	C
Burgers / DeepONet / SGD-M	0.3	22	0.1007	0.1177	C
Burgers / DeepONet / Balance-only	0.3	762	4.93×10^{-4}	5.85×10^{-4}	C
Burgers / DeepONet / QuotientOpt	0.3	910	3.62×10^{-4}	4.37×10^{-4}	F
Matrix completion / AdamW	0.3	2000	3.13×10^{-9}	2.98×10^{-9}	C
Matrix completion / SGD-M	0.3	2000	1.10×10^{-8}	1.17×10^{-8}	C
Matrix completion / Balance-only	0.3	2000	4.87×10^{-9}	4.25×10^{-9}	C
Matrix completion / QuotientOpt	0.3	2000	4.67×10^{-9}	4.08×10^{-9}	C
CiteSeer / GCN / AdamW	1	4	2.9122	2.9363	C
CiteSeer / GCN / SGD-M	1	26	1.3674	1.3193	C
CiteSeer / GCN / Balance-only	1	43	1.2690	1.2168	C
CiteSeer / GCN / QuotientOpt	1	43	1.2671	1.2163	C
SDF / SIREN / AdamW	1	67	0.0774	0.0727	C
SDF / SIREN / SGD-M	1	–	—	—	F
SDF / SIREN / Balance-only	1	1980	1.36×10^{-4}	1.32×10^{-4}	C
SDF / SIREN / QuotientOpt	1	1995	1.40×10^{-4}	1.73×10^{-4}	C
Burgers / DeepONet / AdamW	1	36	0.1007	0.1177	C
Burgers / DeepONet / SGD-M	1	300	0.1007	0.1177	C
Burgers / DeepONet / Balance-only	1	29	0.1007	0.1177	C

Task / method	LR	E	Val. score	Test loss	Status
Burgers / DeepONet / QuotientOpt	1	1	0.1332	0.1559	F
Matrix completion / AdamW	1	2000	5.79×10^{-7}	5.05×10^{-7}	C
Matrix completion / SGD-M	1	6	2.67×10^{-5}	2.70×10^{-5}	F
Matrix completion / Balance-only	1	2000	4.69×10^{-7}	1.98×10^{-7}	C
Matrix completion / QuotientOpt	1	2000	3.97×10^{-7}	1.87×10^{-7}	C
CiteSeer / GCN / AdamW	3	6	12.0411	14.8568	C
CiteSeer / GCN / SGD-M	3	23	1.3106	1.4144	C
CiteSeer / GCN / Balance-only	3	32	1.3635	1.3001	C
CiteSeer / GCN / QuotientOpt	3	32	1.3628	1.2997	C
SDF / SIREN / AdamW	3	429	0.0775	0.0730	C
SDF / SIREN / SGD-M	3	–	—	—	F
SDF / SIREN / Balance-only	3	1	0.3028	0.3246	F
SDF / SIREN / QuotientOpt	3	1	0.3035	0.3098	F
Burgers / DeepONet / AdamW	3	35	0.1007	0.1177	C
Burgers / DeepONet / SGD-M	3	549	0.1007	0.1177	C
Burgers / DeepONet / Balance-only	3	30	0.1007	0.1177	C
Burgers / DeepONet / QuotientOpt	3	1	77.1926	76.3920	F
Matrix completion / AdamW	3	2000	6.52×10^{-6}	6.32×10^{-6}	C
Matrix completion / SGD-M	3	2	1.41×10^{-4}	1.36×10^{-4}	F
Matrix completion / Balance-only	3	2000	1.69×10^{-6}	1.39×10^{-6}	C
Matrix completion / QuotientOpt	3	2000	1.21×10^{-6}	1.04×10^{-6}	C
CiteSeer / GCN / AdamW	10	7	147.0064	185.7595	C
CiteSeer / GCN / SGD-M	10	16	1.2514	1.2508	C
CiteSeer / GCN / Balance-only	10	24	1.3988	1.3532	C
CiteSeer / GCN / QuotientOpt	10	24	1.3975	1.3524	C
SDF / SIREN / AdamW	10	1249	0.0774	0.0730	C
SDF / SIREN / SGD-M	10	–	—	—	F
SDF / SIREN / Balance-only	10	–	—	—	F
SDF / SIREN / QuotientOpt	10	–	—	—	F
Burgers / DeepONet / AdamW	10	1583	0.1005	0.1175	C
Burgers / DeepONet / SGD-M	10	40	0.1007	0.1177	C
Burgers / DeepONet / Balance-only	10	37	0.1007	0.1177	C
Burgers / DeepONet / QuotientOpt	10	1	16.4258	16.4429	F
Matrix completion / AdamW	10	1941	9.48×10^{-5}	9.59×10^{-5}	C
Matrix completion / SGD-M	10	1	0.0011	0.0011	F
Matrix completion / Balance-only	10	1133	4.10×10^{-17}	4.14×10^{-17}	C
Matrix completion / QuotientOpt	10	1197	5.03×10^{-17}	4.65×10^{-17}	C

Stopped-checkpoint sensitivity: Burgers / DeepONet / QuotientOpt. The minimum-validation-score checkpoint when stopped trials are eligible occurs at learning rate 0.3 and epoch 910, with test loss 4.37×10^{-4}. The run status is failed. Its score is retained as a sensitivity control; it is not presented as a completed full-horizon run.

P. Adjusted Fixed-Step Recovery Study

The original 208-attempt four-task search contains 22 numerical stops. For each stopped configuration, the follow-up retains the same task, model, rank, data, initialization, momentum coefficient, damping, balance interval, non-factor learning rate, and 2000-epoch horizon. The factor learning rate is lowered using completed runs in the same task/method regime, with a separate protocol and run identifier. The parent failure record and its finite validation checkpoint remain in the original search. Every adjusted run begins from the identical parent initial model, verified against its saved epoch-zero checkpoint.

The preflight evaluates three training epochs for each of nine distinct task/method/step-scale groups. All 22 formal trajectories subsequently contain exactly 2000 recorded epochs. The model within each run and the reported retry configuration within a task/method group are chosen by minimum validation MSE. Test measurements do not enter either choice. No test measurements are present in the per-epoch trajectories. Table 20 records every adjusted learning rate and validation-selected outcome. The separate five genuinely failed rank-one sensing SGD-M configurations have complete 1000-step Armijo

recovery controls in Appendix N.

Table 20. All 22 adjusted fixed-step runs. Every row completes 2000 epochs. The parent learning rate produced a recorded numerical stop, and the adjusted learning rate uses the original initialization. Validation MSE determines the checkpoint; test MSE is measured at that selected checkpoint.

Task / method	Parent LR	Retry LR	Epoch	Val. MSE	Test MSE
Burgers / DeepONet / QuotientOpt	0.01	0.00255	1780	0.0021	0.0025
Burgers / DeepONet / QuotientOpt	0.03	0.0021	1792	0.0031	0.0037
Burgers / DeepONet / QuotientOpt	0.1	0.00165	1695	0.0031	0.0036
Burgers / DeepONet / QuotientOpt	0.3	0.00255	1992	0.0029	0.0034
Burgers / DeepONet / QuotientOpt	1	0.00225	1995	0.0029	0.0035
Burgers / DeepONet / QuotientOpt	3	0.00195	1993	0.0028	0.0033
Burgers / DeepONet / QuotientOpt	10	0.00165	1995	0.0031	0.0037
SDF / SIREN / SGD-M	0.03	0.0085	1785	1.12×10^{-4}	1.09×10^{-4}
SDF / SIREN / SGD-M	0.1	0.0055	1934	1.68×10^{-4}	1.85×10^{-4}
SDF / SIREN / SGD-M	0.3	0.0085	1981	2.17×10^{-4}	2.52×10^{-4}
SDF / SIREN / SGD-M	1	0.0075	1997	2.26×10^{-4}	2.49×10^{-4}
SDF / SIREN / SGD-M	3	0.0065	1977	2.53×10^{-4}	2.81×10^{-4}
SDF / SIREN / SGD-M	10	0.0055	1994	2.73×10^{-4}	3.00×10^{-4}
SDF / SIREN / Balance-only	0.1	0.0255	146	0.0018	0.0018
SDF / SIREN / Balance-only	3	0.85	1974	1.20×10^{-4}	1.35×10^{-4}
SDF / SIREN / Balance-only	10	0.55	1998	1.36×10^{-4}	1.45×10^{-4}
SDF / SIREN / QuotientOpt	0.1	0.0255	163	0.0014	0.0015
SDF / SIREN / QuotientOpt	3	0.85	1984	1.41×10^{-4}	1.40×10^{-4}
SDF / SIREN / QuotientOpt	10	0.55	1957	1.29×10^{-4}	1.40×10^{-4}
Matrix completion / SGD-M	1	0.255	2000	2.24×10^{-8}	2.48×10^{-8}
Matrix completion / SGD-M	3	0.21	2000	3.77×10^{-8}	4.06×10^{-8}
Matrix completion / SGD-M	10	0.165	2000	3.69×10^{-8}	4.08×10^{-8}

Q. Conditional Initialization Confirmations

We additionally freeze the validation-selected learning rates from the original Cora seven-learning-rate loss-based analysis (Table 13) and the complete SDF nine-learning-rate analysis (Table 16). Each method then runs three model initializations, 42, 43, and 44, without learning-rate retuning across initializations. Cora uses 1000 epochs and SDF uses 2000. This adds 24 full-horizon attempts across all four methods.

Data generation, split masks, and shuffle RNG remain at seed 42. Only the model initialization changes. A paired initialization has identical initial model states across the four methods; the three initializations have distinct model states. A protocol audit checks these properties against saved initial checkpoints, data tensors, split indices, and RNG states. Initialization 42 is rerun on CPU so all three confirmation runs use the same execution setting. This audit fixes data, initial models, and RNG states; it does not assert bitwise-identical floating-point optimization trajectories.

Within each run the earliest minimum-validation-loss checkpoint is selected, with test metrics excluded. Every individual result and failure is retained. Mean and sample SD are reported only when all three full-horizon runs for that method complete. The SD describes conditional initialization variability on one fixed data split; these checks do not constitute independent-data replications or a population-risk confidence interval.

Table 21. Conditional initialization confirmation. Data and shuffle seed stay fixed at 42; only model initialization changes over 42, 43, 44. Each method uses its frozen parent-selected learning rate, with no retuning across initializations. Test loss is measured at the validation-selected checkpoint. Mean ± sample SD is shown only for three completed runs. This SD does not describe independent-data uncertainty.

Task	Method	LR	Complete/3	Test loss mean ± SD
Cora / GCN	AdamW	0.01	3/3	1.1376 ± 0.0793
Cora / GCN	SGD-M	0.01	3/3	0.8741 ± 0.0562
Cora / GCN	Balance-only	0.01	3/3	0.8807 ± 0.0551
Cora / GCN	QuotientOpt	0.01	3/3	0.8807 ± 0.0552
SDF / SIREN	AdamW	0.003	3/3	$(6.21 \pm 0.17)\times10^{-5}$
SDF / SIREN	SGD-M	0.003	3/3	$(2.62 \pm 0.62)\times10^{-4}$
SDF / SIREN	Balance-only	0.001	3/3	$(1.18 \pm 0.07)\times10^{-5}$
SDF / SIREN	QuotientOpt	0.001	3/3	$(1.10 \pm 0.26)\times10^{-5}$

Table 22. Every initialization outcome. F is a stopped run; its finite validation-selected checkpoint is retained when available. No initialization is dropped from the record.

Task / method	Init.	LR	Epoch	Val. loss	Test loss	Status
Cora / GCN / AdamW	42	0.01	15	1.1728	1.0525	C
Cora / GCN / SGD-M	42	0.01	36	0.9079	0.8579	C
Cora / GCN / Balance-only	42	0.01	37	0.9141	0.8646	C
Cora / GCN / QuotientOpt	42	0.01	37	0.9141	0.8646	C
Cora / GCN / AdamW	43	0.01	12	1.2179	1.1510	C
Cora / GCN / SGD-M	43	0.01	35	0.8879	0.8278	C
Cora / GCN / Balance-only	43	0.01	35	0.8807	0.8355	C
Cora / GCN / QuotientOpt	43	0.01	35	0.8806	0.8355	C
Cora / GCN / AdamW	44	0.01	13	1.3324	1.2093	C
Cora / GCN / SGD-M	44	0.01	29	1.0943	0.9367	C
Cora / GCN / Balance-only	44	0.01	29	1.0953	0.9420	C
Cora / GCN / QuotientOpt	44	0.01	29	1.0955	0.9422	C
SDF / SIREN / AdamW	42	0.003	1130	4.90×10^{-5}	6.03×10^{-5}	C
SDF / SIREN / SGD-M	42	0.003	1947	2.47×10^{-4}	2.42×10^{-4}	C
SDF / SIREN / Balance-only	42	0.001	1956	1.02×10^{-5}	1.18×10^{-5}	C
SDF / SIREN / QuotientOpt	42	0.001	1993	1.25×10^{-5}	1.39×10^{-5}	C
SDF / SIREN / AdamW	43	0.003	978	5.77×10^{-5}	6.37×10^{-5}	C
SDF / SIREN / SGD-M	43	0.003	1994	2.99×10^{-4}	3.32×10^{-4}	C
SDF / SIREN / Balance-only	43	0.001	1997	1.09×10^{-5}	1.10×10^{-5}	C
SDF / SIREN / QuotientOpt	43	0.001	1975	8.90×10^{-6}	8.84×10^{-6}	C
SDF / SIREN / AdamW	44	0.003	1111	5.94×10^{-5}	6.25×10^{-5}	C
SDF / SIREN / SGD-M	44	0.003	1980	1.88×10^{-4}	2.12×10^{-4}	C
SDF / SIREN / Balance-only	44	0.001	1968	1.19×10^{-5}	1.25×10^{-5}	C
SDF / SIREN / QuotientOpt	44	0.001	1993	1.05×10^{-5}	1.04×10^{-5}	C

Matrix-completion target confirmation. An additional 12 runs use initializations 42, 43, and 44 with the same planted matrix and disjoint masks. Each method's step size is frozen from its earliest parent validation hit at the relative-MSE target 10^{-4}; SGD-M, which never reaches it in the parent search, uses its minimum-validation-loss completed configuration. The initial model changes while data generation and mask RNG remain fixed. The common initial validation MSE is recorded separately for each paired initialization. No step size is retuned across these checks. Every failure and unreached target is retained; initialization means are reported only for three observed target hits.

Table 23. Matrix-completion initialization confirmation with frozen step sizes selected by the parent validation target. Data and masks stay fixed; model initializations are 42, 43, 44. The target is 10^{-4} of the common initial validation MSE of each paired initialization. Every initialization is retained; an unreached target is censored at 2000 epochs. Mean $\pm$ initialization SD is shown only if all three runs reach the target.

Method	LR	Complete/3	Reached/3	Epochs mean $\pm$ SD
AdamW	0.03	3/3	3/3	118.7 ± 16.9
SGD-M	0.3	3/3	2/3	—
Balance-only	10	3/3	2/3	—
QuotientOpt	10	3/3	2/3	—

Table 24. Every matrix-completion initialization: first epoch reaching the target. A dash denotes an unreached target, not an omitted trial. All full-horizon and failure records are retained.

Init.	AdamW	SGD-M	Balance	QO
42	138	—	117	107
43	107	1234	—	—
44	111	1851	105	171

R. Additional Complete Loss Trajectories

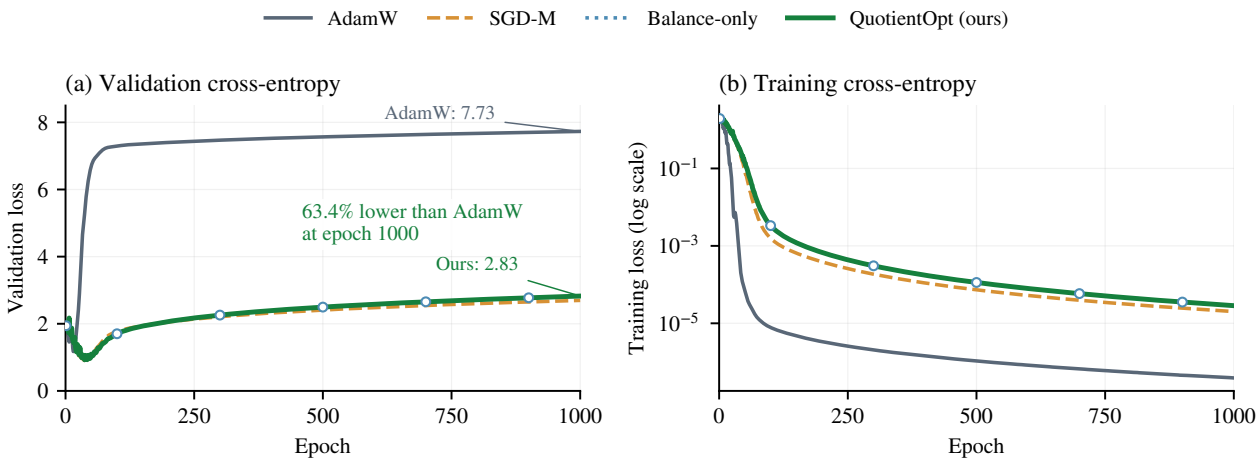


Figure 6. **Core loss curves: lower validation loss than AdamW.** Cora/GCN input adapter, rank 4, seed 42, $\beta = 0.9$. The complete seven-learning-rate block selects $\eta = 10^{-2}$ for every method using validation accuracy; test metrics do not enter selection. All 1000 raw epoch observations are shown without smoothing or interval truncation. (a) Validation cross-entropy: QuotientOpt's final loss is 63.4% below AdamW's, while SGD-M is slightly lower and balance-only nearly overlaps QuotientOpt. Open green circles identify the overlapping balance-only trajectory. (b) Training cross-entropy on a logarithmic scale: AdamW attains the smallest training loss.

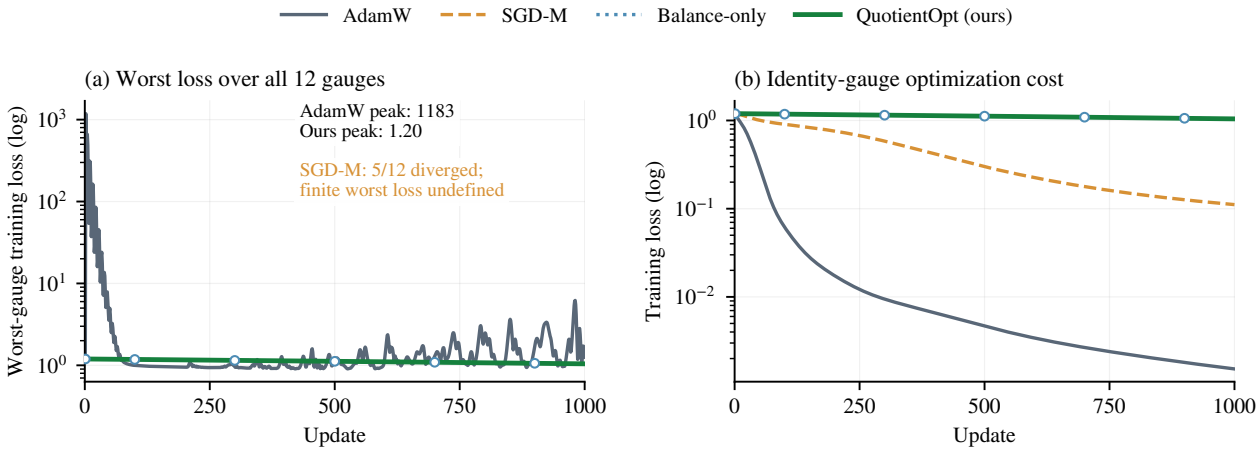


Figure 7. **Core sensing loss curves: robustness and optimization cost.** True rank-one sensing, 128×128, rank 16, 19,200 observations, seed 42, $\eta = 10^{-3}$, $\beta = 0.9$; all 1000 raw observations and logarithmic loss axes. (a) At each update, the maximum training loss over every one of the 12 prescribed gauges. AdamW and both balanced arms have all 12 completed trajectories. The five confirmed SGD-M failures preclude a finite 12-gauge envelope; an envelope of only its successful cases is not substituted. (b) The identity-gauge reference: AdamW fits this problem much better at the fixed learning rate. Both panels retain the overlapping balance-only control. The gain in (a) is gauge robustness, not faster identity-gauge recovery or an accuracy gain uniquely due to projection.

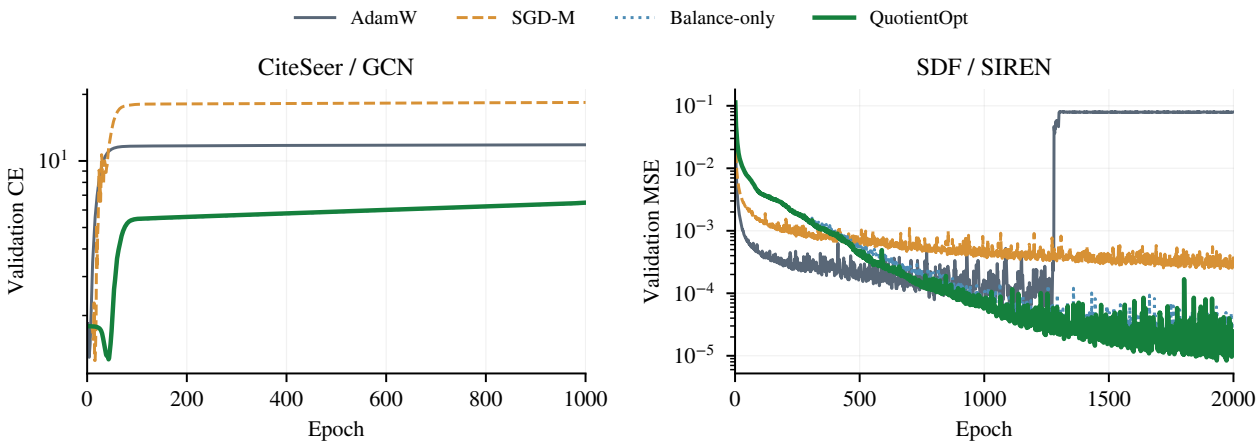


Figure 8. Exploratory validation-loss case studies from completed new-task blocks. The presentation rule requires at least 1% lower minimum validation loss for a full-horizon QuotientOpt run than AdamW, including its best finite stopped-trial checkpoints; test outcomes do not select panels. Every method receives the same thirteen-setting optimizer budget and every selected trajectory is shown in full. All four methods are retained. This selected-case figure does not establish improvement over every baseline. All tasks, test metrics, and unfavorable outcomes are retained in Table 17 and Appendix O.

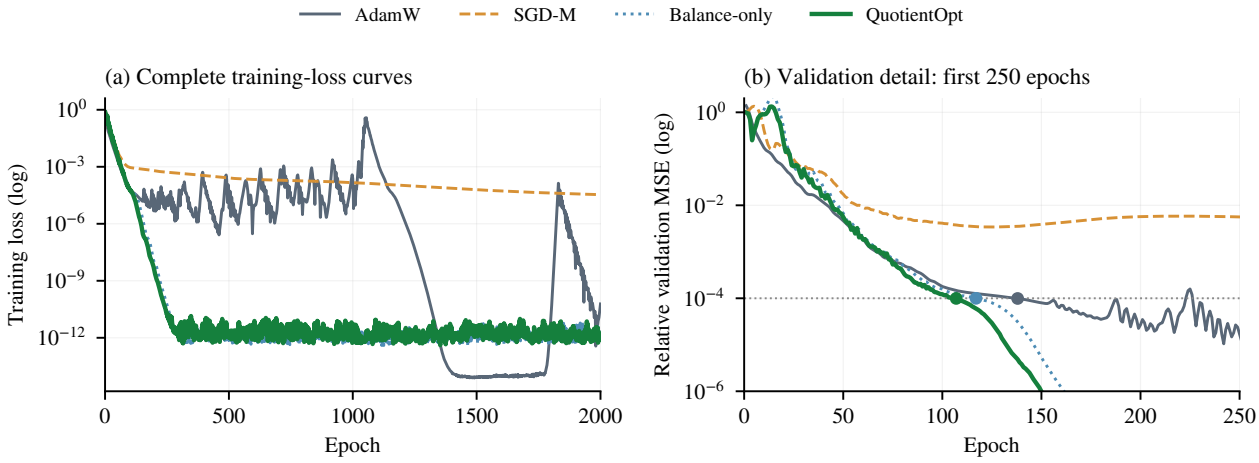


Figure 9. **Additional matrix-completion loss curves.** Rank 8, planted condition 100, 128×128, seed 42. Each method selects its setting by the earliest validation epoch reaching 10^{-4} of the common initial validation MSE, over the same 13-setting budget. For SGD-M, which never reaches the target, the minimum-validation-loss full-horizon setting is shown. (a) Complete raw training-loss observations, without smoothing. (b) A common first-250-epoch validation zoom, with first-hitting points and the target line. These are actual selected-run trajectories. Figure 5 separately reports tuning envelopes; the full record includes every setting and stopped prefix.

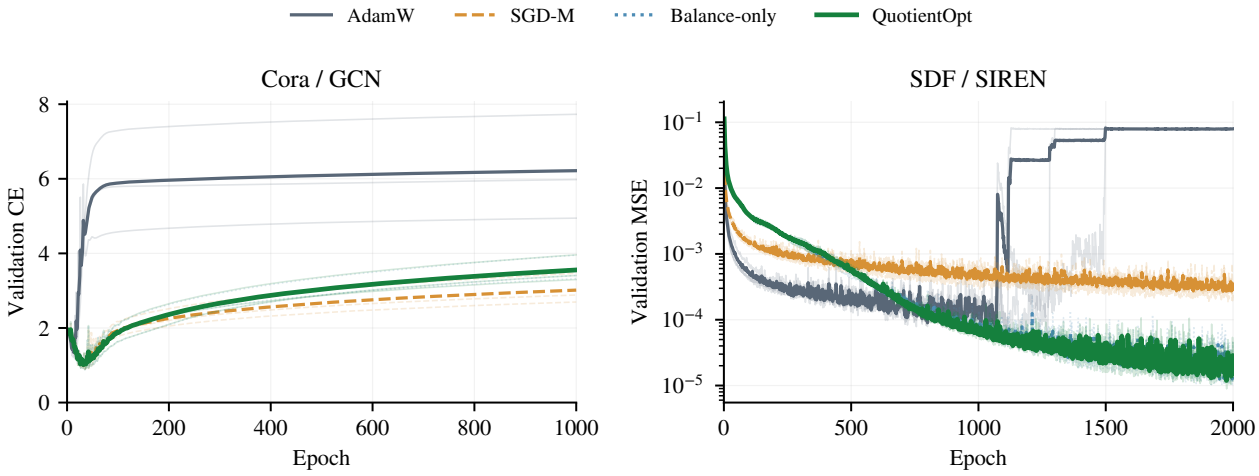


Figure 10. **Conditional initialization loss checks.** Data and shuffle seed 42 are fixed, with model initializations 42, 43, and 44. Parent-selected learning rates are frozen per method, with no per-initialization retuning. Thin lines show every initialization; thick lines show their arithmetic mean validation loss at each epoch. All raw epochs are retained without smoothing. The mean curves are descriptive summaries, not confidence bands. Test outcomes and all four controls are retained in Table 21.

S. Initial Central-Setting Comparisons

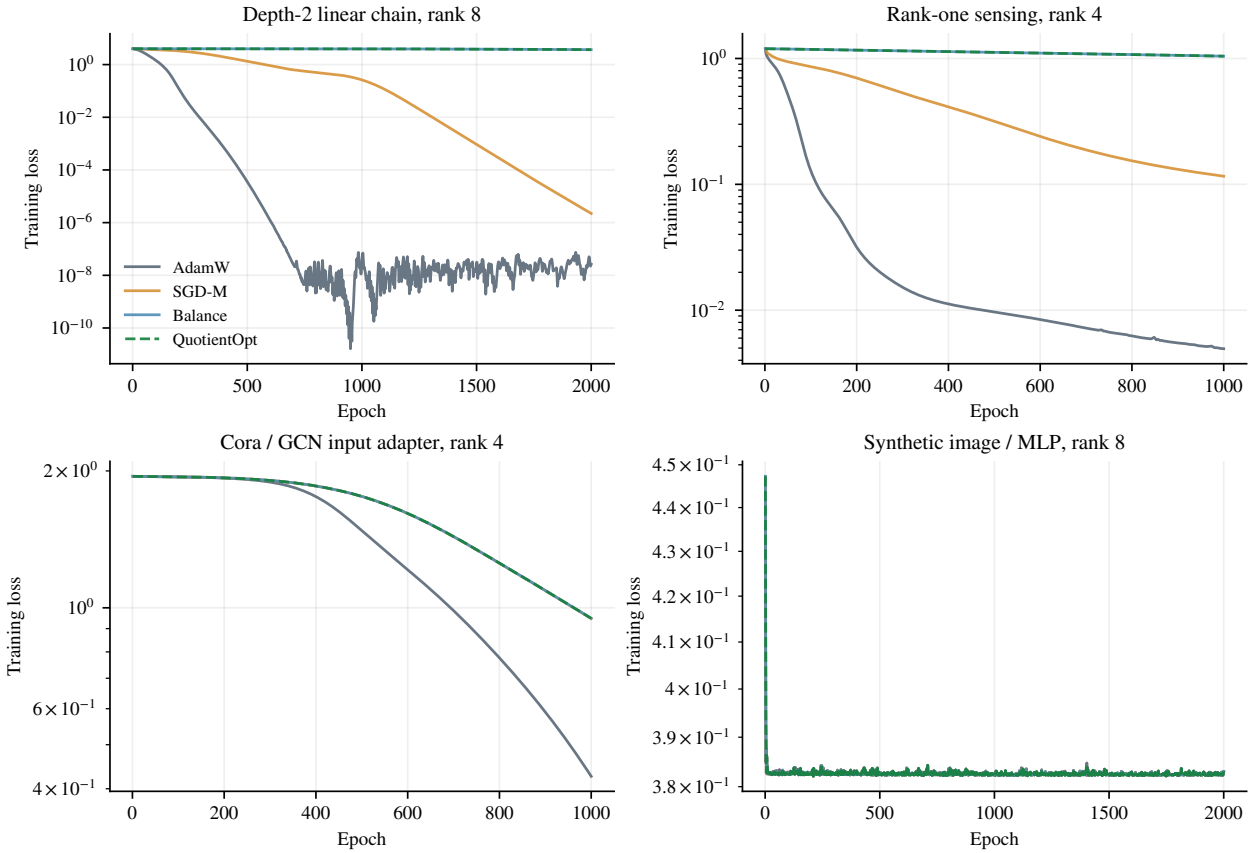


Figure 11. Initial completed central-setting replacement cohorts, before exhaustive tuning. The four panels show all four families with a matched completed cohort at the figure snapshot: 2000 epochs for depth-2 chains, sensing, and INR; 1000 epochs for Cora. The balanced momentum methods nearly overlap. AdamW optimizes the chain and sensing objectives much better at this setting; lower training loss on Cora does not imply higher test accuracy. These trajectories do not establish a best-tuned-method comparison.

Central-setting outcomes across families. At the displayed central setting, Cora/GCN's final test accuracy is 69.9% for balance-only and QuotientOpt, versus 66.5% for AdamW and 70.2% for SGD-M. This is a positive comparison against AdamW, not against the stronger SGD-M endpoint. On the initial rank-one sensing and depth-2 chain cohorts, AdamW has substantially smaller final reconstruction error than the balanced momentum arms. The initial INR/MLP cohort has nearly tied held-out error across all four methods. Figure 11 reports the corresponding complete training trajectories. The search remains incomplete, so these are scoped central-setting outcomes, not conclusions from the full learning-rate/momentum/rank grid.

T. Completed Replacement Experiments

Only complete runs with the configured horizon and a saved result record enter these tables. Runs from the two-epoch implementation checks are excluded. This snapshot is generated from the experiment artifacts and can grow as the full grid executes. A cohort requires all four methods at a matched task/rank/hyperparameter setting; partial cohorts remain in the progress record without a winner label.

Table 25. Completed matched main-experiment four-method cohorts. Gauge and ablation runs are reported separately. Values are final task metrics: planted-operator reconstruction error for matrix/chain tasks, held-out metrics for other tasks. Lower error is better; higher accuracy/PSNR is better. Only the first 30 cohorts are typeset here; all records remain in the machine-readable progress artifact.

Task	Metric	AdamW	SGD-M	Balance	QO
poisson/pinn r16, lr=0.001	Rel. L_2	0.06223	0.07767	0.05577	0.05384
synthetic/chain r8/L2, lr=0.001	Rel. error	0.06798	0.06799	0.9579	0.9579
cora/gcn r4, lr=1e-05	Accuracy	0.452	0.4	0.401	0.401
synthetic/chain r8/L2, lr=0.001	Rel. error	0.01359	0.01363	0.959	0.959
cifar10/mlp r32, lr=0.001	Accuracy	0.4942	0.4782	0.4723	0.4724
image/mlp r8, lr=0.001	PSNR	4.536	4.539	4.539	4.539
sensing/factorization r16, lr=0.001	Rel. error	0.02616	0.2498	0.9277	0.9277
image/mlp r32, lr=0.001	PSNR	4.538	4.542	4.542	4.542
sensing/factorization r8, lr=0.001	Rel. error	0.02687	0.2578	0.9675	0.9675
cora/gcn r4, lr=0.001	Accuracy	0.665	0.702	0.699	0.699
image/mlp r16, lr=0.001	PSNR	4.543	4.545	4.545	4.545
cifar10/mlp r16, lr=0.001	Accuracy	0.4885	0.4808	0.4771	0.4776
cifar10/mlp r4, lr=0.001	Accuracy	0.477	0.4684	0.4744	0.4721
cora/gcn r4, lr=0.0001	Accuracy	0.631	0.711	0.712	0.712
cora/gcn r4, lr=0.01	Accuracy	0.665	0.73	0.698	0.698
cifar10/mlp r8, lr=0.001	Accuracy	0.4837	0.4719	0.4767	0.4704
cora/gcn r4, lr=3e-05	Accuracy	0.603	0.621	0.622	0.621
poisson/pinn r8, lr=0.001	Rel. L_2	0.05951	0.07489	0.07868	0.07868
cora/gcn r8, lr=0.001	Accuracy	0.745	0.709	0.703	0.7
wikitext2/10m r8, lr=0.001	Perplexity	1.104e+08	1.135e+16	2.833e+14	3.499e+14
wikitext2/10m r4, lr=0.001	Perplexity	1.864e+09	1.777e+15	6.815e+13	5.979e+13
cora/gcn r16, lr=0.001	Accuracy	0.734	0.71	0.701	0.703
sensing/factorization r4, lr=0.001	Rel. error	0.03213	0.2173	0.8936	0.8936
cora/gcn r4, lr=0.003	Accuracy	0.646	0.711	0.702	0.702
synthetic/chain r8/L2, lr=0.001	Rel. error	7.996e-05	0.000742	0.9592	0.9592

Task	Metric	AdamW	SGD-M	Balance	QO
cora/gcn r4, lr=0.0003	Accuracy	0.64	0.7	0.701	0.702

T.1. RU01 completions and narrowed continuation

The formal RU01 report increased from 139 to 158 full-horizon results between the recorded snapshots, and from 20 to 26 complete four-method cohorts. Table 26 includes every new completion in that interval, without selecting rows by test performance. Final-epoch values in this table are distinct from the validation-selected checkpoints of the additional four-task search.

Table 26. **Nineteen full-horizon RU01 completions since the 139-run snapshot.** Fixed-setting, final-epoch held-out endpoints at learning rate 10^{-3} and seed 42; no test-based row selection. All nineteen are listed, including unfavorable outcomes. CIFAR-10/MLP rank 32 and rank-16 matrix sensing are complete four-method cohorts led by AdamW. Cora/GCN hidden rank 4 has only an AdamW–QuotientOpt pair at this snapshot.

Task / setting	Method	Metric	Final test
CIFAR-10 / MLP, $r = 32$	AdamW	Accuracy (%)	49.42
CIFAR-10 / MLP, $r = 32$	SGD-M	Accuracy (%)	47.82
CIFAR-10 / MLP, $r = 32$	Balance-only	Accuracy (%)	47.23
CIFAR-10 / MLP, $r = 32$	**QuotientOpt**	Accuracy (%)	47.24
Cora / GCN, hidden, $r = 4$	AdamW	Accuracy (%)	77.10
Cora / GCN, hidden, $r = 4$	**QuotientOpt**	Accuracy (%)	77.70
WikiText-2 / 10M decoder, $r = 16$	AdamW	NLL	17.230
WikiText-2 / 10M decoder, $r = 16$	SGD-M	NLL	36.777
WikiText-2 / 25M decoder, $r = 4$	AdamW	NLL	36.610
Poisson / DeepONet, $r = 8$	SGD-M	Relative L_2	0.4047
Synthetic image / SIREN, $r = 8$	Balance-only	MSE	0.04949
Matrix sensing / factorization, $r = 32$	SGD-M	Relative error	0.21516
Deep linear chain / depth-2 chain, $r = 8$	Balance-only	Relative error	0.98476
Synthetic image / MLP, $r = 32$	SGD-M	MSE	0.35142
Matrix sensing / factorization, $r = 16$	**QuotientOpt**	Relative error	0.92767
Matrix sensing / factorization, $r = 16$	AdamW	Relative error	0.02616
WikiText-2 / 10M decoder, $r = 8$	**QuotientOpt**	NLL	33.489
Deep linear chain / depth-2 chain, $r = 8$	SGD-M	Relative error	0.06799
Cora / GCN, input, $r = 16$	Balance-only	Accuracy (%)	70.10

Frozen Poisson/PINN follow-up. The original grid workers were paused after the 158-run snapshot; four in-progress job trees retain checkpoints and trajectories. The separate follow-up fixes the Poisson/PINN model, rank 16, factor learning rate 10^{-3}, momentum 0.9, damping 10^{-6}, and 2000 epochs from the completed seed-42 four-method cohort. Data and shuffle seed remain 42. Model initializations 43 and 44 are crossed with AdamW, SGD-M, balance-only, and QuotientOpt, for eight new runs with no per-method retuning. All test outcomes and numerical stops enter the follow-up report; results are not pooled into the original grid counts. All eight new runs complete the prescribed 2000 epochs. The earliest minimum validation-MSE checkpoint is evaluated on held-out data; the complete per-initialization records and final-horizon endpoints are in Table 27. The eight saved selected checkpoints were independently re-evaluated and agree with their reported test metrics.

Table 27. **All Poisson/PINN initialization outcomes.** Best epoch minimizes validation MSE; the selected test and final-horizon columns are held-out relative L_2 errors. Seed 42 is the pre-existing original-grid run; seeds 43 and 44 are the separately predeclared follow-up.

Init.	Method	Best epoch	Best val. MSE	Selected test	Final test
42	AdamW	1945	0.000203	0.06143	0.06223
42	SGD-M	1989	0.000331	0.07777	0.07767
42	Balance-only	1995	0.000147	0.05022	0.05577
42	QuotientOpt	1986	0.000148	0.04994	0.05384
43	AdamW	1993	0.000191	0.05875	0.05945
43	SGD-M	1969	0.000226	0.06438	0.06500
43	Balance-only	1992	0.000129	0.04655	0.04733
43	QuotientOpt	1987	0.000128	0.04663	0.04737
44	AdamW	1915	0.000229	0.06472	0.06732
44	SGD-M	1986	0.000297	0.07450	0.07526
44	Balance-only	1996	0.000155	0.05255	0.05726
44	QuotientOpt	1996	0.000155	0.05255	0.05731